

Q-CRAFT Explorer Companion Guide

A course in climate-aware fiscal projection

Teal Insights & NatureFinance

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Preface

The whole model in two minutes

Q-CRAFT, the Quantitative Climate Risk Assessment Fiscal Tool, is how the IMF Fiscal Affairs Department projects public finances under climate warming. It takes a country's published macroeconomic and fiscal data, adds assumptions about demography, productivity, prices and fiscal policy, and projects the [debt-to-GDP ratio](#) to 2099 under a baseline and [six warming scenarios](#).

Underneath sits one equation fed by three numbers. Each year, the [debt dynamics equation](#) takes last year's debt ratio, grows it by the interest rate, shrinks it by economic growth, and subtracts the primary balance. Warming reaches the projection through two of those numbers, by two named channels. Hotter years drag on productivity, so growth slows. Spending stays rigid while the tax base slows with it, so the primary balance weakens. Each year's climate dent is small, and the model's subject is what seventy years of small dents compound into.

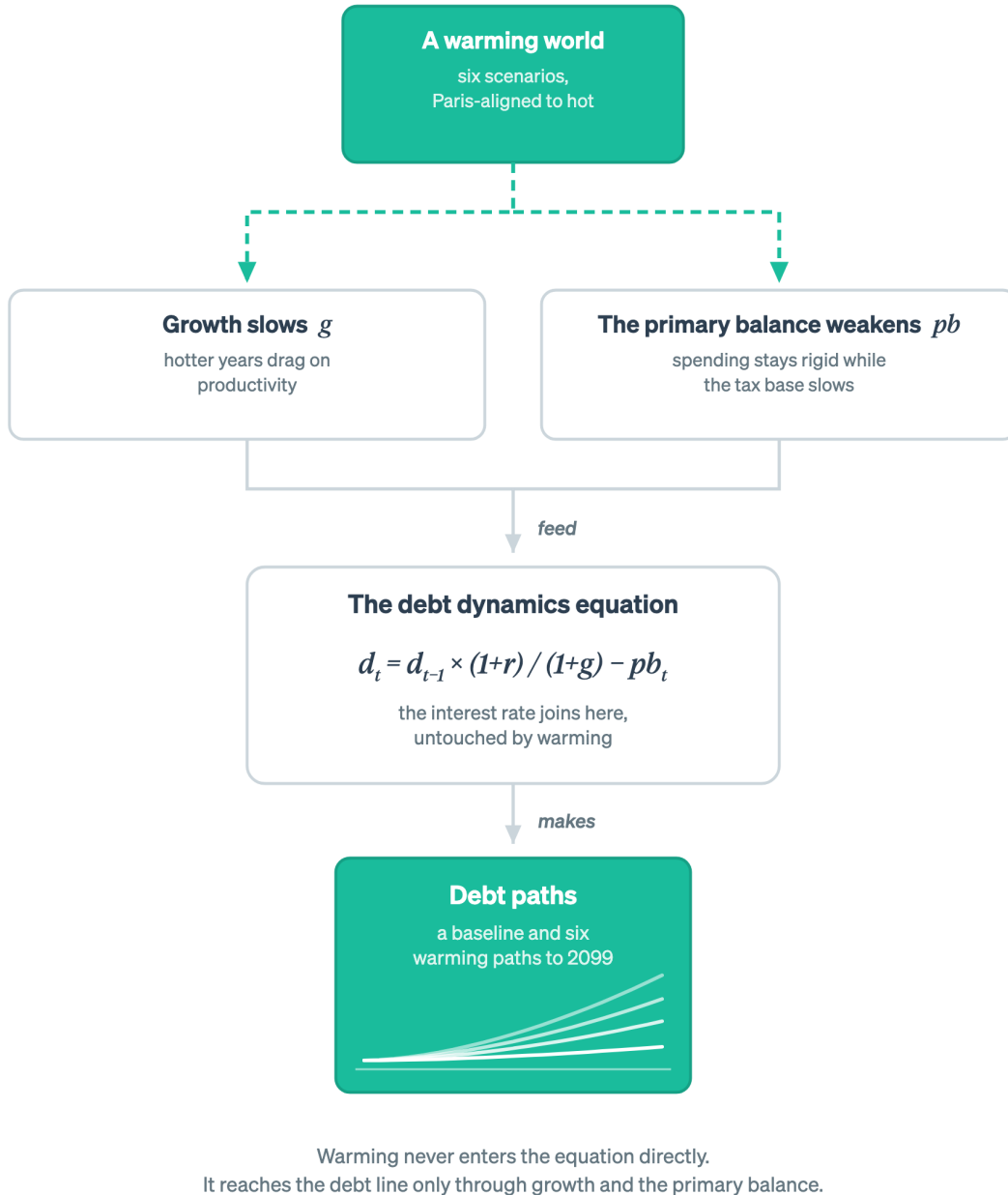


Figure 1: The whole model in one picture. Warming slows growth and squeezes the primary balance, and the debt dynamics equation turns those two dents into debt paths to 2099.

That design buys coverage and speed. The tool runs on already-published data: the IMF’s [WEO](#) macro series, the UN’s population projections, and a short list of choices you set. It needs no bespoke country dataset, no new survey and no waiting for a mission, so it runs for most of the world (the climate estimates cover 171 economies), and a team can have a first pass by week’s end.

The same design keeps the model to one slice of climate change, so read the results as a lower bound, not a central estimate. The scenarios carry a single channel, the slow effect of rising temperature on productivity. They exclude tipping points, nonlinearities, natural disasters,

sea-level rise, the public spending adaptation takes, and whatever channel is specific to your own country ([User Guide](#), pp. 5-6). This is the looking-under-the-streetlight trade, made with open eyes: the model looks where the light is because the lit ground is what nearly every country has published. What the beam misses is left out of the number, not out of the world. Chapter 6 sets out the exclusions with the sources.

A tool that runs anywhere on published data cannot carry the channel only your country has. The strengths and limitations follow from that one design choice. This guide teaches what the tool does, why it exists, and what it leaves out. It is the first in a series asking those three questions of the models behind sovereign climate-fiscal analysis.

The results reach published fiscal-risk documents

The Fiscal Affairs Department built Q-CRAFT for its own climate technical assistance, and finance ministries carry the results into fiscal-risk documents they already publish. The cycle has run end to end in print: Uganda’s Fiscal Risk Statement for FY 2024/25 has a chapter on climate fiscal risks whose figures are sourced to “QCRAFT (2023)”, tracing back to a five-day workshop written up as *Uganda: PFM Climate Assessment* ([IMF, 2024](#)). Kenya, Thailand and Ethiopia carry the course’s teaching examples.

i This is not an IMF product

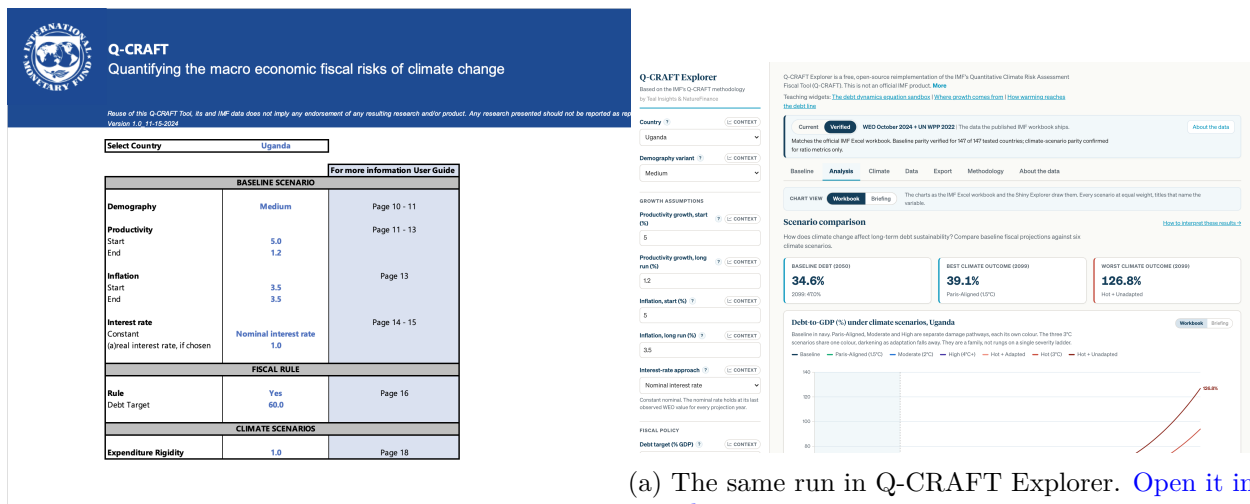
Q-CRAFT Explorer and this course are an independent project by [Teal Insights](#) and [NatureFinance](#). Neither is an official IMF product and neither carries IMF endorsement. Both are meant to be complementary to the IMF’s own training materials and to the Q-CRAFT User Guide ([Tim and Rahman, 2024](#)), which remain the authoritative references. Where this guide explains a concept it cites the User Guide section behind it, so you can always check the teaching against the source.

The workbook and the Explorer run the same model

This course teaches you to run Q-CRAFT both ways: in the official IMF Excel workbook the Fund publishes on its website, and in Q-CRAFT Explorer, an open-source web tool. The modules teach on the Explorer. The appendix [Running the official workbook](#) is the quick orientation to operating the workbook, and the IMF’s User Guide remains its official manual.

Teal Insights and NatureFinance build the Explorer, its code permissively licensed open source: free to use, copy and improve (the MIT license). The economics is the IMF’s on either surface. The interface, the automatic data loading and the record of what you assumed are ours. Under the interface sits a calculation engine that is its own open-source package, separate enough that a ministry can run the whole tool on its own machines, and checked against the workbook cell by cell, with baseline parity exact for 147 of 147 tested countries and climate-scenario parity confirmed for ratio metrics only. Chapter 2 sets out what that claim covers and what it does not.

The workbook is where Q-CRAFT lives officially, and Excel is a deliberate choice. Every finance ministry has Excel, staff already know it, and a Fund tool must run in nearly two



(a) The same run in Q-CRAFT Explorer. [Open it in your browser.](#)

(a) The official workbook's Dashboard, Uganda selected. [Download it from the IMF.](#)

hundred countries without assuming data or infrastructure that may not be there. Q-CRAFT is designed simple so it runs pretty much anywhere.

The Explorer runs that same economics on the web, built so your time goes on the analysis rather than on moving numbers by hand. In practice:

1. **Up-to-date data.** A posted workbook ships with the WEO vintage it was built on, and that vintage ages between releases. We keep the Explorer's bundled data current.
2. **Guidance where you need it.** Help for choosing a parameter sits beside the parameter, at the moment of choosing. In a deadline week, that is the moment the help gets read.
3. **Fewer steps between the question and the chart.** The data is already loaded and the scenarios already drawn, so more of the hour goes on the policy thinking.
4. **The export packet.** Assumptions, written remarks on them and presentation-ready output come out together, so less of the write-up gets copied by hand.

Chapter 2 argues these four advantages in full. After that the course mostly stops arguing them and shows them, one at a time, in the flow of the work.

The questions it answers

Q-CRAFT answers a narrow set of questions, each by comparing two runs of the same model. Run the projection with no climate damage and you have the baseline. Switch one warming scenario on and run it again. The gap between the two paths, read at the year you care about, is the answer.

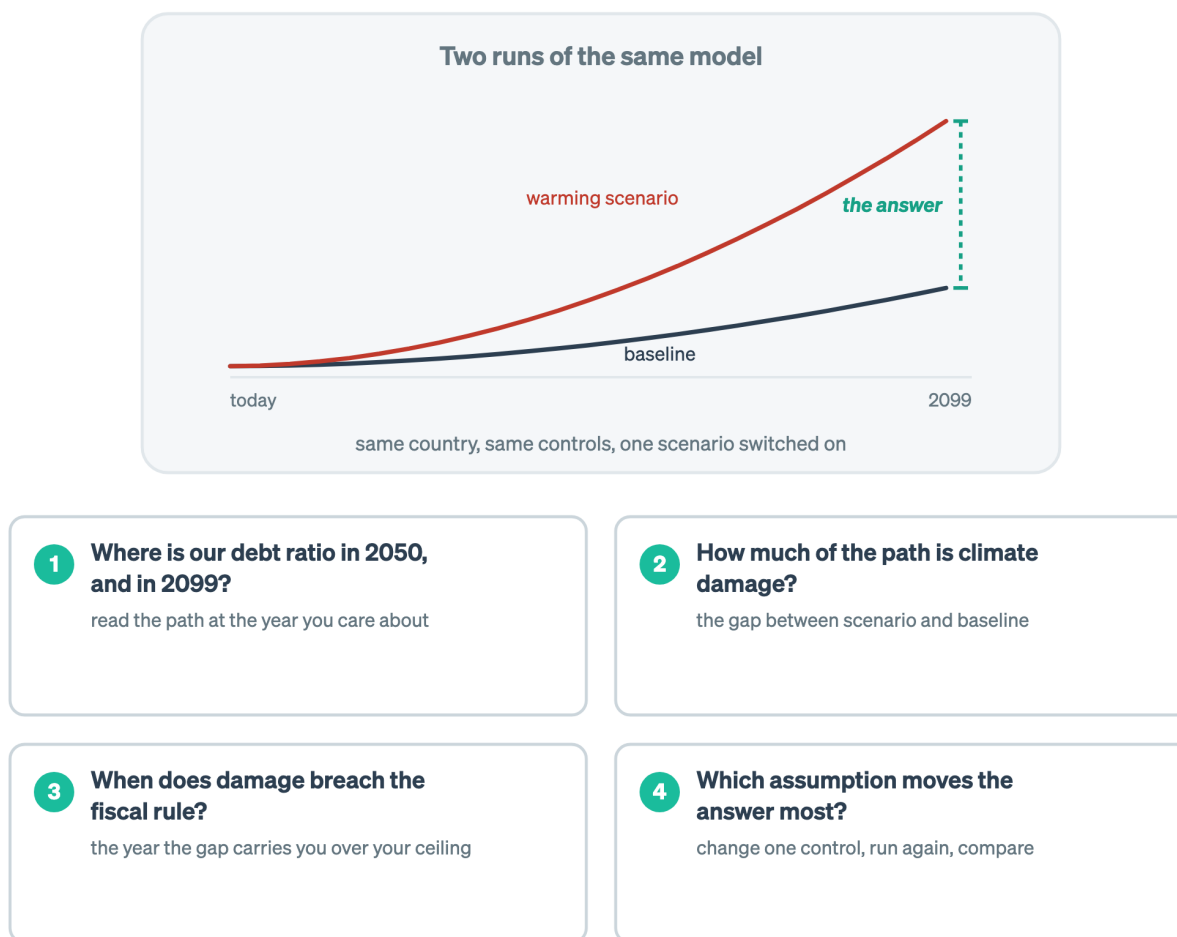


Figure 4: The tool answers four questions, and it answers each one the same way: two runs of the same model, and the gap between them.

None of the four is a forecast. Each is a difference between two projections built on assumptions you chose, which is why the assumptions are worth arguing about and worth writing down.

Who this is for

This course is designed for intelligent, busy people who must run this class of fiscal projection tool, teach it, or read its output. Ministry of finance economists are its primary readers: they run the tool, then defend the numbers to the officials who sign them off. It equally serves IFI and technical assistance staff who build capacity around such tools, and researchers and analysts who read the output and must decide what it lets them claim.

What you will be able to do

The course sets three objectives and weights them equally:

1. **Run the tool, on both surfaces.** Select a country, set the parameters, read the charts and export the results, in the IMF workbook and in the Explorer.
2. **Understand what you are doing when you run it.** Trace the one equation and its three inputs, and see which of your assumptions is moving the answer.
3. **Interpret the output appropriately.** Say what a result supports and what it does not, with its strengths and its limits attached.

The first objective takes an afternoon. The second and third are the ones that hold up in a meeting.

How the course is organised

The course has seven modules, each built around something you can do at the end of it.

Module	What it gets you
0. How to use this course	The capstone stated up front, a self-assessment, and the path that fits your background
1. What Q-CRAFT does and why it exists	A full projection in ten minutes, and the map: three numbers, one equation
2. The model, in three steps	An equation, a growth engine, a climate overlay: the whole tool, traceable end to end
3. Choosing the parameters	Every parameter set and defended, with the rationale written down
4. A worked example, end to end	One country from assumptions to a fiscal risk paragraph, then two more with less help
5. What the tool tells you and where it stops	The strengths, the documented exclusions, and which questions belong to a different tool
6. The capstone	The capstone brief and rubric, and a look back at where you started

The tool covers most of the world, so the examples move country by country and the mechanism picks the country. Demography reads clearest in a country whose workforce is already shrinking, set against one still growing fast. Productivity convergence reads clearest in an economy catching up to the frontier. The climate channel reads clearest where exposure is high. Module 4 runs Uganda the whole way through, because it is the tool's [golden-master](#) verification country and every walkthrough figure can be checked against published documents. The exercises after it run on other countries, and the capstone runs on yours.

Start at Module 0. It takes about ten minutes and tells you which parts of the later modules you can read quickly. Module 2 is the spine: it assembles the whole tool in three steps, and everything after it either chooses the inputs, runs the thing or reads the output.

The [Glossary](#) defines key terms and the [References](#) section annotates the primary sources behind the methodology.

Two appendices sit outside the learning path. [Running the official workbook](#) covers the Excel half of the both-ways promise. The [appendix on co-design and the SovTech vision](#) is about the project behind the tool rather than about using it. Read it to shape what the next version does.

What this course defers to

This is an educational companion to the IMF's User Guide ([Tim and Rahman, 2024](#)), the authoritative methodology reference, and to the IMF's own training materials, the authoritative teaching materials. Where this guide and the User Guide ever differ, the User Guide is right.

The official materials

The official sources are one click away throughout.

- The IMF's [Q-CRAFT page](#), which posts the [User Guide \(PDF\)](#) and the [Excel workbook](#) itself
- The climate methodology: [Kahn et al. \(2021\)](#), and [IMF How-To Note 2025/009](#), the public documentation of the FADCP Climate Dataset
- The published worked case: the [Uganda PFM Climate Assessment summary](#) (IMF, 2024) and [Uganda's Fiscal Risk Statement FY 2024/25](#) (MoFPED, 2024)
- The data: the [IMF World Economic Outlook Database](#) and the [UN World Population Prospects](#)
- The wider assessment framework: the [C-PIMA Handbook](#) (IMF, 2025)

Each entry on the [References](#) page carries a note on what the source is and how this course uses it.

 This is an initial version

Q-CRAFT Explorer is a starting point rather than a finished product. We want to co-design the next version with you. See the [co-design appendix](#) for how to get involved.

 Try the App

[Open Q-CRAFT Explorer](#) to follow along as you read. The source code is on [GitHub](#).

Colophon

This course is permissively licensed open source, free to use, copy and improve, and it builds completely from its repository. It sets in Inter, IBM Plex Serif and IBM Plex Mono, three faces under the SIL Open Font License, bundled with the book. Clone the repository, run `quarto render docs/companion-guide`, and you get this book in the same three faces on a machine with no network access. Opening a page fetches no font from anyone's server, which matters in a ministry network that blocks outside hosts.

Teal Insights also publishes a house edition of the same content, set in Klim Type Foundry faces where the license permits. Those font files are not in the repository, because their license covers tealinsights.com. Where they are absent, every font stack falls through to the open faces and the book still sets. No word, number or figure in the course depends on which edition you are reading.

The figures are built from the repository too. `scripts/build_course_map.py` draws the course map that opens every module, `scripts/build_preface_figures.py` draws the two

figures in this preface, and `scripts/build_parameter_context.py` draws the source-data figures in Chapter 4 from the same Parquet files the Explorer runs on. The screenshots are real captures: `scripts/build_app_screenshots.py` shoots the Explorer for the modules, and `scripts/capture_both_ways_shots.py` shoots the preface pair and the workbook appendix from the live deployment and the posted v10 workbook file.

Chapter 1

How to use this course

In this module

You will see the deliverable the whole course builds toward, the yardstick for every later module. You will calibrate your starting point in about ten minutes, then pick the path that fits, so nobody sits through material they do not need.

Fast path

Short on time? Answer the three questions in Section 1.5 and read the path table in Section 1.6. About ten minutes, and it tells you which later modules you can skip.

1.1 A ministry has already published this deliverable

This course teaches you to write an ordinary ministry-of-finance product. Fiscal risk statements, budget framework papers, medium-term debt reports: most finance ministries publish in this genre every year. The climate section is the recent arrival. At least one published fiscal risk statement already carries a climate section built with the tool this course teaches, figures sourced to the tool by name.

Chapter 5 opens that document and works through it, page by page, against numbers you can check. Until then, the target is a real published section rather than an exercise.

You will produce two things:

1. **An export packet.** The parameters you chose, a written rationale for each judgment call, and the polished charts and CSV that Q-CRAFT Explorer exports.
2. **A two-paragraph draft** in the register of your ministry's fiscal-risk documentation. If your ministry publishes none yet, the published section in Chapter 5 is the model.

That packet is the capstone. You defend it to the senior officials who sign the document off. Chapter 7 contains the brief and rubric it is marked against. Every module in between either feeds that packet or tells you when not to trust it.

i This course defers to the User Guide

The IMF's User Guide ([Tim and Rahman, 2024](#)) remains the authoritative methodology reference. This course explains the concepts behind the tool and cites the User Guide section for each, so you can go deeper.

1.2 By the end of this module you can

- **Locate** your starting point against the three self-assessment tables
- **Diagnose** which of three concept-inventory items you answered from intuition rather than from the [debt dynamics equation](#)
- **Select** the path (A, B or C) matching your background, and justify the choice
- **Describe** the capstone deliverable in one sentence to a colleague who has not taken the course

1.3 Where you are in the course

The map below opens every module. You meet it here first.

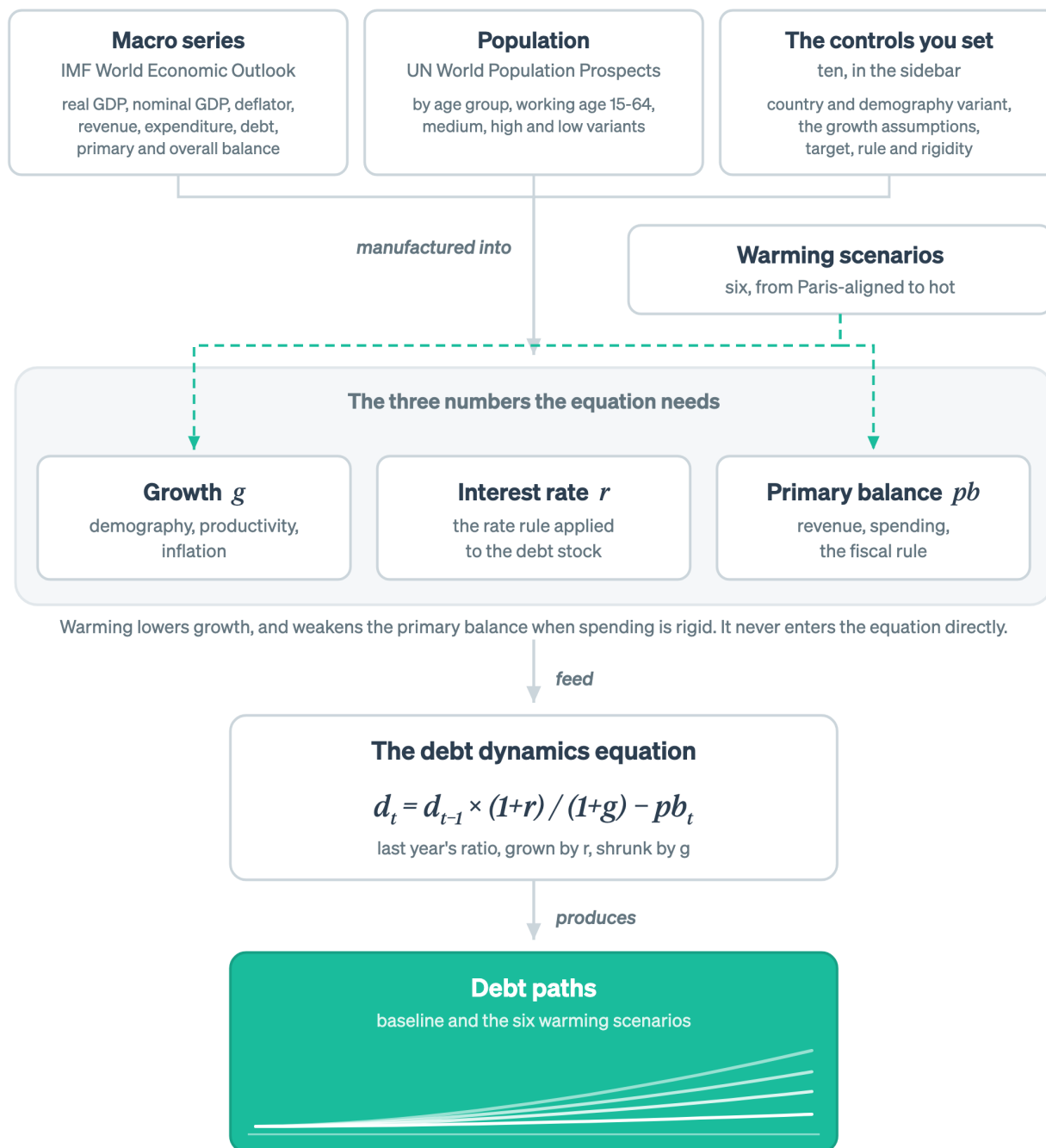


Figure 1.1: The course map. Two published data sources and the controls you set are manufactured into the three numbers the debt dynamics equation needs. This module fixes the destination on the right.

The course map lights only the destination, the part to hold on to now. Later modules light their own nodes.

1.4 Self-assessment: your starting point in three areas

Rate yourself once in each of the three areas below, picking the description closest to true. No one scores or sees your answers. The point is to route you and give you something to compare against in Chapter 7.

On long-term fiscal projection:

	Description
A1	I have not worked with a debt sustainability analysis. I know what the debt-to-GDP ratio is and roughly why it matters.
A2	I have read DSAs and can follow the tables. I could not build the projection myself without help.
A3	I have produced or reviewed a DSA for a country. I could set up a baseline projection this week if asked.

On climate-fiscal analysis:

	Description
B1	I work on climate policy or climate finance. I have not connected it to a debt projection.
B2	I have seen climate scenarios used in fiscal analysis and can describe what they are for.
B3	I have run or interpreted a climate-scenario fiscal projection for a country.

On the tool itself:

	Description
C1	I have never opened Q-CRAFT, in Excel or on the web.
C2	I have opened it and clicked around. I could not defend the parameter settings I used.
C3	I have run Q-CRAFT for a country and explained the output to someone else.

Write your three answers down. You will be asked for them again at the end of the course.

1.5 Three questions before you start

These three questions check which intuitions you carry in rather than what you have memorised. Answer each one, commit to it in writing, then open the answer.

! DRAFT FOR TEAL: concept-inventory question 1 (interest-growth differential)

Question. A country's government runs a **primary balance** of exactly zero: revenue equals non-interest spending, to the last unit of local currency. The nominal interest rate on its debt is 9 percent. Nominal GDP growth is 6 percent. Over the next year, the debt-to-GDP ratio will:

- (a) Fall, because the government is not adding to its debt.
- (b) Stay flat, because a zero primary balance is by definition debt-stabilizing.
- (c) Rise, even though the government borrowed nothing new for programmes.

Answer

(c) Rise. Debt grows at the interest rate. The economy it is measured against grows more slowly. So the ratio climbs by roughly $(9 - 6) / 1.06$ in the first year, about 2.8 percent of the starting ratio. A country starting at 50 percent of GDP ends the year near 51.4 percent with no policy change.

If you picked (b), you hold the most expensive misconception in this material. A zero primary balance stabilizes debt only when the interest rate equals growth. Every other time, the interest-growth gap is doing work someone has to pay for. Step 1 of Chapter 3 is built around that gap.

If you picked (a), you may be thinking of the deficit rather than the ratio. The stock of debt is flat in local currency. The ratio is not, because the denominator, GDP, moved.

! DRAFT FOR TEAL: concept-inventory question 2 (expenditure rigidity)

Question. In Q-CRAFT Explorer you set **expenditure rigidity** to 1.0 and run a climate scenario. What have you assumed?

- (a) Spending falls in step with GDP, holding the expenditure-to-GDP ratio at its baseline level.
- (b) Spending stays at its baseline level in local currency even as GDP falls, so the whole revenue loss lands on the deficit.
- (c) Spending is capped by the fiscal rule, whatever happens to GDP.

Answer

(b). Rigidity 1.0 means spending is sticky. The wage bill, pensions and existing commitments do not move when growth disappoints, so the whole climate-driven revenue shortfall becomes borrowing. That is the worst case, and the tool makes it the default on purpose.

If you picked (a), you have the scale inverted. That is rigidity 0.0. The inversion flips the sign of your headline risk number. It is the easiest way to publish a wrong figure from this tool.

If you picked (c), you have merged two separate controls. The fiscal rule and expenditure rigidity are different parameters acting on different scenarios. Chapter 4 separates them.

! DRAFT FOR TEAL: concept-inventory question 3 (what the output is)

Question. A Q-CRAFT run shows debt-to-GDP at 66 percent in 2099 under the Hot scenario, one of the tool’s six [warming scenarios](#), against 47.5 percent in the baseline. A colleague asks what that means. Which answer is defensible?

- (a) The country’s debt will be about 66 percent of GDP in 2099 if the world warms on that path.
- (b) Under stated assumptions, climate damage of this magnitude would add about 18.5 percentage points of GDP to the debt ratio by 2099 relative to the same projection without climate damage.
- (c) Climate change will cost the country 18.5 percent of GDP.

Answer

(b). The number that survives scrutiny is the *difference between two runs of the same model*, stated with its assumptions attached. Q-CRAFT compares scenarios. It does not forecast.

Answer (a) treats a 2099 projection as a prediction. Nobody forecasts 2099, and the tool’s own User Guide says the results are not forecasts ([pp. 5-6](#)).

Answer (c) converts a debt-ratio gap into a cost figure, which it is not. The gap is the extra borrowing implied by lower growth under stated assumptions about how spending responds, and it leaves out disasters, sea-level rise and adaptation spending entirely. Chapter 6 is about that class of error.

The 47.5 and 66 percent figures are real. They come from a five-day IMF workshop with one finance ministry’s staff in September 2023, reported in the [C-PIMA](#) high-level summary ([IMF, 2024](#)), and Chapter 5 runs the case they came from.

1.6 Pick your path

The three paths cover the same tool, differing only in how much of the economics they rebuild along the way.

Path	Choose it if
A. Guided	Common if you arrive from the climate side without the debt arithmetic (mostly A1/B1/C1 answers).
B. Standard	Common if you know debt dynamics already and the tool is the new part (mostly A2/B2/C1 or C2).
C. Fast	You have built projections and want the tool’s specifics and the judgment calls (A3, plus C2 or C3).

The figure below lays the three routes across the seven modules, showing what each abridges.

Every path reads every module, the fast path three of them in part

Every path arrives at the same capstone. Only the guided path rebuilds the debt dynamics equation from scratch.

YOUR ROUTE	M0	M1	M2	M3	M4	M5	M6	TIME
A. Guided	read	read	read	read	read	read	read	5 to 6 hours
B. Standard	read	read	in part	read	read	read	read	3 to 4 hours
C. Fast	read	in part	in part	in part	read	read	read	2 hours

the only module a path drops

Route yourself with the self-assessment above. Every module also carries its own fast path, for the day you have twenty minutes.

Figure 1.2: What the faster paths abridge is always the same thing: material that rebuilds economics you may already have. Nothing about running the tool, reading its output or defending an assumption is dropped on any route.

No route drops a module. The faster paths abridge only the material that rebuilds economics you may already own.

Nobody should skip Chapter 3, the only place the whole tool comes together in one sitting. If you already own the debt dynamics equation, its Step 1 will be about twenty minutes of revision, with a collapsible depth layer available. Steps 2 and 3 are new material for almost everyone.

Every module carries a fast-path marker near the top, telling you the shortest useful route through it.

i Budget three to four times your estimate

Three to four times is the standard multiple by which experts underestimate other people's learning time. The ten-minute run in Chapter 2 does take ten minutes. The rest of the course does not match its time estimate.

1.7 What you should have now

- The capstone in one sentence: an export packet with documented rationale, plus a two-paragraph fiscal risk draft.
- Three self-assessment answers written down for comparison in Chapter 7.
- Three concept-inventory answers, with a note on which came from intuition rather than the debt dynamics equation.
- A path.

On your desk this week: read the section of your ministry's most recent fiscal risk documentation that would carry this analysis. If your ministry publishes none, read the published section Chapter 5 works through. Either way, that is the genre you are writing into.

Chapter 2

What Q-CRAFT does and why it exists



In this module

You will run a full projection in ten minutes, before anything is explained. Then the model is built in front of you in two stages. First the base machine, a long-term fiscal projection model with no climate in it, resting on one equation and the three numbers it needs: growth, the interest rate and the primary balance. Then the warming block bolts onto that finished machine at two points and the same equation becomes climate-aware. The module closes with the parity evidence, so you can decide now whether to trust the numbers: baseline parity is exact for 147 of 147 tested countries.



Fast path

Short on time? Do Section 2.4 (the ten-minute run), Section 2.5.2 (the three-number table) and the map at the top of Section 2.6 (where warming attaches). Skip the rest and come back to it.

2.1 By the end of this module you can

- **Run** a full projection in Q-CRAFT Explorer for a country of your choice and export the results
- **Name** the three numbers the debt dynamics equation needs and **identify** what builds each one
- **Distinguish** what the base projection model does from what the warming block adds to it
- **Match** any parameter you are asked about to the number it moves, without searching
- **Explain** to a colleague why a Python reimplement of an Excel tool needs a parity test, and what “147 of 147” does and does not cover

2.2 Where you are in the course

The map below is the same one from Chapter 1, with a different part lit.



Figure 2.1: This module covers the middle of the chain: the three numbers, and the debt dynamics equation they feed.

Four nodes are lit: growth, the interest rate, the primary balance, and the equation that turns them into a debt path. That group is the base machine this module builds first. The warming block at the bottom of the map arrives second, and the same picture then comes back with only that block lit.

2.3 Q-CRAFT runs in a published Excel workbook, and in this Explorer

The IMF publishes Q-CRAFT as an Excel workbook, downloadable from its website. The choice was deliberate and sound: every ministry has Excel, staff know it, and a Fund tool must run in nearly two hundred countries without assuming data or infrastructure that may not be there. The workbook is well structured, clearly documented, and covers 197 economies. Three counts recur in this course, measuring different layers: the workbook’s macro data covers those 197 economies, the climate dataset behind the scenarios carries 171, and the Explorer’s dropdown holds the 175 present in all its input datasets.

Q-CRAFT Explorer runs that same economics on the web, built so more of your hour goes on the analysis and less on carrying numbers by hand. Four things are different:

1. **Up-to-date data.** A posted workbook ships with the WEO vintage it was built on, which ages between releases. The Explorer ships two data modes. Current, the default, carries the latest bundled release, WEO April 2026 as this is written. Verified carries the workbook’s own WEO October 2024 vintage for checking the tool against the original. Refreshing the data is a release rather than a download and a rebuild.
2. **Guidance where you need it.** The help for choosing a parameter sits next to the parameter. That is the difference between reading about expenditure rigidity and choosing a value while looking at what your own country’s spending has actually done, which Chapter 4 has you do.
3. **Faster to the analysis.** Fewer manual steps sit between the question and the chart.
4. **The export packet.** Your assumptions, your written remarks on them, and presentation-ready output come out together, so the record of what you decided travels with the numbers. You build one in Chapter 4, export it in Chapter 5, and hand it over in Chapter 7.

The engine underneath the Explorer is a standalone Python package with extensive automated testing, checked against the workbook cell by cell (Section 2.7). The code is permissively licensed open source, so anyone can examine, verify, and extend it.

i The workbook has its own walkthrough

The modules teach on the Explorer. The appendix [Running the official workbook](#) is the quick orientation to the workbook itself: the download, the Dashboard, each control mapped to the concept it sets, and where the outputs live. The IMF’s User Guide remains the official manual.

2.4 Run a country in ten minutes

Run the tool before you read how it works. The explanation lands better on something you have already seen move. Six steps follow, about ten minutes in all.

The figure below is the shape of those ten minutes, one panel per stop.

Six moves produce a projection and two capstone CSVs

Run the tool before you read how it works. The two CSVs from step 6 are the first page of your export packet.



About ten minutes, five clicks, and nothing explained yet. The explanation lands better on something you have already seen move.

Q-CRAFT Explorer, the sidebar and four of its seven tabs. The parameters you leave alone in this run are the subject of Module 3.

Figure 2.2: Nothing here is explained, on purpose. Both prediction and explanation land better on a chart you have already watched move, which is why the reading starts after the clicking.

Two moves happen in the sidebar and one in each of four tabs, and the last of them produces a file you keep. Now do it.

Zero to a projection

Open Q-CRAFT Explorer and follow these steps:

1. **Select a country** from the sidebar dropdown. Pick the one you work on, or pick Uganda for a run checkable against a published result: it is the tool's verification country and the worked case in Chapter 5. Either way, WEO macroeconomic data and UN population projections load automatically. Nothing to download, nothing to paste.
2. **Leave every control at its engine default, and read the data banner.** The sidebar footer confirms the defaults, and the banner above the tabs names what they run on: Current mode, WEO April 2026 and UN population projections 2024. That vintage is the first thing a reviewer will ask about.
3. **Open the Baseline tab.** This is the country's fiscal trajectory with no climate damage. The shaded band through the WEO horizon is the published record and forecast: if the latest debt-to-GDP figure there does not square with what you know about the country, work out why before going further. Then note the debt-to-GDP number at 2099.
4. **Open the Climate tab.** Six warming scenarios, their effect on GDP. Note where the scenarios separate, and in which year.
5. **Open the Analysis tab.** Baseline and all climate scenarios on one debt chart. The vertical gap between the baseline line and the Hot + Unadapted line is the number your fiscal risk paragraph will be about.
6. **Open the Data tab and download both CSVs,** the selected scenario and all scenarios together. That is the first piece of your capstone export packet. Keep it.

If you picked Uganda, you have done with five clicks the analysis ministry staff produced in a five-day workshop with a visiting IMF team in September 2023. The workshop's finding, published in the [C-PIMA](#) high-level summary: GDP loss by end of century could pass 4 percentage points, with debt rising to 66 percent of GDP against 47.5 percent in the baseline ([IMF](#),

2024).

Your numbers will not match theirs exactly, even on the same country. That workshop used the 2023 WEO vintage and its own parameter choices, and the Explorer’s default Current mode runs WEO April 2026. Which vintage and which parameters were used is precisely what gets lost, and precisely what the export packet is for.

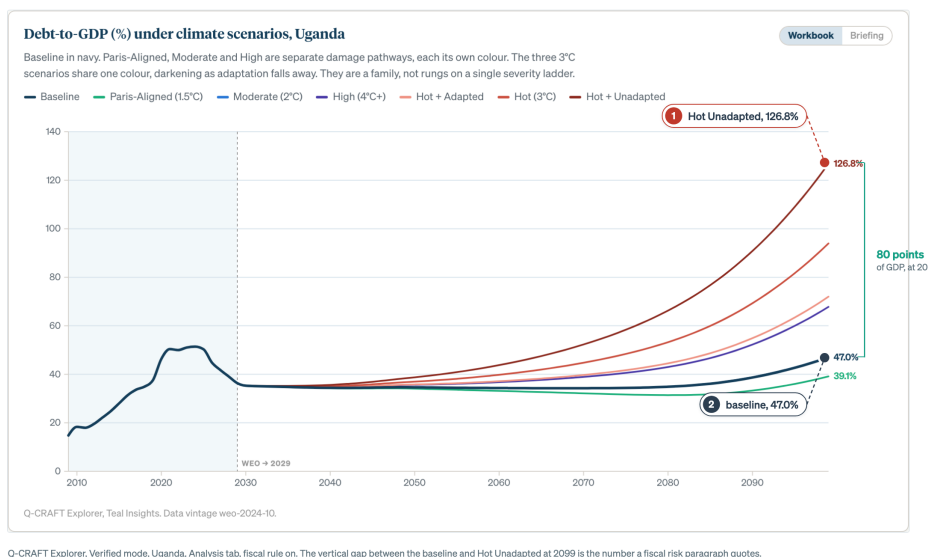


Figure 2.3: The Analysis tab, Uganda, in Verified mode. Six scenarios and a baseline, and the one measurement a fiscal risk paragraph quotes is the vertical distance between the outer two at 2099: 47.0% under the baseline against 126.8% under Hot Unadapted, which is 80 points of GDP.

2.5 The base machine, before any climate

Before Q-CRAFT is a climate tool it is a long-term fiscal projection model. Give it a country and it projects the debt-to-GDP ratio to 2099 with no warming in the picture at all. It is the same machine underneath every debt sustainability analysis you have read.

Nothing below mentions climate. Warming arrives in the next section as one block that bolts on.

2.5.1 One equation produces the whole debt path

Next year’s debt ratio is this year’s, grown by the interest rate, shrunk by economic growth, less whatever the government paid down.

That sentence is the whole model. In shorthand:

$$d_t = d_{t-1} \times \frac{1 + r_t}{1 + g_t} - pb_t$$

Symbol	Meaning
d_t	Debt-to-GDP ratio in year t
r_t	Effective nominal interest rate on government debt
g_t	Nominal GDP growth rate
pb_t	Primary balance as a share of GDP (revenue minus non-interest spending)

Three of those four are inputs. The fourth, d , is the answer and also last year's input, so the tool walks the projection forward one year at a time.

The sign of r minus g decides whether the ratio climbs on its own. When interest rates exceed growth, the debt ratio rises automatically, and the government must run a primary surplus even to hold debt still. When growth exceeds interest rates, the ratio declines even with modest deficits. Step 1 of Chapter 3 works this through with numbers, and Steps 2 and 3 build the rest of the tool on top of it.

i In numbers: one year of the debt dynamics equation

Start a country at a debt ratio of 60 percent of GDP, effective interest at 8 percent, nominal growth at 6 percent, and a primary deficit of 1 percent of GDP.

The amplifier goes first. 1.08 over 1.06 is 1.019, so last year's ratio carries forward at a little over one for one:

$$60 \times \frac{1.08}{1.06} = 61.1$$

Then the primary balance. A deficit of 1 percent of GDP is $pb = -1$, and subtracting a negative adds:

$$61.1 - (-1.0) = 62.1$$

Next year's ratio is 62.1 percent of GDP. The government borrowed one point of GDP and the ratio rose by 2.1, because the interest-growth gap supplied the other 1.1 with nobody deciding anything.

Sensitivity. Raise growth by one point, to 7 percent, and next year's ratio lands at 61.6 instead of 62.1. Half a point in one year looks like nothing. Hold the same two settings for ten years and the paths land at 83.2 and 76.3 percent of GDP. That is this course's whole climate story in advance: warming takes a fraction of a point off g , and the equation compounds it for seventy years.

2.5.2 Where the three numbers come from

Growth has three suppliers, the interest rate has one rule, and the primary balance has two sides. The table below is the thing to remember. The sections after it are for the number you happen to be arguing about.

The number	What builds it	What you control
Growth, g	Working-age population, output per worker, and the price path, combined into nominal GDP growth	The UN population variant, and the productivity and inflation paths, a start and a long-run value for each
Interest rate, r	One of three rate rules, applied to the debt stock	The interest-rate approach: which of the three rules runs
Primary balance, pb	Revenue, which tracks nominal GDP, minus primary expenditure, which grows with productivity, inflation and total population, plus any fiscal rule adjustment	The debt target, and the fiscal rule on or off

Every one of the Explorer’s ten sidebar controls, country selection included, lands on exactly one of the three numbers. Every control starts at an engine default. Choosing them, and writing down why, is Chapter 4’s whole job.

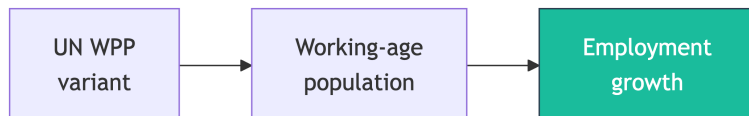
⚠ WIDGET-TODO: three-numbers-to-one-equation

Embed the intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). The widget was specified as “seven-modules-to-one-equation” before the teaching frame changed. Rebuild it on the chain above, data and assumptions producing g , r and pb , with “seven modules” demoted to an architecture aside. Iframe under Quarto for now, native React once the site moves to Astro. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

2.5.2.1 Growth, g , is three things multiplied together

Growth is the number with the most suppliers, and the one your assumptions move most.

Demography sets how fast the workforce grows.



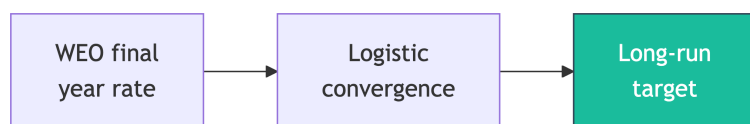
The workforce sets how fast the economy can grow. Q-CRAFT takes population by age group (children, working age, elderly) from UN World Population Prospects, in three variants: medium, high, and low. Growth in the working-age population, ages 15 to 64, drives employment growth after the WEO forecast horizon, and employment growth drives potential GDP. Ethiopia’s working-age population is projected to keep growing for the rest of the century, pushing potential growth up and debt ratios down. Thailand’s has already turned and is falling, which does the opposite before any climate damage is applied.

Productivity sets output per worker, and it moves gradually rather than in a step.



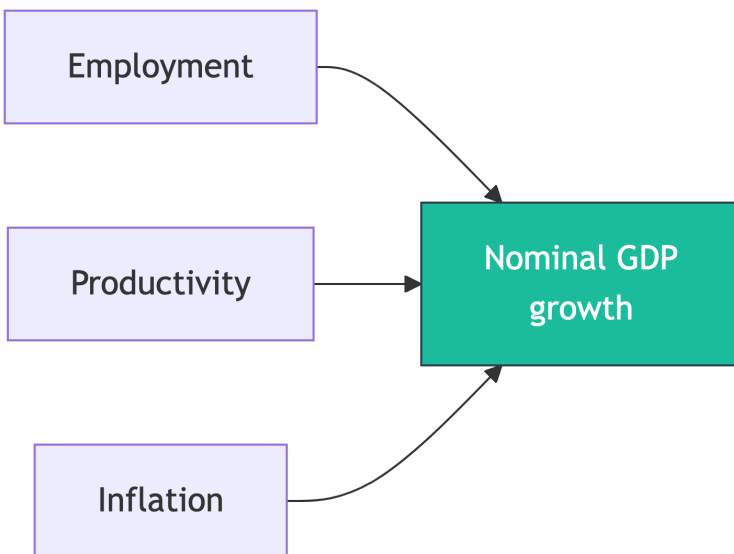
The measure is GDP per employed person, from World Bank data. You give the model a starting growth rate and a long-run rate, and it slides between them along an S-shaped path, which economists call a logistic curve. Vietnam makes the shape concrete: a catch-up economy raising output per worker quickly now, with the rate expected to ease as the level approaches the frontier. The resulting productivity level relative to the OECD is a useful realism check: assumptions that imply a low-income country overtaking OECD productivity by 2099 deserve another look.

Inflation turns real growth into the nominal growth the debt ratio actually responds to.



Inflation follows the same gradual S-shaped path as productivity. The starting rate should reflect the WEO projection for the final forecast year, the ending rate the long-run target: typically 2 to 3 percent for advanced economies and 3 to 6 percent for emerging and developing economies. Where the central bank publishes an explicit target, that number is the reference point.

The three combine into nominal GDP growth.



The three multiply rather than add. Q-CRAFT uses the IMF's own projections through the WEO forecast period, which the tool caps at 2029. Beyond that horizon the model takes over, using the assumptions set in the dashboard. Where a country's WEO series ends earlier, the handover happens at that earlier year instead.

i In numbers: the growth decomposition

Put employment growth at 2 percent, productivity growth at 3 percent, and inflation at 5 percent.

Real growth is the first two multiplied together:

$$1.02 \times 1.03 = 1.0506$$

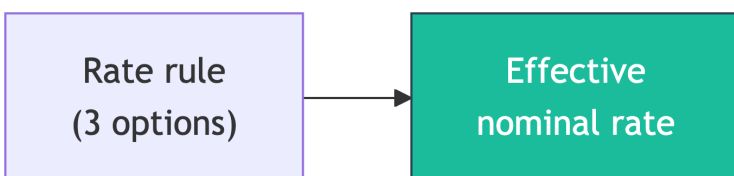
which is 5.06 percent. Nominal growth puts the price path on top of that:

$$1.0506 \times 1.05 = 1.1031$$

which is 10.31 percent. That is the g the debt dynamics equation uses. Adding the three rates instead gives 10.00 percent, close enough to be tempting and wrong by three tenths of a point every year.

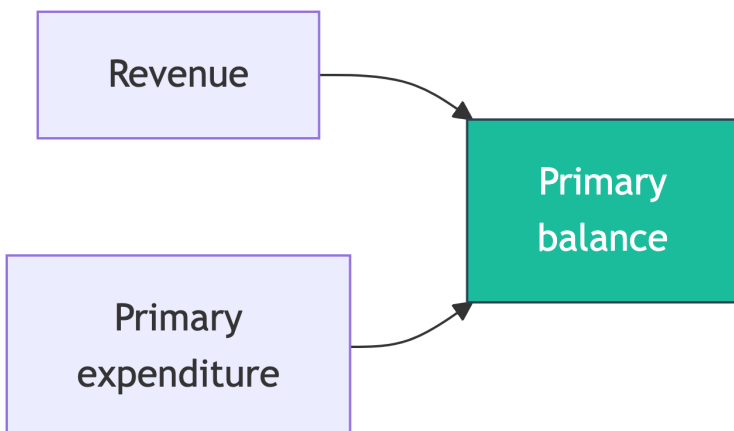
Sensitivity. Take one point off productivity growth, down to 2 percent, and nominal growth lands at 9.24 percent instead of 10.31. The loss is a little larger than the point you removed, because the terms multiply.

2.5.2.2 The interest rate, r , is one rule applied to the debt stock



This number is what carrying the debt costs. Q-CRAFT offers three rules: a constant nominal rate, a constant gap between the interest rate and growth, or a constant real rate. Country expertise matters: the choice should reflect concessional lending terms, sovereign risk premia, and the maturity structure of domestic financial markets. For a country moving from concessional to commercial borrowing, this assumption does more work than any other. Zambia's shift into Eurobond issuance through the 2010s is the standard example, and the debt-service consequences arrived faster than any demographic or productivity trend could have.

2.5.2.3 The primary balance, pb , is revenue minus non-interest spending



The two sides grow on different rules. Revenue grows with nominal GDP, holding the revenue-to-GDP ratio constant. Primary expenditure, meaning spending excluding interest payments, grows multiplicatively with productivity, inflation, and total population: $(1 + \text{productivity growth}) \times (1 + \text{inflation}) \times (1 + \text{population growth})$, plus any fiscal rule adjustment. When the fiscal rule is enabled, it adjusts spending to steer debt toward a target ratio.

i In numbers: why the expenditure formula multiplies

Put productivity growth at 3 percent, inflation at 5 percent, and population growth at 2 percent. Primary expenditure of 100 units next year is

$$100 \times 1.03 \times 1.05 \times 1.02 = 110.3$$

Adding the three rates instead would give 110.0. Three tenths of a unit in the first year, which is nothing.

Sensitivity. The two versions keep separating, because each year's gap compounds on the last. Run both for seventy years and the multiplicative path ends about 22 percent above the additive one: a different debt projection rather than a rounding difference. The engine implements the multiplicative form.

2.5.3 The base machine is complete

Two published data sources feed three manufactured numbers, and one equation turns them into the debt path. Every sidebar control except expenditure rigidity works on this machine. Rigidity waits for the warming block. The machine works for any country in the tool, and nothing in it knows that the climate exists.

! DRAFT FOR TEAL: the same chain as a kitchen analogy

The pantry holds the ingredients somebody else bought: the WEO macroeconomic series and the UN population projections. You cannot change them, and every cook in every country works from the same shelves.

The seasoning is the sidebar controls, and it is yours: the demography variant, the growth assumptions, the interest-rate approach, the debt target, the fiscal rule, expenditure rigidity. They come in small quantities with large effect, and are why two cooks working from one pantry produce different dishes.

Everything the tool does before the final step is preparing growth, the interest rate and the primary balance.

The one recipe is the debt dynamics equation. The same method runs for every country, every scenario, every year.

The analogy breaks in two places: feedback and prediction. A cook tastes and adjusts. This model does not: nothing downstream feeds back into the ingredients, so cutting spending inside the model never slows growth, the partial-equilibrium limit stated below. A recipe is a claim about what comes out of the oven. A projection to 2099 is not. It is a comparison between two dishes made from deliberately different seasoning.

i Two properties of the base machine worth knowing

It is partial equilibrium. The tool does not let fiscal policy feed back into GDP growth or interest rates ([User Guide, p. 5](#)). That is the reason a Q-CRAFT result is a comparison rather than a forecast.

It is seven functions. Under the hood the engine composes one pure function per box in the sections above into a single call. That is a fact about the code rather than the economics. The code lives in the [repository](#).

2.6 The warming block snaps on

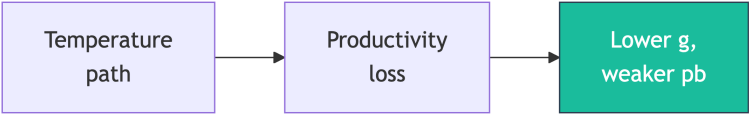
Now add one block.



Figure 2.4: The same map, with the base chain greyed out and the warming block highlighted. The block docks onto growth and the primary balance. Nothing else in the chain changes.

The warming scenarios do not add a term to the equation and do not change how any of the three numbers is built. They reach into the finished machine at two nodes. Growth falls, and where spending is rigid the primary balance weakens with it. The equation then does what it already did.

2.6.1 Warming lowers growth, and weakens the primary balance when spending is rigid



Lowering growth is the only way climate enters the equation. The damage estimates are empirical, drawn from the IMF Fiscal Affairs Department’s FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), building on the Kahn et al. (2021) methodology for long-term macroeconomic effects of climate change. Six scenarios run from Paris-Aligned, below 2°C warming, to Hot + Unadapted, high warming with a slow policy response. Impacts begin in 2030, and before that year all scenarios match the baseline (User Guide, p. 19). The channel is productivity: reduced GDP growth means lower revenue, higher expenditure-to-GDP ratios, and accelerating debt.

The weakened primary balance is the channel people miss. Revenue tracks nominal GDP, so a smaller economy collects less. Spending does not automatically shrink to match. The expenditure rigidity control in Chapter 4 sets how much of it does. At full rigidity the whole revenue shortfall becomes borrowing. That is why the same warming scenario produces a much wider debt fan in one country than another.

The number	Under warming	What you control
Growth, g	Damage applied from 2030, rising with the temperature path	The scenario you read against baseline
Primary balance, pb	Revenue falls with GDP; spending falls only as far as rigidity allows	Expenditure rigidity, 0.0 to 1.0
Interest rate, r	Untouched by the climate block	Nothing

The channel is temperature, so hot low-latitude economies take the largest modelled hit. Bangladesh is exactly that kind of economy, and the one that shows what the model leaves out: sea-level rise is not in this model at all (Chapter 6).

i The six climate scenarios			
Scenario	Warming (wrt present)	Based on	Description
Paris-Aligned (1.5°C)	+0.7°C	SSP1-2.6	International commitments met, with significant emission cuts
Moderate (2°C)	+1.6°C	SSP2-4.5	Current trends continue, with no new aggressive mitigation

High (4°C+)	+2.5°C	SSP3-7.0	Countries scale back mitigation, response fragmented
Hot (3°C)	+3.5°C	SSP3-7.0 (90th pctile)	Same emissions as High, using the 90th percentile of temperature increases
Hot + Adapted	+3.5°C	SSP3-7.0 (90th pctile)	Same temperatures as Hot, with faster adaptation
Hot + Unadapted	+3.5°C	SSP3-7.0 (90th pctile)	Same temperatures as Hot, with very slow adaptation

Warming values are IPCC best estimates for 2081-2100 relative to present ([User Guide Table 1](#)). Adding about 1.1°C for present-to-pre-industrial warming gives the familiar above-pre-industrial framing: Paris below 2°C, Moderate about 2.7°C, High about 3.6°C, Hot about 4.6°C. The parenthetical numbers in the scenario names are the workbook’s own labels rather than realized warming, which is why High (4°C+) lands at +2.5°C while Hot (3°C) lands at +3.5°C: read severity from the Warming column.

Know one naming difference before a meeting. A published fiscal risk statement may report a subset of these under its own labels. The worked case in Chapter 5 carries five scenarios and calls the last one “Vulnerable” rather than “Hot + Unadapted”. Same scenario, different label, and this course uses the Explorer’s naming.

These scenarios are conservative in ways that matter for how you write up the results. Chapter 6 sets out what they leave out.

💡 Self-check: where does it live?

A colleague sends you five questions in one email. For each, name which of the three numbers it moves, and where the tool builds that number. Commit to your five answers before opening the panel.

1. “Our fiscal rule ceiling is 50 percent of GDP. Does the model know that?”
2. “The projection assumes we keep growing at 6 percent forever. Does it?”
3. “Why does the debt path only start diverging across scenarios in 2030?”
4. “We are shifting from IDA credits to Eurobonds. Where does that show up?”
5. “Spending per person keeps rising in the projection. Who decided that?”

DRAFT FOR TEAL: self-check answers

Four of the five land on two of the three numbers. Five questions in one email are usually two problems.

1. **The primary balance.** The debt target and the fiscal rule on/off switch both sit on the spending side, and both are yours to set.
2. **Growth.** It is built from employment, productivity and inflation along a gradual path, so it does not sit at the WEO rate forever.
3. **Growth, and the primary balance behind it.** The climate scenarios

move both, and impacts start in 2030 by construction, with every scenario equal to the baseline before that ([User Guide, p. 19](#)).

4. **The interest rate.** The rate rule is where concessional-to-commercial transitions live. The sidebar's interest-rate approach chooses among the three rules: run all three and report the spread in your write-up.
5. **The primary balance.** Primary expenditure grows with productivity, inflation and total population, multiplicatively. Nobody decided it per person. It falls out of that formula.

If you missed 3, reread the warming section above. If you missed 5, the expenditure growth formula is the thing to hold: $(1 + a) \times (1 + b) \times (1 + c)$, never $a + b + c$.

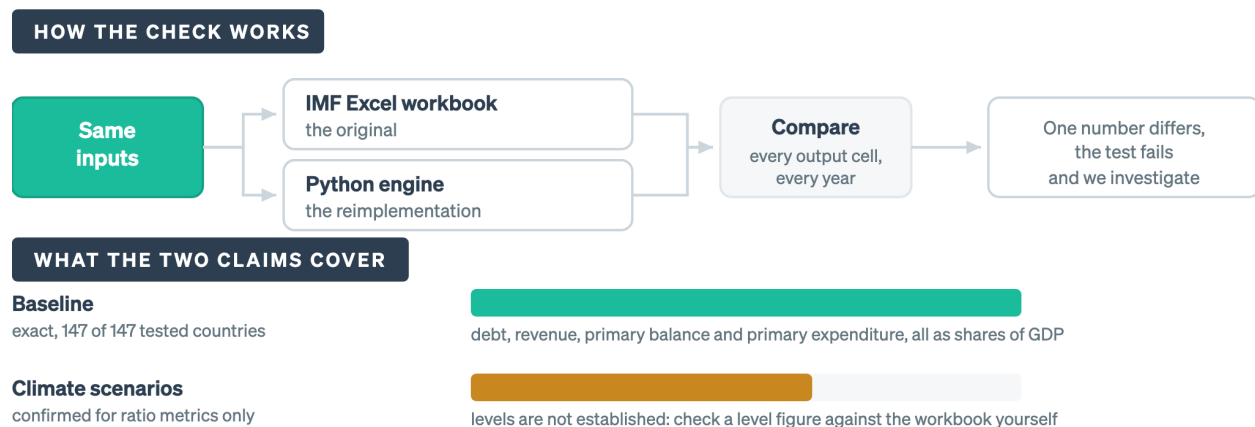
2.7 The Python engine is checked against the Excel workbook, cell by cell

A reimplementation is worth nothing if it quietly disagrees with the original. The course tells the verification story once, here, because you should decide now whether to trust the numbers.

The figure below shows the check on top and, underneath, the two claims it produces, drawn to scale against each other.

The baseline claim is wider than the climate claim

Identical inputs go through both engines and every output cell is compared. What survives the comparison is what you may quote.



Source: the golden-master test suite in this repository, run on every change. The bars show coverage of the claim, not a pass rate.

Figure 2.5: The bars are drawn to the reach of each claim rather than to a pass rate. Both claims passed. One of them was simply asked a narrower question.

Identical inputs go through both the original Excel workbook and the Python engine, and every output cell for every projection year is compared. The two bars differ in length because the baseline claim covers four series exactly and the climate claim stops at the ratios.

Think of the comparison as an auditor reconciling two copies of the same ledger. If a single number differs, the test fails and we investigate.

Baseline parity is exact for 147 of 147 tested countries: zero percentage-point deviation

on debt-to-GDP, revenue, primary balance and primary expenditure as shares of GDP. An additional 25 parameter sensitivity combinations were tested and passed. Climate-scenario parity is confirmed for ratio metrics only, meaning debt-to-GDP, revenue and expenditure as percent of GDP, across the tested countries and scenarios.

The climate claim covers the ratios you would quote in a Fiscal Risk Statement, not every level series in the workbook. If your analysis turns on a climate-scenario figure in levels rather than as a share of GDP, check it against the Excel tool yourself.

The test suite is public and runs on every change. You can run it yourself.

2.8 What you can now do

- You have run a full projection and exported the data. That file is capstone material.
- You can name the three numbers the equation needs and say what builds each one.
- You can separate the base projection model from the warming block, and name the two nodes the block attaches to.
- You can say what the parity claim covers (baseline, 147 of 147, exact) and what it does not (climate levels).

The common error at this stage is treating the tool as seven things to learn. It is one equation with three inputs, the inputs have suppliers, and warming is one block on top. If you can answer the “where does it live” self-check, you have the map.

On your desk this week: when someone hands you a fiscal projection, ask which of r , g and pb their headline number moves. The question works on projections that have nothing to do with climate.

Chapter 3

The model, in three steps

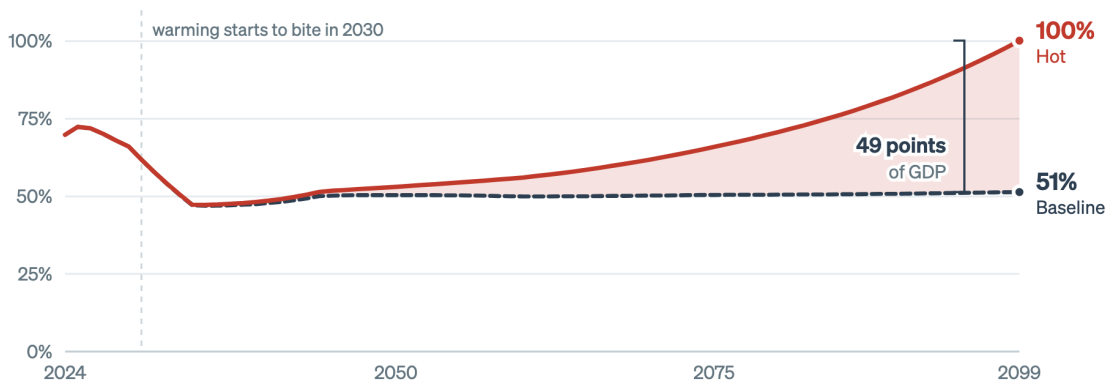


In this module

Q-CRAFT is three things stacked on each other: an equation that turns three numbers into a debt path, a projection model that manufactures those three numbers year after year, and a climate overlay that changes one of them. Read the three in order and you can trace every result the tool produces.

By 2099 the hot scenario puts Kenya's debt 49 points of GDP higher

Same country, same starting debt, same equation. This chapter is where that number comes from.



Source: this repository's engine, Kenya at the Explorer's shipped defaults, from figures/series/m2-debt-paths.csv. The two runs are identical until 2030.

Figure 3.1: Both runs load the same country data and start from the same debt stock. The gap is what a single assumption about warming does once it has seventy years to compound, and every step of this chapter is a piece of it.

Forty-nine points of GDP is a large number to take on trust. Both runs: same Kenyan data, same debt stock, same equation. One assumes a hot warming path, and seventy years later the debt ratio is nearly double.

This module takes that number apart. Three steps, and they build:

1. **An equation.** Three numbers a year, and it gives you a debt path.
2. **A growth engine.** Where the three numbers come from: most of what Q-CRAFT is.
3. **A climate overlay.** One block that changes one of the three, and the fan opens.

By the end you can point at any part of the chart above and say which step put it there.

3.1 By the end of this module you can

- **Compute** next year's debt ratio from this year's, given r , g and the primary balance
- **Trace** each of the three numbers back to the data source or the assumption that produced it
- **Explain** how a temperature path becomes a productivity growth rate, and why the estimate is not country-specific
- **Distinguish** the part of a climate result that comes from the damage estimate from the part that comes from your own spending assumption

i Fast path

No route skips this module. If you already own the debt dynamics equation, treat Step 1 as about twenty minutes of revision, then read Steps 2 and 3 in full: new material for almost everyone.

3.2 Step 1. The equation

i **You are here.** Step 1 of 3: the equation. It needs three numbers a year and nothing has supplied them yet.

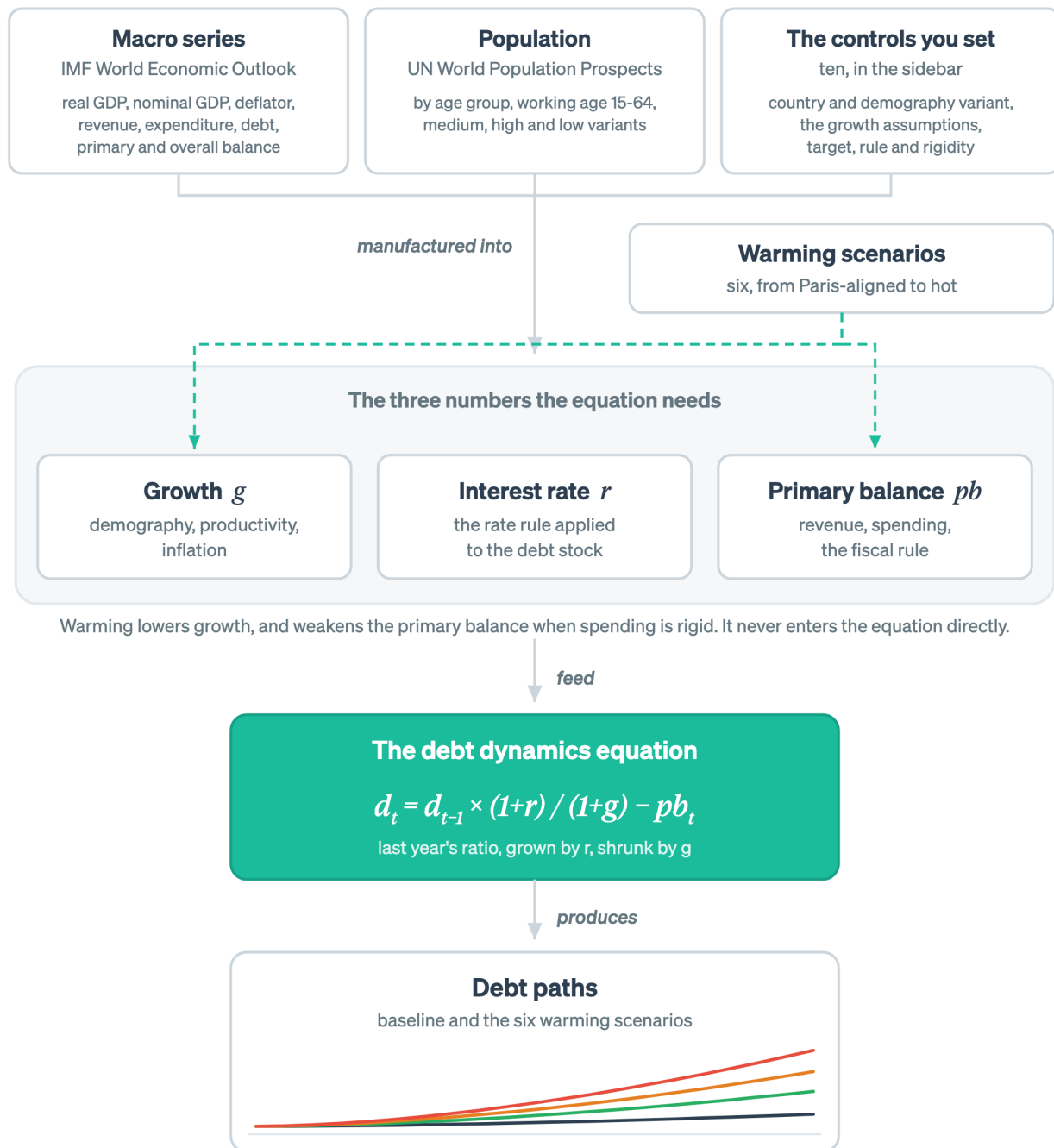


Figure 3.2: Step 1 is the equation node on its own. It needs three numbers a year, and nothing on this map has supplied them yet.

3.2.1 The identity is a stock and two flows

A government's debt is a stock, and next year's is this year's plus interest owed and new borrowing.

$$\text{Debt next year} = \text{debt this year} + \text{interest on it} + (\text{non-interest spending} - \text{revenue})$$

That last bracket is the **primary deficit**. Flip its sign and it is the primary balance. Nothing controversial has happened yet.

Now divide by **GDP**: a debt stock in currency units means **nothing until set against the economy carrying it**. The numerator grows at the interest rate, the denominator at the rate of nominal GDP. Two things are racing:

$$d_t = d_{t-1} \times \frac{1 + r_t}{1 + g_t} - pb_t$$

Give me these three numbers every year and I will give you the debt path

The equation needs g , r and pb . The rest of the model exists to supply them.

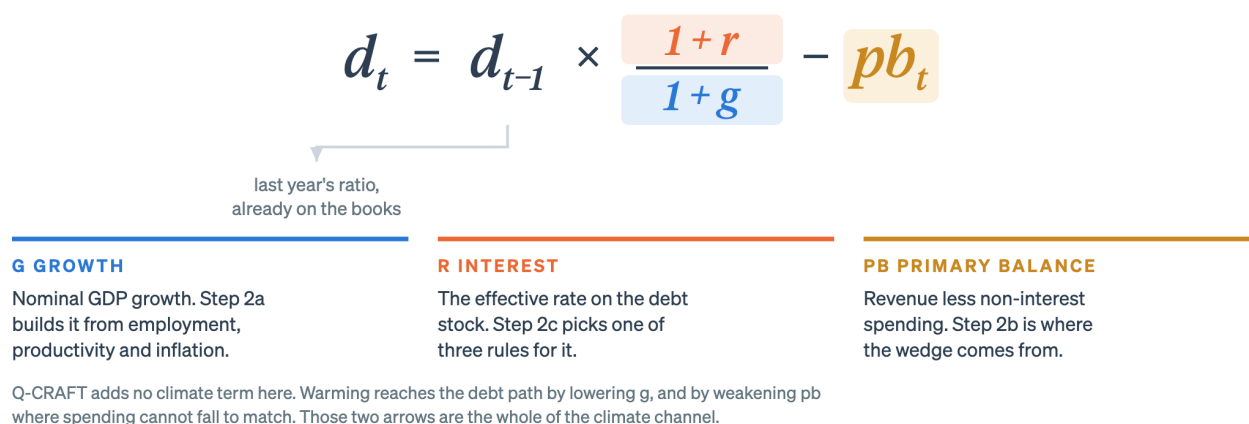


Figure 3.3: The first term is not one of the three. Last year's ratio is a stock you inherit, so the only things a projection has to manufacture are the growth rate, the interest rate and the primary balance.

The first term is not a decision. Last year's ratio is a stock the government inherits. A projection must manufacture the other three, and Step 2 is the manufacturing.

3.2.2 One year forward: the amplifier adds more than the borrowing

Start a country at 60 percent of GDP. Give it an effective interest rate of 8 percent, nominal growth of 6 percent, and a primary deficit of 1 percent of GDP. (The same year as the Chapter 2 depth layer.)

Multiply by 1.08 and divide by 1.06. That is a factor of 1.0189, so the ratio carries forward at 61.1. Add the point of GDP the government borrowed: 62.1 percent.

Only one of the 2.1 points came from a decision. One point from borrowing, which everyone expects. The other 1.1 points from the amplifier, which nobody voted for.

Run it three years and the amplifier's contribution grows: it applies to a bigger stock each time.

End of year	Ratio carried forward	Plus borrowing	Debt ratio
1	61.1	+1.0	62.1
2	63.3	+1.0	64.3
3	65.5	+1.0	66.5

Borrowing contributes exactly 1 point a year, every year. The amplifier contributes 1.1 points in year one, 1.2 in year three, and 3.5 points across the three years against borrowing's 3.0.

One percentage point of growth, held for three years, is worth 1.8 points of debt ratio. Run the identical three years with growth at 7 percent instead of 6 and the ratio lands at 64.7 rather than 66.5. Held for seventy, that point is worth a different fiscal policy. That arithmetic is the entire reason a climate model that only touches growth can move a debt path this far.

Three percentage points of differential, held for ten years

The same zero primary balance ends at 66, at 50 or at 38, depending only on which of r and g is larger.

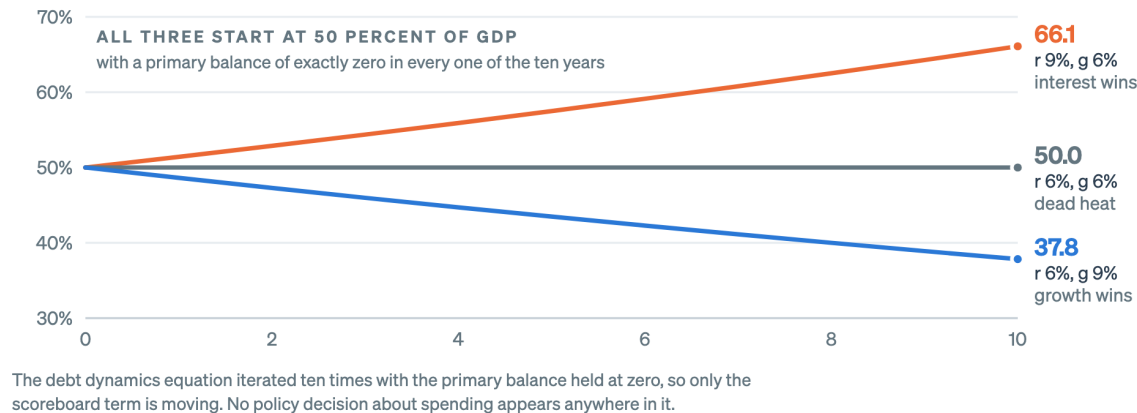


Figure 3.4: The middle line is the only place on this chart where a zero primary balance holds the ratio still, and it needs r and g to be exactly equal. Every other position moves the ratio with nobody borrowing for a new programme.

⚠ WIDGET-TODO: the debt sandbox

Embed the debt sandbox widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The widget lets a reader set r , g and pb and watch ten years of the equation redraw. The prose above carries the full idea alone, because an interactive's default state must work for readers who never touch it.

i Going deeper: the derivation, and what the interest-growth gap does

Building the ratio from the stock. Write the stock recursion in local currency, with D for debt, PB for the primary balance and Y for nominal GDP:

$$D_t = D_{t-1}(1 + r_t) - PB_t$$

Divide both sides by Y_t , and use $Y_t = Y_{t-1}(1 + g_t)$:

$$\frac{D_t}{Y_t} = \frac{D_{t-1}(1 + r_t)}{Y_{t-1}(1 + g_t)} - \frac{PB_t}{Y_t}$$

which is the equation above with $d = D/Y$ and $pb = PB/Y$. The only substantive step, dividing by a denominator that is itself growing, is where the whole interest-growth story comes from.

The amplifier has a name. $\frac{1+r}{1+g}$ is usually reported as a single number, the [interest-growth differential](#) $(r - g)/(1 + g)$. Its sign is the most consequential fact about a country's debt position, and in the short run no minister chooses it.

Take a debt ratio of 50 percent of GDP and a primary balance of exactly zero, so only the amplifier is left.

r	g	After 1 year	After 10 years	What is happening
9%	6%	51.4	66.1	Interest wins. The ratio climbs with no new programme borrowing.
6%	6%	50.0	50.0	Dead heat. A zero primary balance stabilizes debt only here.
6%	9%	48.6	37.8	Growth wins. The ratio falls while the primary account is balanced.

Three percentage points of differential, held for a decade, is the difference between 66.1 and 37.8 percent of GDP. Nothing in that table is a decision about spending.

The balance that holds the ratio still. Set next year's ratio equal to this year's and solve for the primary balance:

$$pb^* = d \times \frac{r - g}{1 + g}$$

That is the debt-stabilizing primary balance, a standard quantity in any debt sustainability analysis. The worse your interest-growth gap and the higher your existing debt, the larger the surplus you need to stand still.

Run it for the country in the table. Debt at 50 percent of GDP, r at 9 percent, g at 6 percent:

$$pb^* = 0.50 \times \frac{0.09 - 0.06}{1.06} = 0.0142$$

That is a primary surplus of about 1.4 percent of GDP, every year, to keep the ratio at 50. A government running a primary deficit of 1 percent in that position is 2.4 points of GDP from standing still.

Take one point off growth, to 5 percent, and the required surplus rises from 1.4 to 1.9 percent of GDP. The bill for standing still went up half a point because a macro variable moved. A Step 3 climate scenario does exactly this and nothing else.

Three claims a colleague might send you. Decide whether you agree and what you would check.

1. "Our debt ratio fell last year, so the fiscal position improved." Not necessarily. A ratio can fall on its denominator. Uganda's own Fiscal Risk Statement on FY2022/23: the ratio fell to 46.9 percent from 48.4, part of the fall attributed to nominal GDP rising on high inflation, with a warning the effect would subside as inflation fell. Check whether the stock of debt fell or the denominator moved.
2. "We should compare our primary deficit to the regional average." Wrong benchmark. The sustainable primary deficit depends on your own r , g and starting debt. The comparison that means something is your actual primary balance against your own pb^* . Two countries with identical deficits can be in opposite positions.
3. "Inflation is helping us, because it inflates away the debt." Half right, and the wrong half is expensive. Unexpected inflation raises nominal g and deflates the ratio once. It then feeds into the effective rate on new and floating-rate borrowing, so r follows g up with a lag. In Q-CRAFT specifically, revenue and primary expenditure both grow with inflation, so a permanently higher inflation assumption largely cancels out of the ratio.

Step 1 lands here: give the equation r , g and pb every year and it gives the debt path. The rest of the model exists to supply those three.

3.3 Step 2. From an equation to a projection model

i **You are here.** Step 2 of 3: the growth engine. It manufactures three numbers a year for seventy years, from published data and nine controls.

Producing r , g and pb for 2030, 2031, and every year to 2099 is the bulk of Q-CRAFT: three separate small machines. This step takes them in the order the equation reads: growth, then the primary balance, then the interest rate.

3.3.1 Step 2a. Growth is built from three published series

Q-CRAFT does not estimate a production function. No capital stock, no factor shares, no total factor productivity residual. It uses an accounting identity, a much weaker and much more portable claim:

$$\text{output} = \text{people working} \times \text{output per person working}$$

That is true by the definition of output per worker. Turn it into growth rates, add prices, and you have nominal GDP growth:

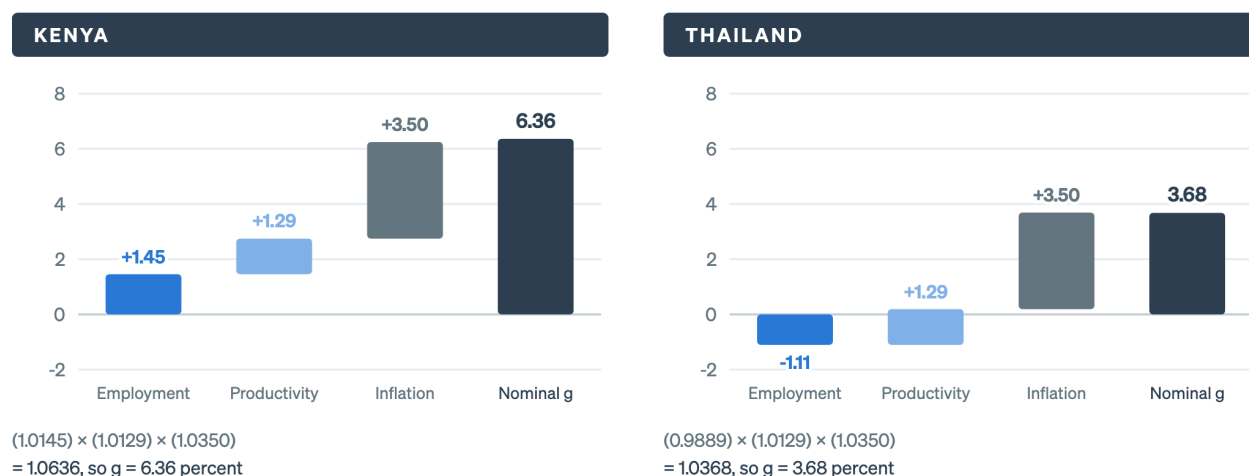
$$1 + g = (1 + \text{employment growth}) \times (1 + \text{productivity growth}) \times (1 + \text{inflation})$$

Each of the three inputs comes from a source you can check.

- **Employment growth** comes from the UN's World Population Prospects, working-age population aged 15 to 64. The only choice is the variant, Medium by default. The numbers arrive with the country.
- **Productivity growth** is GDP per employed person. You set the start and the end, and the tool slides between them along a logistic path.
- **Inflation** is the GDP deflator, on the same start-and-end treatment.

Demography is the first term in the growth identity

Both countries in 2050, on identical productivity and inflation assumptions. Only the workforce differs.



The three parts multiply rather than add, so each bar starts where the one before it finished and the last bar is computed rather than summed. Source: this repository's engine at the Explorer's shipped defaults, from figures/series/m2-growth-parts.csv. Employment growth is the UN working-age projection.

Figure 3.5: The productivity and inflation assumptions are the tool's defaults and are identical across the two panels, so the whole of the difference in the final bar is the working-age population.

The toy numbers put Kenya's 2050 nominal growth at 6.36 percent. Employment grows at 1.45 percent, productivity at 1.29, inflation at 3.50. Multiply: $(1.0145)(1.0129)(1.0350) = 1.0636$, so g is 6.36 percent.

Thailand in the same year has the same productivity and inflation, both tool defaults left unchanged. Its working-age population is falling at 1.11 percent a year. Multiply: $(0.9889)(1.0129)(1.0350) = 1.0368$, so g is 3.68 percent.

Two and a half points of nominal growth separate them, and the entire difference is demography. Fed back into Step 1's amplifier, it is the difference between a debt ratio that drifts down and one that climbs.

3.3.1.1 The World Economic Outlook hands over around 2030

The three parts above do not drive the projection from the first year: the tool uses the IMF's own numbers, and starts its own arithmetic only once those run out.

The World Economic Outlook hands the wheel to the model in 2030

Before that year the projection is the Fund's. After it the projection is yours, and it runs for seventy years.



Every scenario is identical before 2030, so the fan in the cold open opens from a single line rather than from the first year of the chart. Source: Q-CRAFT User Guide (Tim and Rahman, 2024), p. 19. This repository's engine carries the same year in its PROJ_START constant.

Figure 3.6: The handover is why a Q-CRAFT result is not a forecast. For the first twenty years you are reading the Fund's projection, and for the seventy after it you are reading your own assumptions compounded.

A Q-CRAFT chart is two different objects glued together: a short WEO forecast, then a long mechanical extrapolation of assumptions you chose. Nobody is forecasting 2087. The tool answers what happens if these rates hold.

The handover year is your country's own. It falls at the last year the release reports the figures the country's projection needs, never later than 2029. For almost every country that is 2029. The model takes over in 2030. For a few the release stopped much earlier: on the October 2024 data Syria's last reported year is 2010, so nearly its whole chart is projection. The shaded band sits at the country's own year, not a fixed 2029, and where the release ran past a country's last reported figures the Explorer puts a notice above the results naming the anchor year. Check that year before you quote a date.

i Going deeper: why an identity rather than a production function

What the choice buys. A production function needs a capital stock. Capital stock series are estimated rather than observed, missing or unreliable for a large share of the world, and built on assumptions about depreciation and initial conditions a user cannot check. Q-CRAFT is designed to run for 171 economies on already-published data, and the identity route needs only population and GDP per worker.

What the choice costs. No capital means no investment channel. A scenario cannot reduce growth by reducing the return to investment, and adaptation spending cannot raise growth by adding to the capital stock. Everything arrives through productivity, which is why Step 3's climate damage shows up as a cut to productivity growth and nowhere else.

A wording note. The IMF User Guide describes the baseline as "grounded in a simple production function and standard debt dynamic equation" (p. 5). The arithmetic the tool actually performs is the identity above: employment times productivity, no capital term, no factor shares. The two descriptions point at the same lines of code. This guide uses "accounting identity" because it more precisely describes what is assumed, and because "production function" leads readers to expect a capital stock that is not there.

The productivity path. Between the start and end rates you set, the tool interpolates along a logistic curve rather than a straight line: growth stays near the start rate for a while, falls quickly through mid-century, then flattens onto the end rate. That shape is a claim: convergence toward the frontier happens fastest when a country is furthest from it. Chapter 4 covers choosing the two endpoints and the evidence you can cite for them.

3.3.2 Step 2b. The primary balance is a wedge between two rules

Revenue and spending in Q-CRAFT are not forecasts of policy. They are rules simple enough to state in two sentences.

Revenue rides nominal GDP. The revenue-to-GDP ratio holds at its last observed value and carries forward. The assumption is unchanged tax policy: no new taxes, no new exemptions, no change in collection efficiency.

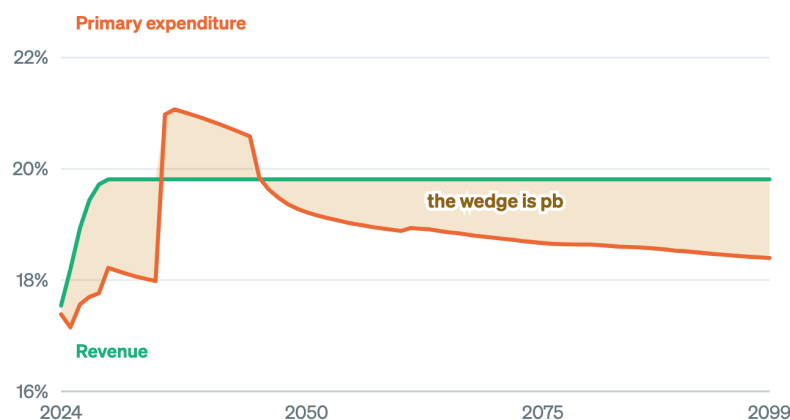
Spending follows people, productivity and prices. Primary expenditure grows by total population, times productivity, times inflation. The assumption is unchanged service levels: the same real provision per person, priced up each year.

The two demographic terms count different people. Growth uses the working-age population: the people who work. Spending uses the total population: the people served. In a country whose workforce shrinks faster than its population, the two pull in opposite directions: the fiscal squeeze demography produces on its own, with no climate anywhere near it.

The gap between revenue and spending is the primary balance, the only one of the three a government sets directly.

Revenue rides nominal GDP, spending follows people and prices, and the gap is pb

Neither line is a forecast of policy. Both are rules, and two dials decide how hard the rules bite.



TWO DIALS ON THE WEDGE

Fiscal rule

on / off

On, the primary balance is pushed toward whatever holds debt at your target. Off, it stays where the data left it.

Expenditure rigidity

1.0 to 0.0

1.0 holds spending at its baseline level when GDP falls, so the ratio rises. 0.0 cuts to hold the ratio. It moves climate runs only.

Source: this repository's engine, Kenya at the shipped defaults, from figures/series/m2-primary-balance.csv. Both series are percent of GDP.

Figure 3.7: Revenue is a fixed share of GDP by assumption, so it cannot rescue a projection on its own. Everything interesting in the wedge happens on the spending line, which is where both dials act.

The toy numbers put the wedge at 0.59 percent of GDP: the primary balance the equation gets for that year. Kenya's revenue settles at 19.81 percent of GDP and stays there for seventy years. Primary expenditure in 2050 is 19.22 percent.

The wedge equals the debt-stabilizing balance to two decimal places, because the fiscal rule is on and holding debt at the target is what the rule does. Step 1 already told you how to check: Kenya's debt-stabilizing primary balance in 2050 is 0.59 percent of GDP.

Two dials sit on the wedge, and both appear again in Step 3.

The fiscal rule decides whether the primary balance is left where the spending rule put it or pushed toward whatever holds debt at your target. It adjusts next year's primary expenditure by this year's fiscal gap: the distance between the actual primary balance and the debt-stabilizing one from Step 1. With the rule on, Kenya's baseline converges toward the 50 percent target, ending at 51. With it off, the ratio goes wherever the spending rule takes it.

Expenditure rigidity decides what happens to spending when GDP disappoints. At 1.0, spending stays at its baseline level in local currency, so a smaller GDP means a higher spending-to-GDP ratio and a worse balance. At 0.0, spending is cut to hold the ratio where the baseline had it. The dial changes nothing in the baseline itself. It bites only when a scenario makes GDP smaller than the baseline, meaning only in Step 3.

i Going deeper: what the fiscal rule adjusts, and where it applies

The rule fires in both directions. It tightens when debt is rising and above target, relaxes when debt is falling and below target, and does nothing in the years when debt is moving toward the target on its own. The adjustment is a level, in local currency, added after the multiplicative growth terms rather than folded into them.

It is a baseline-only device. The climate scenarios run on the baseline's spending path with no further consolidation. The asymmetry is deliberate: they ask what happens if warming arrives and policy does not respond, the right question for a fiscal risk statement and the wrong one for a policy plan.

That asymmetry also means a large part of any baseline-to-scenario gap is the rule, not the damage. Step 3 puts a number on that for Kenya.

A run with the rule off looks alarming and often is not. Turn it off for Kenya and the baseline debt ratio falls to zero: the spending rule leaves a persistent primary surplus and the amplifier works in the country's favour for most of the projection. The baseline is floored at zero and the climate scenarios are not, so the climate runs go through zero into negative territory and the chart inverts. That floor asymmetry comes from the workbook's own formulas, and Chapter 6 covers it. Treat a baseline sitting on zero as a signal to check assumptions, not a finding.

Revenue cannot rescue anything. Because the revenue-to-GDP ratio is fixed by assumption, better collection or a broader base cannot improve the debt path in this model. If your question is about revenue mobilisation, this is the wrong tool and Chapter 6 says which one is right.

3.3.3 Step 2c. The interest rate is a choice between three rules

The effective interest rate is the average rate the government pays on its whole stock of debt. It is not a market yield. Q-CRAFT does not model one. Its three rules for projecting the rate forward are three different views about what stays constant.

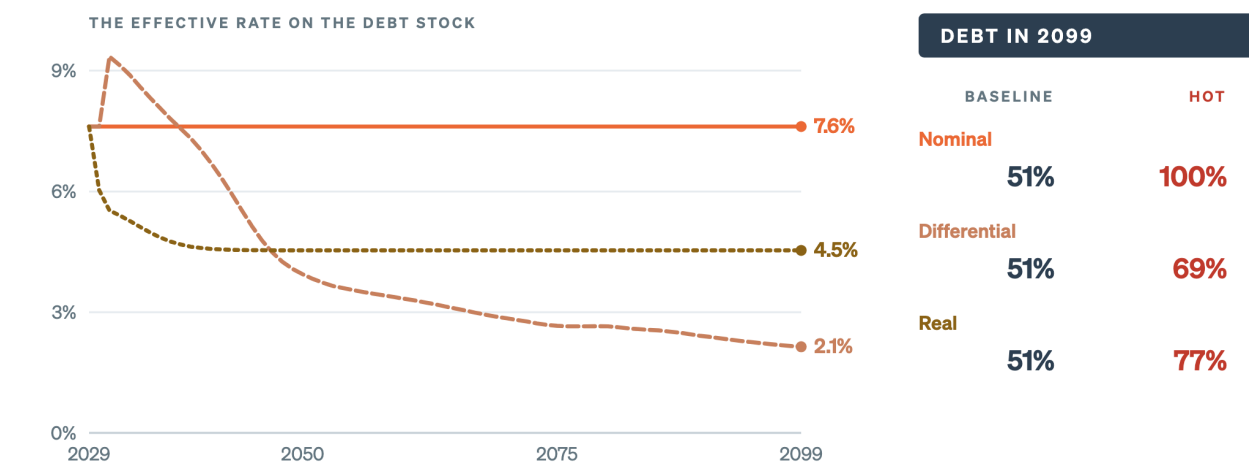
Hold the nominal rate flat. Take the last observed effective rate and carry it to 2099. The view: borrowing costs are what they are and nothing systematic changes them.

Hold the interest-growth differential flat. Preserve the last observed gap between r and g , so the interest rate follows nominal growth down as the economy matures. The view: rates and growth move together.

Hold the real rate flat. Fix a long-run real rate and let the nominal rate follow inflation. The view: the real cost of borrowing is the structural quantity.

The three rules for r disagree about the end of the century

Kenya on identical assumptions everywhere else. Only the rule that projects the interest rate changes.



Three defensible readings of the same debt stock, and a thirty-point spread in what the hot scenario costs.

Source: this repository's engine, Kenya at the shipped defaults with the rate rule varied, from figures/series/m2-interest-rules.csv. The three rules are the ones in User Guide Section IV.A.

Figure 3.8: The rule is the assumption that moves the climate answer most, and the Explorer exposes it as the interest-rate approach control, constant nominal by default. If the answer matters, run all three approaches and report the spread.

For Kenya the three rules give an effective rate in 2099 of 7.6, 2.1 and 4.5 percent. All three baselines end at 51 percent of GDP, because the fiscal rule does the work in each. Under the Hot scenario no rule intervenes. They end at 100, 69 and 77.

One dropdown creates thirty-one points of GDP of disagreement, with every other assumption identical. It is the single assumption that moves a climate answer most. The Explorer exposes it in the sidebar as the interest-rate approach control, constant-nominal by default. If the answer matters, run all three and report the spread.

i Going deeper: why the differential rule is so much gentler

Under the flat-nominal rule Kenya pays 7.6 percent in 2099 while nominal growth is 4.6 percent, so the amplifier is above one and every point of debt compounds. Under the differential rule the interest rate is tied to growth, so when climate damage lowers g it lowers r with it, and the amplifier barely moves.

That is not the rule being clever: it assumes away part of the risk, by construction. A country whose borrowing is mostly concessional and long-dated has an effective rate that responds slowly to anything, arguing for the flat-nominal rule. A country refinancing at market rates every few years has a rate that moves with conditions, arguing for one of the other two. The choice is a claim about your debt portfolio, worth a sentence in your export packet either way. Chapter 4 sets out what to look at before choosing.

3.3.4 Step 2 lands here: a baseline with no climate in it

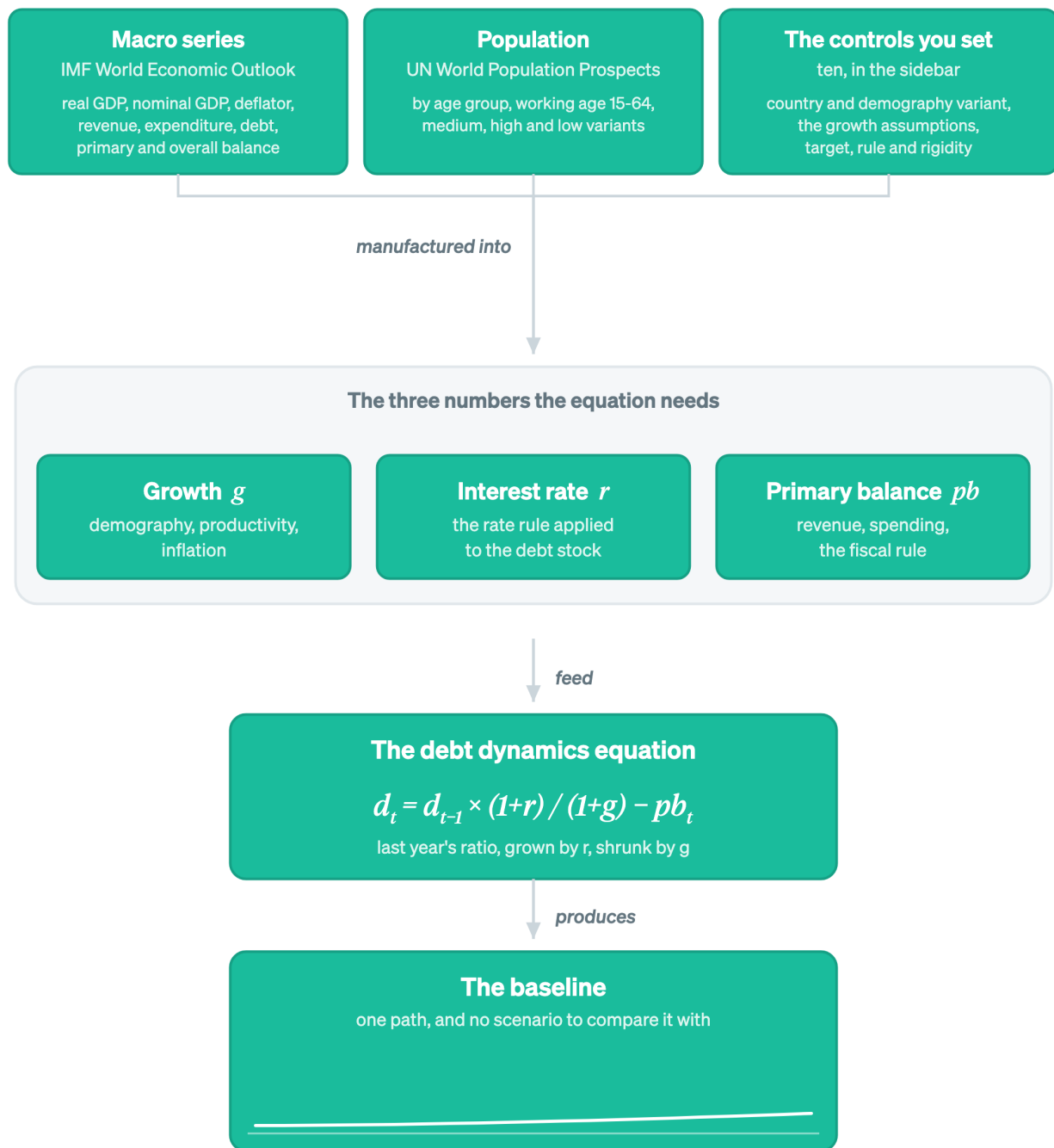


Figure 3.9: The end of Step 2: a complete long-term fiscal projection model. The warming block is absent from this drawing because it has not been built yet. Run this chain and you get the baseline.

Two published data sources and nine of the ten controls feed three small machines, and the machines supply the one equation. The chain gives a baseline: a complete long-term fiscal projection model. The tenth control, expenditure rigidity, waits for Step 3. Everything to this point would work identically in a world with a stable climate.

💡 Predict, observe, explain: run the baseline before the fan opens

Predict. Take a country you know. Write down whether its baseline debt ratio in 2099 is above or below where it starts, and one sentence on why, naming r and g .

Observe. Open the [Q-CRAFT Explorer](#), select that country and read the Baseline tab. Note the debt trajectory, then check the Fiscal Balances chart for whether the primary balance is in surplus or deficit.

Explain. Reconcile the two. If the primary balance is in deficit and the ratio is not exploding, the amplifier is working in that country's favour: nominal growth is beating the effective interest rate, the normal position for a country borrowing largely on concessional terms. If you expected a deficit to mean rising debt, that is the misconception from Chapter 1, now watched failing on real data.

Then do it again with a country that borrows commercially, and watch the same chart tell the opposite story. The arithmetic did not change between them. The sign of r minus g did.

The debt ratio: +3.3 points across the deficit years



Q-CRAFT Explorer, Verified mode, Kenya, Baseline tab, engine defaults. Kenya runs a primary deficit every year from 2035 to 2045, and over those eleven years the debt ratio moves +3.3 points, from 47.0 to 50.2 percent of GDP. A deficit does not have to raise the ratio, because nominal growth is working on the denominator at the same time. That is the reconciliation this chapter's arithmetic predicts, on the chapter's own country, so every plotted point is checkable against `figures/series/m2-primary-balance.csv` and `m2-debt-paths.csv`.

The primary balance: in deficit from 2035 to 2045

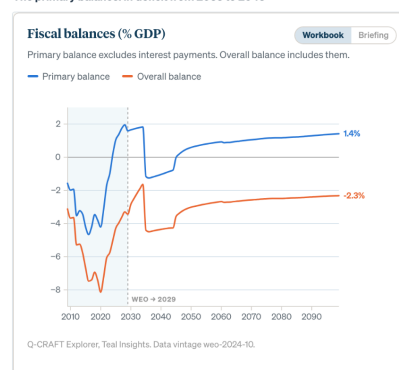


Figure 3.10: Kenya on the Baseline tab, with the debt ratio beside the fiscal balances that produce it. The primary balance is in deficit every year from 2035 to 2045, and across those 11 years the debt ratio moves +3.3 points. A primary deficit does not have to raise the ratio, which is the whole content of the equation in Step 1.

3.4 Step 3. Snapping on the climate block

i You are here. Step 3 of 3: the overlay. It adds one block and two arrows, with no new equation.

The climate part of Q-CRAFT is smaller than most people expect. It is a table of numbers, one per country per scenario per year: how much smaller GDP is than it would otherwise have been.

3.4.1 Step 3a. The econometrics measures deviation from a country's own norm

The estimates behind every climate number in Q-CRAFT trace to one paper: [Kahn, Mohaddes, Ng, Pesaran, Raissi and Yang \(2021\)](#), “Long-term macroeconomic effects of climate change”, published in *Energy Economics*.

It used a long global panel: 174 countries with annual data from 1960 to 2014. Neither a simulation of a warmer world nor a small sample of hot countries, it set fifty-five years of every country's growth against fifty-five years of its weather.

The variable that does the work is deviation, not heat. The regressor is the gap between this year's temperature and that country's own historical norm, a moving average of the past thirty years. A country at 25 degrees that has always been at 25 is at its norm, and the model expects no effect. A country at 8 degrees that has always been at 6 is two degrees above its norm, and the model expects one.

The paper's stated reason: temperature is trended in almost every country, so putting the level into a growth regression forces a trend into growth that the data does not show. Deviations have no trend, so they can be used.

A sustained deviation costs output every year it lasts, because the estimated effect is on the growth rate rather than the level. The headline estimate: a persistent above-norm rise of 0.01 degrees a year lowers per-capita real output growth by 0.0543 percentage points a year. Precipitation deviations showed no statistically significant effect and drop out.

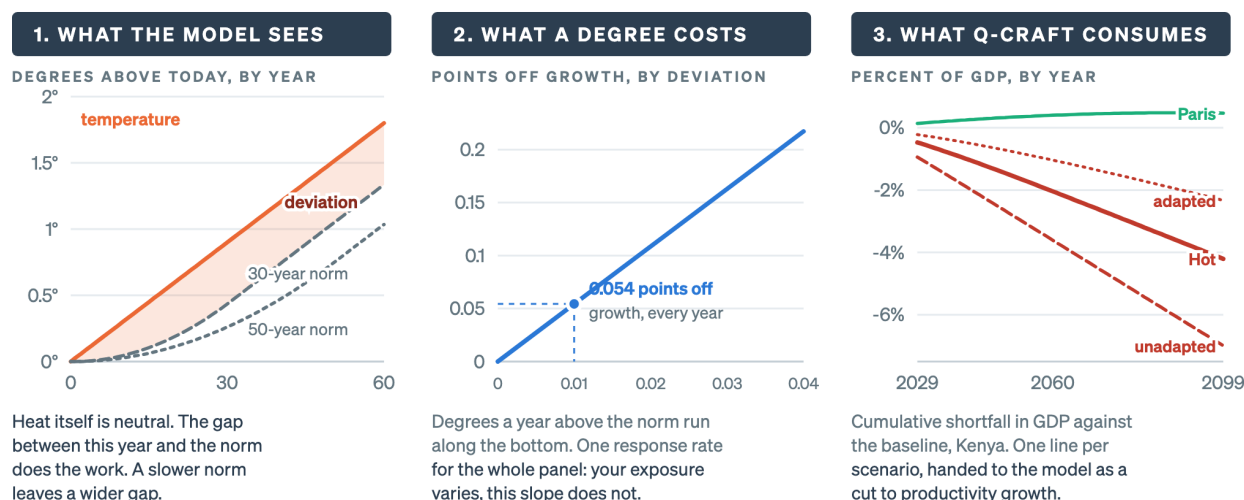
Adaptation is how fast the norm catches up. This is where the tool's adapted and unadapted scenarios come from.

The norm is a trailing average, so it chases a rising temperature and always lags. The paper works this out exactly: under a steady warming trend, the average gap between this year's temperature and an m -year trailing norm settles at $\frac{m+1}{2}$ years' worth of warming.

So a thirty-year window leaves a gap of about 15 years' warming. A fifty-year window leaves about 25 years' worth. Same temperature path, a gap two thirds larger, and a proportionally larger drag on growth every year. A short window is an economy that quickly treats the new temperature as normal. That is the only thing adaptation means in this model.

A deviation, one response rate, and a shortfall path for every scenario

The three moves that turn a global panel regression into a number your own projection can use.



Sources: Kahn, Mohaddes, Ng, Pesaran, Raissi and Yang (2021), Energy Economics 104, 105624, for panels 1 and 2. Panel 3 is the FADCP-derived climate table in this repository, Kenya, from m2-climate-drag.csv.

Figure 3.11: Panel two is the one to argue with. A single response rate for every country is what makes the estimate portable, and it is also what makes it a floor rather than a country-specific answer.

The response is pooled across countries. One number, 0.0543 percentage points per 0.01 degrees a year, applies to every country in the panel. The regression has a country fixed effect, so each country gets its own average growth rate, but a single long-run coefficient on temperature deviation covers all 174. The IMF User Guide says so plainly: “The authors do not estimate country-specific impact of temperature on GDP. Rather, temperature projections vary between countries, leading to different GDP impacts” (Box 4, p. 34).

That pooling is the single most important thing to know about how conservative a Q-CRAFT climate result is, and Step 3d comes back to it.

The adaptation window sets Kenya’s 2099 Hot cumulative GDP shortfall: 2.33 percent with a twenty-year window, 4.20 with the default thirty-year window, 6.99 with a fifty-year window.

The window arithmetic predicts those numbers almost exactly. Twenty years leaves 10.5 years of warming in the gap against thirty years’ 15.5, a ratio of 0.68. Fifty years leaves 25.5, a ratio of 1.65. The tool’s own numbers give 0.55 and 1.66: the unadapted case lands almost exactly, the adapted case does slightly better than the window width alone predicts.

i Going deeper: what the paper found, and what changed between drafts

The global counterfactuals. The paper runs its estimates forward against IPCC temperature projections. Under an unmitigated path of 0.04 degrees a year, world real GDP per capita is more than 7 percent lower by 2100 than in a reference case where temperature keeps rising along its own 1960 to 2014 trend. Under the Paris path of 0.01 degrees a year, about 1 percent.

The published abstract adds that losses reach 13 percent if the variability of climate conditions rises alongside the average.

The reference case is not a world that stopped warming. Q-CRAFT inherits that convention, so a Q-CRAFT baseline is a continued-historical-warming world rather than a no-climate-change world. In a few countries this makes the mitigation scenarios look mildly positive, which is why Kenya’s Paris line sits slightly above zero in the third panel above rather than at it.

The losses move a long way with the window. The working paper’s own counterfactual table gives world losses by 2100 under the unmitigated path of 4.44 percent with a twenty-year window, 7.22 with thirty, and 9.96 with forty. The paper concludes that adaptation reduces the loss and is highly unlikely to eliminate it.

One thing to know if you go to the working paper. The freely available 2019 working papers (Dallas Fed Globalization Institute WP 365, NBER 26167) are an earlier draft of the 2021 journal article and differ in two places that matter. The working paper reports point estimates of 7.22 and 1.07 percent where the published abstract says “more than 7 percent” and “about 1 percent”. More substantively, the working paper concludes that the findings “apply equally to poor or rich, and hot or cold countries”, while the published abstract states that the marginal effects of temperature shocks vary across climates and income groups. Cite the published article for anything load-bearing.

3.4.2 Step 3b. The IMF turns the response rate into a per-country table

The IMF’s Fiscal Affairs Department built the temperature side. The FADCP Climate Dataset (Massetti and Tagklis, 2023) assembles country-level historical temperature from the CRU observational record and country-level projected temperature from CMIP6 climate models, for each emissions pathway.

It re-ran the counterfactuals. Centorrino, Massetti and Tagklis (2024), “Climate Effects on GDP Growth: Updated Estimates of Kahn et al. (2021)”, applies the Kahn method to that dataset and produces, for 171 economies, a per-country per-scenario per-year path of GDP impact. That table is what Q-CRAFT reads. The re-run is also why Q-CRAFT’s numbers do not exactly reproduce the published paper’s: the paper used CMIP5-generation RCP pathways, and the FADCP dataset uses CMIP6-generation SSP pathways.

Two of the three ingredients are country-specific, and the third is not.

	Country-specific?	Why
Your warming path	Yes.	Country-level CMIP6 projections for your emissions pathway
Your historical norm	Yes.	Your own observed record, so your deviation is your own
The response to a degree of deviation	No.	One pooled coefficient, the same for every country

Exposure is yours. The response rate is the world’s. Two countries with the same projected deviation get the same estimated growth drag, whatever their income, institutions, sectoral mix or existing heat.

The six scenario names carry one trap: Hot is more severe than High, not less. Paris is SSP1-2.6. Moderate is SSP2-4.5. High is SSP3-7.0. Hot uses the same SSP3-7.0 emissions as High but takes the 90th percentile of the climate models rather than the median. Hot + Adapted and Hot + Unadapted are the same Hot temperature path with the norm window set to 20 and 50 years. Kenya’s 2099 shortfalls run: Paris +0.47 percent, Moderate −0.06, High −1.94, Hot −4.20, Hot + Unadapted −6.99. The dataset deliberately excludes SSP5-8.5, on the ground that its emissions are implausibly high.

3.4.3 Step 3c. The drag enters through g, and leans on pb

Everything above produces one number per year: how much smaller GDP is than the baseline said it would be. Q-CRAFT converts that into a cut to productivity growth and hands it to Step 2a. Impacts start in 2030.

Climate damage enters the equation as a smaller g

The hot run is the same equation with a smaller growth rate, every year from 2030 onward.

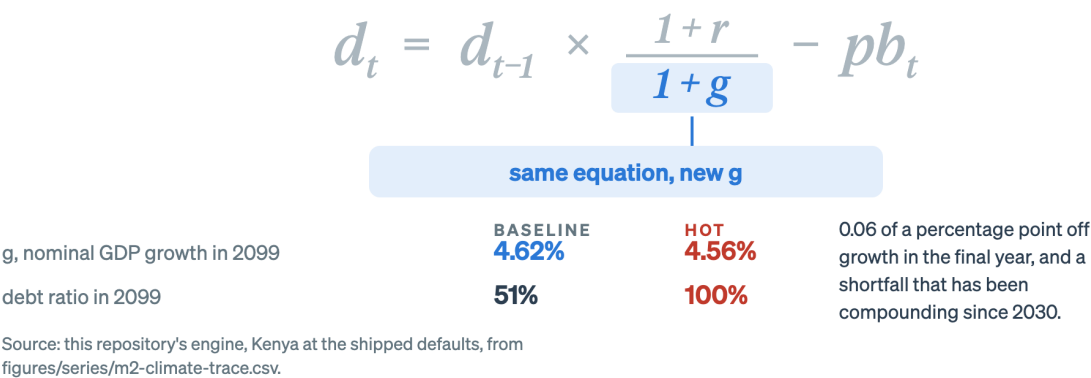


Figure 3.12: Nothing structural changes between a baseline and a climate run. The model recomputes the same line with a smaller growth rate, which is why reading a climate result always means comparing two runs rather than reading one.

The first channel changes one symbol and nothing else. The equation is not modified. No climate term is added. Employment is unchanged, because warming does not move the UN population projection. Inflation is unchanged. The interest rate is unchanged. One symbol is smaller.

The second channel is the primary balance. It only exists because of a dial. Revenue is a fixed share of GDP, so when GDP falls revenue falls with it and the revenue ratio does not move. Spending at rigidity 1.0 stays where the baseline put it in local currency, so smaller GDP means a higher spending ratio, a narrower wedge and a worse balance.

The Hot run halves Kenya’s primary balance by 2099. In 2050 the baseline primary balance is 0.59 percent of GDP and the Hot run’s is 0.40. By 2099 the baseline is 1.41 and the Hot run is 0.71.

The Hot run falls more than two points short of what Step 1 says it needs. The baseline needs 1.47 to hold its ratio still and runs 1.41, roughly stable. The Hot run needs 2.86 and runs

0.71, short by more than two points of GDP every year on a debt stock already twice as large. That is what a spiral looks like. It is why the cold-open fan bends upward rather than drifting.

Most of the 49 points come from the rigidity assumption.

Expenditure rigidity	Kenya's debt in 2099, Hot	Gap over the baseline
1.0, spending fully sticky	100	49 points
0.5	78	26 points
0.0, spending fully flexible	56	4 points

Forty-five of the forty-nine points depend on assuming a government watches GDP fall for seventy years and never adjusts. That assumption is defensible for a fiscal risk statement, whose job is to describe what happens if nothing is done. It is not a forecast. Presenting it as one will not survive the first serious question. Say which dial you used.

i Going deeper: the mechanics, and how to check them

How the shortfall becomes a growth rate. The climate table holds a cumulative GDP index shortfall per year. The engine adds its year-on-year percent change to that year's baseline productivity growth. For Kenya in 2099 the baseline productivity growth is 1.200 percent and the Hot run's is 1.142, a difference of 0.058 percentage points. That single year's cut is almost invisible. Seventy of them compounded, against an interest rate that does not move, is the entire result.

Why the numbers get large slowly. In 2050 Kenya's cumulative GDP shortfall under Hot is 1.5 percent and the debt gap is 2.7 points. In 2099 the shortfall is 4.2 percent and the gap is 49 points. The shortfall roughly tripled and the gap grew about eighteen-fold. Three things drive the divergence: the primary balance falls further behind what it needs every year, the interest-growth differential widens as growth slows, and the baseline has a fiscal rule pulling it back to target while the climate scenarios have none.

One thing you can check for yourself, and one to know. Open the Data tab and export both runs. Revenue as a percent of GDP should be identical in the baseline and every scenario, to the decimal, because revenue is not supposed to respond to warming at all. Employment growth is equally untouched, though the export carries no employment column: warming reaches growth through productivity alone.

3.4.4 Step 3d. A climate result is a lower bound through one channel

The same design choices that let Q-CRAFT run anywhere also cap what it can see. Read the two lists together: the costs are part of knowing the tool, not a caveat to bolt onto a result.

The design buys empirical grounding, discipline, and reach.

- **It is empirical.** The damage estimate comes from fifty-five years of observed data across 174 countries, not from a calibrated guess about the future.
- **It is disciplined.** Warming touches growth, and reaches the primary balance through a dial you set. There is no free parameter for how bad you feel climate change will be.

- **It runs anywhere.** Needing only published temperature records, climate model output, WEO series and UN population, it works for 171 economies without a bespoke country dataset or a mission.

The same design costs country specificity, every faster damage channel, and all feedback.

- **One response rate for the world.** The pooled coefficient cannot know that your agriculture is rain-fed, your capital sits on a floodplain, or your labour force works outdoors. Countries with concentrated exposure get the average response.
- **Only the slow productivity channel.** The scenarios contain no natural disasters, no sea-level rise, no tipping points, no non-market damages, and no adaptation spending. Each is a cost the result leaves out.
- **No feedback in either direction.** Damage does not reduce investment: there is no capital. Spending cuts under low rigidity do not reduce growth: nothing feeds back into productivity. The tool is a partial-equilibrium set-up and the User Guide says so.

Put together: a Q-CRAFT climate result reads as a lower bound on fiscal impact through the modelled channel, rather than as a central estimate of total impact. Chapter 6 works through what that licenses you to write and what it does not, including the specific cases where the arithmetic can mislead.

3.4.5 Step 3e. The three steps rebuild the 49 points

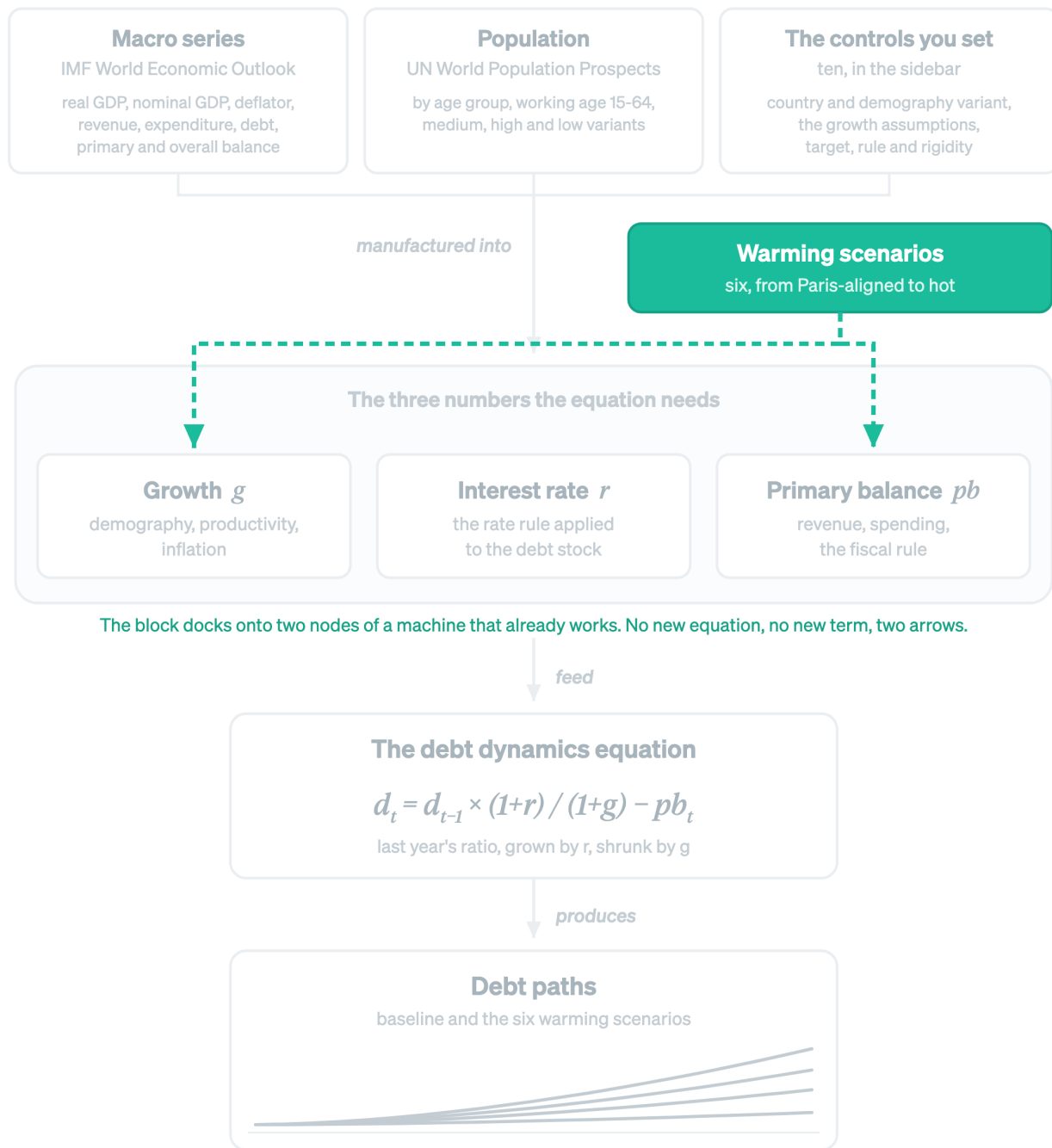


Figure 3.13: Step 3 adds one block and two arrows. The equation is untouched, the sources are untouched, and the interest rate never hears about the weather.

Take Kenya at 2050, halfway along the cold open, and walk the whole model in one paragraph.

The Hot scenario's temperature path leaves Kenya further above its thirty-year norm

than the baseline's does. The pooled response rate turns that gap into a cut to productivity growth, accumulated by 2050 into a GDP level 1.5 percent below the baseline. Employment and inflation did not change, so g falls from 6.36 to 6.30 percent. Revenue falls with GDP and holds its ratio. Spending does not fall, so the primary balance drops from 0.59 to 0.40 percent of GDP. Feed the smaller g and pb into Step 1's equation with an unchanged r of 7.6 percent, and the debt ratio comes out at 53.1 against the baseline's 50.4.

Twenty years in, the gap is already two and a half points. Forty-nine more runs of the same arithmetic build the 49 points: each on a slightly larger stock, with a primary balance falling further behind what the equation needs, against a fiscal rule that holds the baseline down but never touches the scenario. You can now point at every part of it.

3.5 Self-check: one question per step

 Three questions, one per step

Answer all three before opening the block below. If you can only do two, the one you missed tells you which step to reread.

1. **Step 1.** A country's debt is 70 percent of GDP, its effective interest rate is 6 percent and nominal growth is 4 percent. It runs a primary surplus of 1 percent of GDP. Is its debt ratio rising or falling next year, and what surplus would hold it still?
2. **Step 2.** Two countries have identical productivity and inflation assumptions. One has nominal growth of 6.4 percent and the other 3.7 percent. Name the input that accounts for the difference, and name the published source it comes from.
3. **Step 3.** Two countries face the same projected temperature path and have the same historical temperature norm. Q-CRAFT gives them different debt outcomes under the Hot scenario. Give two reasons that have nothing to do with the climate estimate.

DRAFT FOR TEAL: self-check answers

1. **Rising, and it needs 1.35 percent.** The amplifier is $1.06/1.04 = 1.0192$, so the ratio carries forward at 71.3 before any borrowing. Subtract the 1 point surplus and it lands at 70.3, up from 70. The debt-stabilizing balance is $0.70 \times (0.06 - 0.04)/1.04 = 0.0135$, so a surplus of 1.35 percent of GDP holds it still. Running 1 percent against a required 1.35 leaves the government a third of a point of GDP a year short. If you answered "falling, because it is running a surplus", that is the misconception Chapter 1 tested for. Step 1 is worth another twenty minutes.
2. **Employment growth, from the UN World Population Prospects.** With productivity and inflation held identical, the only remaining term in $(1+e)(1+p)(1+i)$ is employment, from the working-age population projection. The tool's only handle is the demography variant, Medium, High or Low. The two chapter countries are Kenya at plus 1.45 percent and Thailand at minus 1.11. Nothing about either country's climate exposure, borrowing or policy enters this answer.
3. **Any two of: starting debt ratio, effective interest rate, revenue and spending ratios, fiscal rule setting, expenditure rigidity setting.** The

climate estimate produces one number, a cut to productivity growth, the same cut for both. Everything after that cut runs through Steps 1 and 2, where the two countries differ in every input. This is the answer that matters most in a meeting, because it is how you answer “why is our number worse than theirs” without reaching for the climate science.

If the third one gave you trouble, that is the diagnostic the whole chapter was built for: the climate block is small, and most of what a climate result contains was already there in the baseline.

3.6 The three steps in one breath

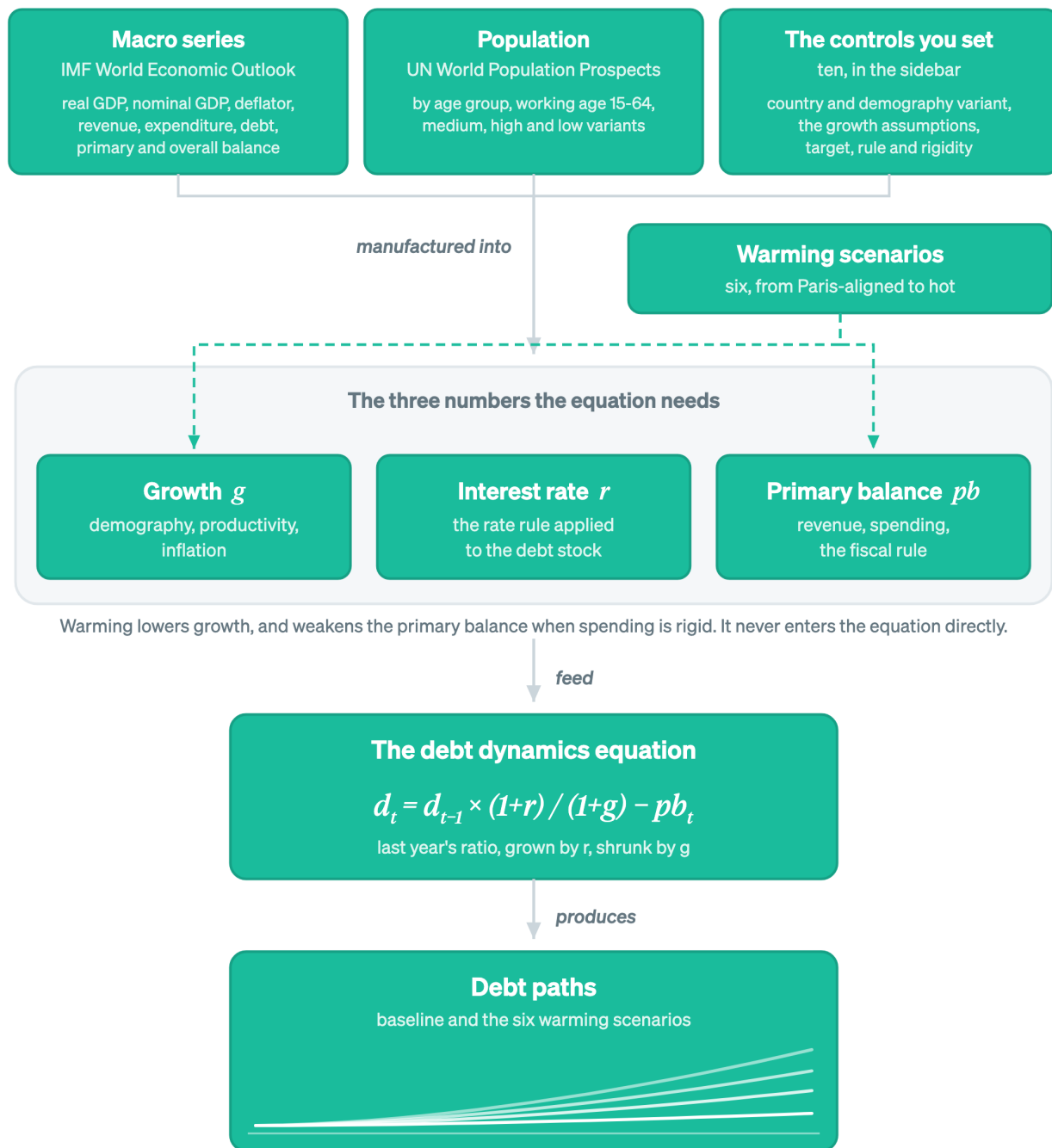


Figure 3.14: The three steps in one picture: an equation, a growth engine that feeds it, and a climate overlay that docks onto two of the three numbers.

An equation carries a debt ratio forward, amplified by interest against growth, less what the government paid down. A growth engine manufactures the three numbers it needs, out of two published data sources and nine controls you set. A climate overlay lowers one of the

three, and leans on a second through your spending assumption.

You can now do four things.

- Compute a debt ratio one year forward from r , g and pb , and say which of the three you could defend to somebody who disagreed.
- Trace any number in a Q-CRAFT output back to the data source, the assumption, or the estimate that produced it.
- Say what “adapted” means in this tool: the width of a moving average and nothing else.
- Separate the part of a climate result that comes from the damage estimate from the part that comes from your own rigidity dial.

Five errors recur on this material. Reading a falling debt ratio as a fiscal improvement. Assuming a balanced primary account stabilizes debt. Drawing climate as a fourth term in the equation. Quoting a scenario gap without saying which rigidity setting produced it. Reading “Hot” as milder than “High”.

The next three modules divide the remaining work. Chapter 4 chooses the sidebar’s ten controls one at a time, with the evidence you would cite for each. Chapter 5 runs a country end to end and writes the result up. Chapter 6 draws the boundary of what any of it supports.

On your desk this week: compute your country’s debt-stabilizing balance and compare it to the actual. Open the Explorer, pick your own country, and read the 2050 row off the Data tab’s grid: the debt ratio and the primary balance. Take your own view of the interest-growth gap from whatever your ministry’s framework assumes. Then compute pb^* from Step 1 and compare. That is one division. The comparison is the sentence most credible fiscal risk write-ups open with.

Chapter 4

Choosing the parameters

In this module

Set the Explorer's controls for a country and write down why you set each one where you did. Ten controls sit in the sidebar. The five carrying the heaviest judgment each get the full treatment, beside the control rather than ahead of it: what it is, why it matters, how to set it, what the underlying source shows, a predict-observe-explain run in the Explorer, and one judgment call to defend. The growth assumptions and the interest-rate approach form the assisted layer, their records one CONTEXT click away. The rationale lines you write here become the assumption section of your export packet. They are what survives you leaving the ministry.

Fast path

Already run the tool? Do the five **Defend your choice** self-checks and the five **Document it** blocks, and skip the explanatory prose. That is about 25 minutes. It is the part the capstone rubric marks.

4.1 Warm-up

From Chapter 2 (the three numbers and what builds them), without looking:

1. Which of the three numbers do the debt target and the fiscal rule move?
2. Expenditure rigidity of 1.0 means spending is what?

(Answers: the primary balance, through the spending side. Sticky, so the full revenue loss lands on the deficit.)

4.2 Why parameter choices need a written record

A year after a projection is built, nobody can say whether 1.2 percent productivity growth was a considered view about the country or a default nobody touched. The analysis is not wrong. It is unusable, because it cannot be defended.

Q-CRAFT Explorer cannot stop you leaving a default in place. What it can do is make the record, so your parameters and your reasons travel with the output.

4.3 By the end of this module you can

- **Set** the five core judgment calls for a country and **justify** each against country evidence
- **Predict** the direction and rough magnitude of each parameter's effect before you move the slider
- **Defend** an expenditure rigidity choice against a colleague who prefers a different one
- **Produce** the assumption-rationale section of the export packet, one entry per parameter

4.4 Where you are in the course

The map below lights the left-hand end of the chain, where this module works.

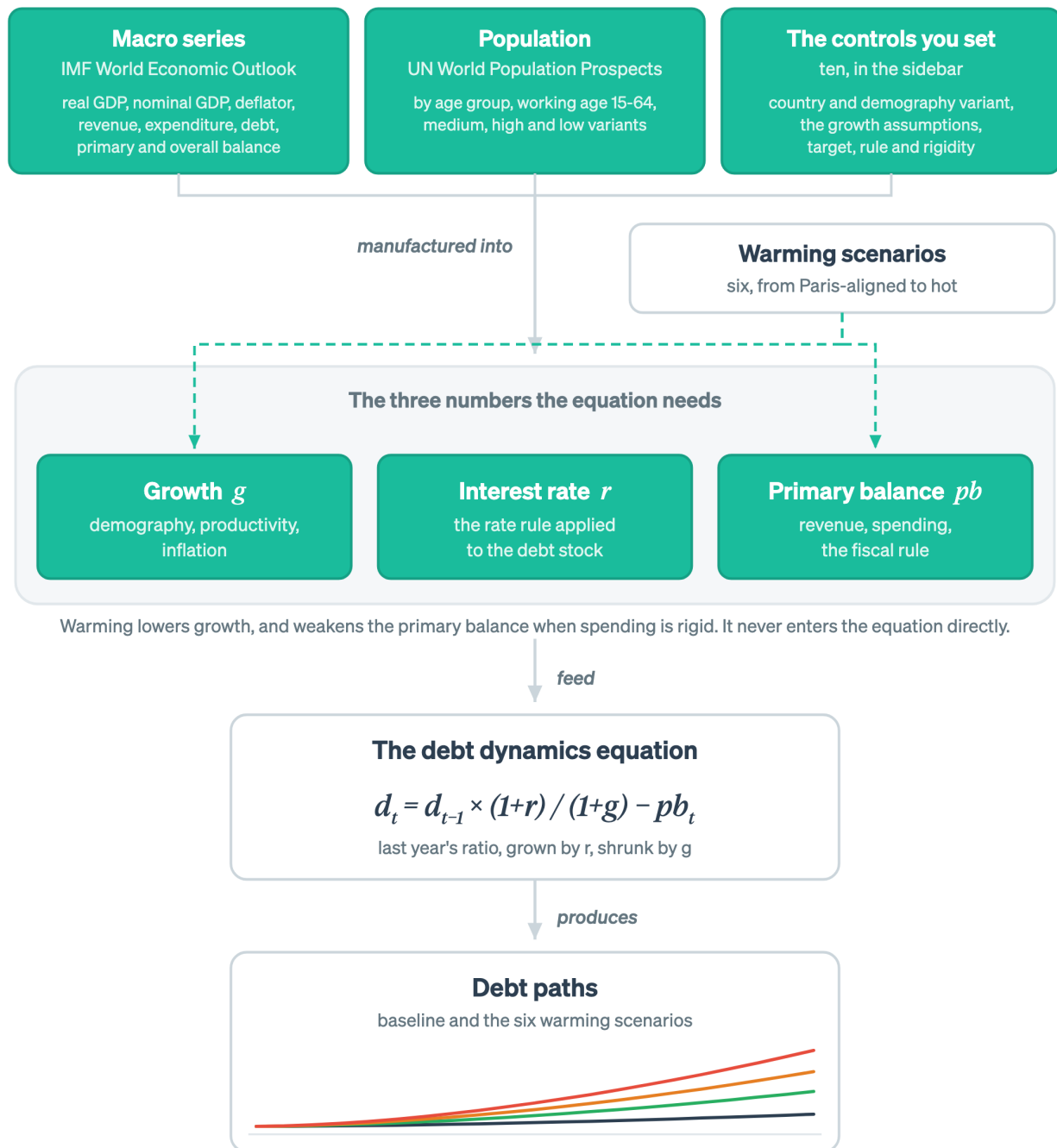


Figure 4.1: This module is the start of the chain: the two sources the country selection loads, the controls you set on top of them, and the three numbers they move.

Six nodes are lit: the two data sources the country selection loads, the controls you set on top of them, and the three numbers they move. Growth takes the demography variant and the growth assumptions, the interest rate takes the approach control, and the primary balance takes the fiscal-policy group.

4.5 Seven tabs, and what each one is for

Seven tabs hold everything the tool shows, and you will move between them and the sidebar constantly. **Baseline** is the no-climate-change projection. **Analysis** overlays baseline and climate debt paths on one chart. **Climate** shows what warming does to GDP, scenario by scenario. **Data** gives the grid and the two CSV downloads. **Export** assembles the export packet. Your assumptions and rationale sit in its annex. **Methodology** and **About the data** carry the model summary and the exact data releases behind each mode. Reading the charts properly is Chapter 5. For now you need only know where to look when a parameter moves.

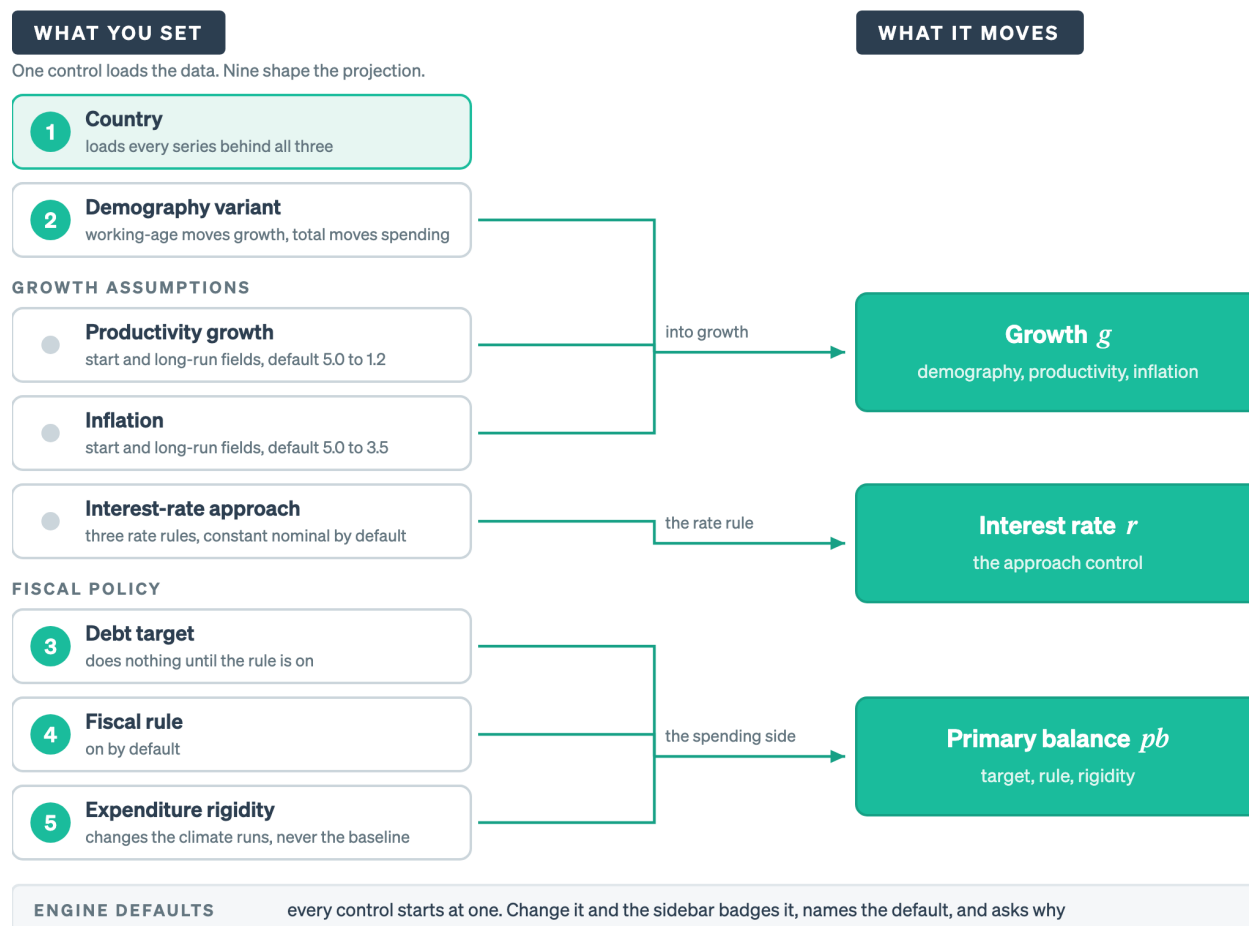
4.6 What the ten controls do

Q-CRAFT Explorer has ten user-facing controls, and every one lands on one of the three numbers. Country selection loads the data everything else runs on, and the demography variant sits beside it. Under **Growth assumptions** sit the productivity and inflation paths, a start and a long-run value each, and the interest-rate approach. Under **Fiscal policy** sit the debt target, the fiscal rule, and expenditure rigidity.

The figure below wires each control to the number it moves. Have that map in your head before you touch any of them.

Every control lands on one of the three numbers

Numbered chips are the five judgment calls, one section each below. The rest are the assisted layer.



Q-CRAFT Explorer sidebar: ten controls in eight rows, productivity and inflation carrying a start and a long-run field each. A result that turns on any of them needs the choice stated in your write-up.

Figure 4.2: Demography is the one control that arrives in two places, because working-age population drives growth while total population drives spending. The numbered five are this module's sections; the unnumbered rows are the assisted layer, each with its record one Context click away.

Five of the ten are the core judgment calls and this module's five sections: country selection, the demography variant, the debt target, the fiscal rule, and expenditure rigidity. Each is a decision no data source can make for you, each taught in full below, each ending in a **Document it** entry the capstone rubric marks. The other five, the growth assumptions and the interest-rate approach, carry published records to read their defaults against. They are the assisted layer.

Every control starts at an engine default. The sidebar keeps the score. Change any control and it gains a CHANGED badge, the engine default printed beside it, and a one-line **Why this value?** field. The export annex prints all ten parameters either way, with your rationale, or with the note that you gave none.

i Guidance where you need it

This module and the Explorer both put guidance beside the parameter, the second of the four reasons in Chapter 2. Every figure in this module is drawn from the same Parquet inputs the Explorer runs on, by `scripts/build_parameter_context.py` in the repository, and sits beside the parameter it informs rather than in a back annex. On screen, where a published source has a view, the **CONTEXT** button beside the parameter opens its record for the loaded country, stamped with the data mode, while you set the control. Two buttons, on country selection and the fiscal rule, open a short note instead, because no source publishes a view on them. The three teaching widgets linked in the Explorer's header build the intuition those records sit on.

4.6.1 Country selection

The country dropdown lists every country present in all four of the Explorer's input datasets. That is a test across sources, not a guarantee that every year of every series is present, so a few listed countries still have a hole where the projection needs a figure. Selecting a country loads macroeconomic data from the IMF's World Economic Outlook and population projections from the UN World Population Prospects. Which releases depends on the data mode: Current, the default, runs WEO April 2026 with the UN's 2024 revision, and Verified runs the workbook's own WEO October 2024 with the 2022 revision.

The country you select determines all historical data and every WEO forecast-period projection. No manual data entry: the tool loads real GDP, nominal GDP, revenue, expenditure, debt, interest rates, and demographic projections automatically.

Choose the country you are analyzing. Then check the starting point where the tool shows it. The Baseline tab's debt chart opens on the WEO record and forecast, shaded through the horizon, and the Data tab's grid carries the same figures year by year. Investigate before proceeding if the latest debt ratio does not match your understanding.

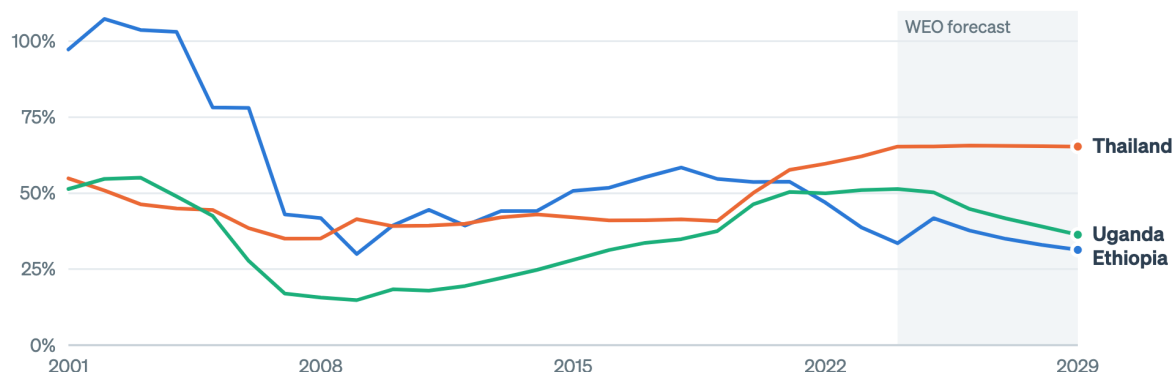
i Country coverage

The dropdown holds 175 countries on each of the two data vintages. Not all 175 produce a projection. Eight or nine, depending on the vintage, are refused outright, because the source lacks something the projection cannot proceed without, most often the primary expenditure figure or the starting debt stock. The Explorer says so on screen rather than drawing a line from a figure that is not there. A few others project from an earlier year than the rest, because the source stopped reporting them before the release ended. For those the tool names the anchor year wherever their results appear. `docs/country-coverage.md` in the repository lists both sets for both vintages, and the IMF User Guide (Tim and Rahman, 2024), Section II.B, describes what is loaded for each country.

The whole projection compounds off the starting debt ratio.

What the tool loads when you pick a country

General government gross debt, percent of GDP. The figure you sanity-check on the Baseline chart is the last of these.



Source: IMF World Economic Outlook, October 2024, as bundled with Q-CRAFT Explorer. Shaded years are the forecast.

Figure 4.3: Three countries the tool already holds data for. Debt ratios differ by more than a factor of three across them, which is why the first thing you do after selecting a country is check the number the Baseline chart reports.

Three countries the tool already holds differ in starting debt ratio by more than a factor of three. Hence the first move after selecting a country: check the number the Baseline chart reports.

💡 Predict, observe, explain

Predict. Before selecting your country, write down the debt-to-GDP ratio you expect the tool to report, from your own knowledge of its latest position.

Observe. Select the country and read the latest figure off the Baseline tab's shaded WEO band, or the same year's row in the Data tab's grid.

Explain. If your number and the tool's differ by more than a point or two, work out which is on a different basis before going further. Three causes are common: gross versus net debt, fiscal year versus calendar year, and the WEO vintage lagging the ministry's own outturn. Each is legitimate. Each will be the first question you are asked about your results.

! Defend your choice: country and vintage

A reviewer asks: "Is this the current debt number?" Take the worked case (the Uganda run Chapter 5 works end to end, the Statement's figures taken as given here). You are running Verified mode, WEO October 2024, and Uganda's FY2024/25 Fiscal Risk Statement reports public debt at 46.9 percent of GDP at June 2023 and projects 49.2 percent at June 2024. What do you say?

DRAFT FOR TEAL: a defensible answer on vintage

Name the vintage, name the basis, and state which direction the difference runs.

"This is WEO October 2024 on a calendar-year basis. The ministry's June 2024 figure is 49.2 percent on a fiscal-year basis. The starting point differs by under a

point, and the climate result is a difference between two runs from the same starting point, so it is not sensitive to that gap.” The last clause ends the conversation. You can say it only because you understand what the tool computes.

i Document it

In your export packet, record: **the data mode and its releases** (every packet file carries the stamp), **starting debt ratio as loaded**, and **any known difference from your ministry’s own figure, with the reason**.

4.6.2 Demography variant

The UN publishes population projections in three variants, **Medium, High and Low**, reflecting different fertility assumptions. This parameter selects which variant Q-CRAFT uses for population and labor force projections through 2100.

Working-age population growth, ages 15 to 64, drives employment growth in Q-CRAFT’s growth identity after the WEO forecast horizon. Higher population growth means a larger labor force, higher potential GDP, and, all else equal, more favourable debt dynamics. Countries facing demographic decline see slower growth and harder fiscal trajectories. Total population growth matters separately. Primary spending grows with it (see Chapter 2 for the expenditure growth formula).

Medium is the default and the standard assumption for most policy analysis:

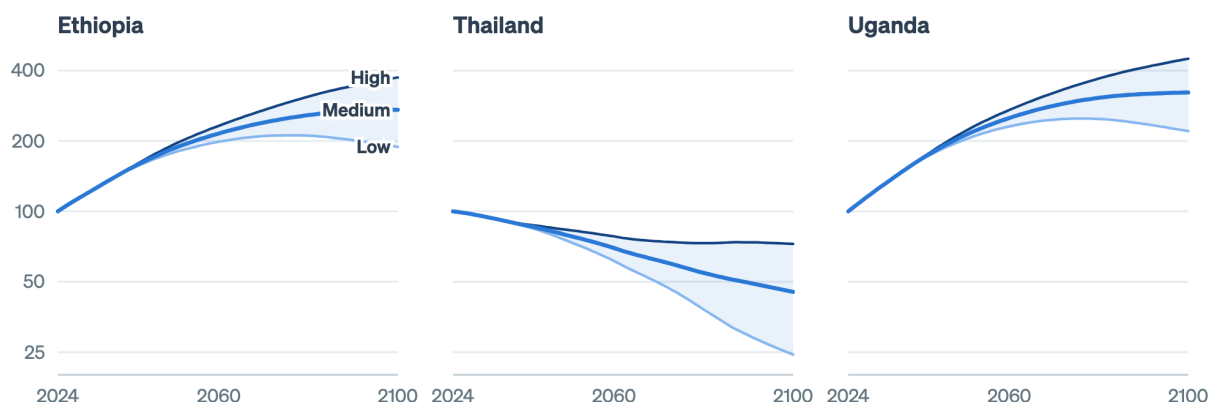
- Use **Medium** unless you have a specific reason to explore demographic scenarios.
- Use **High** and **Low** where fertility trends are uncertain or where you expect demographic policy to change.
- For countries in rapid demographic transition, with a declining working-age population, compare Medium against Low to see how sensitive the fiscal path is to the demographic outcome.

Detailed methodology: [the IMF User Guide, pp. 10-12](#) and Section IV.A on demography and employment.

The three variants are the UN’s own account of how uncertain its projection is, and how far apart they run depends on where the country is in its demographic transition.

The three UN variants, on three different demographics

Population aged 15 to 64, indexed to 100 in 2024, on a log scale. The gap is the uncertainty you choose inside.



Source: UN World Population Prospects 2022, as bundled with Q-CRAFT Explorer. Medium is the tool's default.

Figure 4.4: The demography control picks one of these three lines. In a country whose working-age population is still climbing the variants separate slowly. In one where it has turned down, the choice between Low and High is the difference between a workforce at a quarter of today's and one at three quarters.

Where the working-age population is still climbing the three variants barely separate. Where it has turned down, the choice between Low and High is the difference between a workforce at a quarter of today's and one at three quarters.

Predict, observe, explain

Predict. Ethiopia has one of the world's youngest populations. Thailand's working-age population has already started to fall. Before touching the control, will switching from Medium to High move Ethiopia's 2099 debt ratio a lot, a little, or barely at all? Write down a direction and a rough size in percentage points, then a second prediction for Thailand.

Observe. Run Ethiopia, switch between Medium and High, and watch the Baseline tab's debt trajectory. Then run Thailand and do the same. Compare the size of the two moves.

Explain. In countries with young, growing populations the variants barely separate over the projection, because the working-age share is already rising in every variant. In countries facing demographic decline the gap widens sharply: slower working-age growth means lower GDP growth and rising expenditure-to-GDP ratios at the same time. Both effects run through g : the expenditure channel through total population, the growth channel through working-age population, and the two populations move differently, which is why the net effect is not obvious in advance.

Defend your choice: demography

A colleague argues for the Low variant, because your country's fertility rate has been falling faster than the UN projected. Do you switch?

DRAFT FOR TEAL: a defensible answer on the demography variant

Probably not for the headline run. Certainly worth a sensitivity run. Medium is the reference variant, and using anything else as your headline leaves you defending a demographic forecast inside a fiscal risk document. The colleague's point is testable in two minutes: run Low, record the difference in the 2099 debt ratio, and put the number in the packet as a sensitivity. A small difference retires the objection with evidence. A large one is worth a paragraph.

i Document it

Record: **variant chosen, why** (one line), and **the 2099 debt ratio under at least one alternative variant** as a sensitivity.

4.6.3 Debt target (% of GDP)

The debt target is the debt-to-GDP ratio the fiscal rule steers toward over time. The range runs from 0 to 200 percent of GDP.

The target does nothing until the fiscal rule is enabled, and then it determines how hard spending has to adjust. A target below current debt calls for consolidation, which cuts primary expenditure. A higher target calls for less adjustment.

Start from the country's actual fiscal framework. Many countries legislate debt targets in their fiscal rules. Use them where they exist. Uganda's FY2024/25 Fiscal Risk Statement reads its climate results against a 50 percent of GDP fiscal rule ceiling and reports public debt passing it in the High, Hot and Vulnerable scenarios (Vulnerable is the Statement's label for the Explorer's Hot + Unadapted, and the naming note in Chapter 2 has the mapping). Where no formal target exists, the debt-stabilizing primary balance is a standard DSA benchmark (see [the IMF User Guide, Section IV.A](#), and Step 1 of Chapter 3).

Common practice offers three rough starting points, none of them authoritative IMF guidance:

- **Low-income countries (LICs):** 40 to 50 percent of GDP is a common range in practice, broadly consistent with IMF-World Bank Debt Sustainability Framework thresholds.
- **Emerging markets (EMs):** 50 to 70 percent of GDP, depending on market access, institutional quality, and fiscal track record.
- **Advanced economies (AEs):** can sustain higher ratios, 60 to 100 percent and above, though the appropriate target depends on growth prospects and market confidence.

Detailed methodology: [the IMF User Guide, pp. 15-18](#) on fiscal rule assumptions and the baseline scenario.

The target has no source figure: a policy choice rather than a published series. The nearest evidence is the country's own record, the debt-ratio figure at the top of this module, read against whatever anchor your fiscal framework already sets.

Predict, observe, explain

Predict. With the fiscal rule enabled, you are about to move the debt target from 40 percent to 80 percent. Write down what happens to primary expenditure in the early years and to the 2099 debt ratio.

Observe. Set the target to 40, read the primary expenditure path and the debt path in the Baseline tab. Set it to 80 and read them again.

Explain. A lower target forces faster consolidation, so expenditure is cut harder to bring debt down. A higher target leaves more spending room. The adjustment is gradual rather than immediate, because it works through the fiscal gap over several years. If you predicted an instant jump in expenditure, correct that: the rule is a feedback loop keyed to last year's debt position, not a switch.

Defend your choice: the target

You are asked to run the projection against a 40 percent target rather than the 50 percent ceiling your fiscal risk documentation already uses, “to be prudent.” What do you do, and what do you say?

DRAFT FOR TEAL: a defensible answer on the debt target

Run both, and say what the target does in the model. In Q-CRAFT the debt target is not a judgment about what is sustainable. It is the anchor the simulated fiscal rule steers toward, so changing it changes the assumed policy reaction rather than the country's risk. Give both numbers: “At the 50 percent anchor, the Hot scenario breaches the ceiling by X. At a 40 percent anchor, staying inside it requires Y percent of GDP a year of additional primary balance.” That answers the request without smuggling a policy preference into a risk assessment.

Document it

Record: **target used**, **its source** (legislated rule, Charter for Fiscal Responsibility, DSF threshold, analyst judgment), and **the alternative target you tested**.

4.6.4 Fiscal rule (Yes / No)

The fiscal rule, when enabled, adjusts primary expenditure to steer debt toward the target ratio. When disabled, spending follows baseline trends whatever the debt level does.

Without a fiscal rule, debt dynamics are purely mechanical. They follow the interest-growth differential and whatever primary balance falls out of revenue and expenditure trends. With the rule on, the model simulates active fiscal policy: the government cuts spending as debt rises or spends more as fiscal space opens.

Run it both ways, rule off first:

- **Start with the fiscal rule off** to see the “no policy change” baseline: is debt stable, rising, or falling under current trends?
- **Turn the fiscal rule on** to answer a different question: how much consolidation would reaching the debt target take?

Most countries have some form of fiscal anchor in practice, though enforcement varies. The IMF User Guide notes Q-CRAFT's fiscal rule is an approximation: it does not model the GDP effects of fiscal consolidation ([User Guide, p. 17, footnote 11](#)).

The adjustment lands on the level of primary expenditure rather than its growth rate. It is applied after the multiplicative growth factors and depends on the prior year's state, so Q-CRAFT computes it recursively, year by year.

Predict, observe, explain

Predict. Run your country with the rule off first. Will turning the rule on change the 2099 debt ratio a lot or a little? Reason from where the off-rule path ends relative to the target.

Observe. Run the same country with the fiscal rule on and off. Watch both the debt path and the primary expenditure path.

Explain. With the rule off and unfavourable dynamics, the interest rate exceeding growth, debt-to-GDP can rise without bound. Turn the rule on and debt converges toward your target, the primary expenditure chart showing the spending cuts required. The gap between the two runs is the fiscal effort your target implies. If the off-rule path already sits near the target, turning the rule on does almost nothing. Report that.

Defend your choice: rule on or off

Which run belongs in a Fiscal Risk Statement: rule on or rule off?

DRAFT FOR TEAL: a defensible answer on rule on or off

Both belong, because they answer different questions. Rule **off** is the risk statement: where we end up if policy does not respond, shown as the sensitivity beside the rule-on headline run (Step 1 of Chapter 5 makes rule on the headline, and when a rule-off baseline hits the zero debt floor the comparison breaks, see Section 6.7). Rule **on** is the policy statement: the effort required to hold the anchor. Publishing only the rule-on run understates the risk, because it assumes the response you are trying to justify. Publishing only the rule-off run invites the objection that no government would sit still for seventy years. Report the pair and say which is which.

Document it

Record: **rule on or off for the headline run, the paired run, and one sentence on which question each answers.**

4.6.5 Expenditure rigidity (0.0 - 1.0)

Expenditure rigidity measures how far government spending resists downward adjustment when climate shocks reduce GDP. It affects climate scenarios only, never the baseline.

Rigidity decides how much of the climate cost becomes debt. When climate change reduces GDP, revenue falls with it, because revenue tracks nominal GDP. What happens to spending settles the rest.

- **1.0 (fully rigid, worst case):** spending stays at its baseline level in local currency even as GDP falls, so the entire climate impact passes through to the primary balance and the debt ratio. This is a government that cannot or will not cut spending: a large civil service wage bill, entitlement programmes, or political constraints.
- **0.0 (fully flexible):** spending falls in proportion with GDP, holding the expenditure-to-GDP ratio at its baseline level. The debt impact of climate change is smaller, because the government absorbs the shock by reducing real spending per person.
- **Values between 0 and 1** represent partial adjustment. Most countries sit in between: some categories are rigid (wages, pensions, debt service) while others can adjust (capital investment, discretionary programmes).

Start from the composition of the budget:

- Countries with large public sector wage bills, universal social programmes, or rigid entitlement commitments: lean toward 0.7 to 1.0.
- Countries with more fiscal flexibility, smaller governments, or well-established expenditure review processes: 0.3 to 0.6.

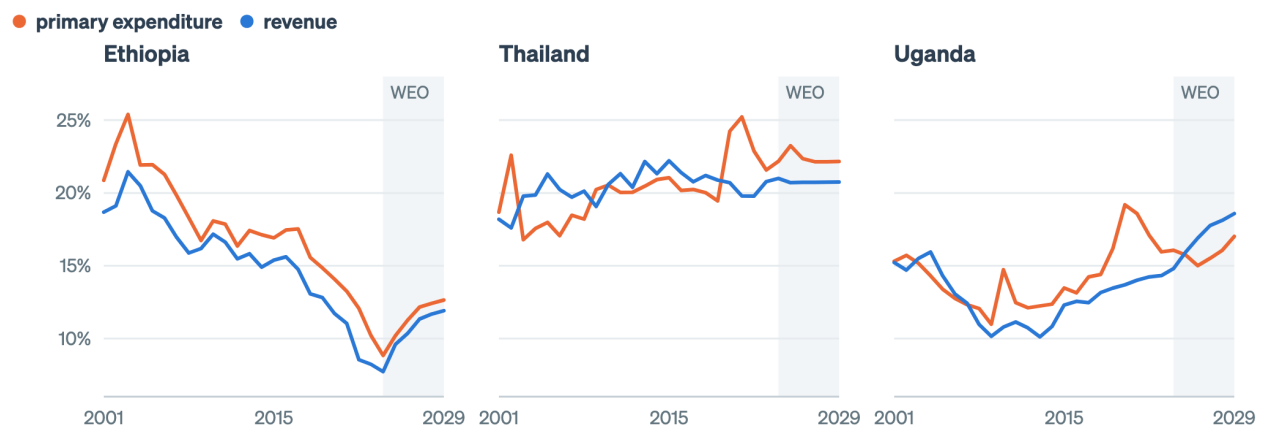
The default of 1.0 is conservative by construction. It shows the worst case, where all climate damage becomes additional borrowing.

Detailed methodology: [the IMF User Guide, pp. 20 and 35-36](#) on the expenditure rigidity parameter and fiscal effects of climate change.

Rigidity is a claim about how spending has behaved, and the WEO record carries evidence on it.

How far spending has actually moved with GDP

Both as a percent of GDP. Where the two ratios drift apart, spending is not tracking the economy.



Source: IMF World Economic Outlook, October 2024. Shaded years are the forecast.

Figure 4.5: Rigidity is a claim about how these two lines move together. Where expenditure holds its share of GDP while revenue falls, the budget has behaved as though rigidity is high.

Read the revenue and expenditure lines against each other rather than separately.

Revenue holds a roughly constant share of GDP by construction in the projection but moves in the historical record, and the question is what expenditure did when it moved. A budget whose

expenditure ratio climbs while revenue falls has behaved as though rigidity is high. That is evidence you can put in the packet, stronger than a view about political will.

⚠ WIDGET-TODO: expenditure rigidity

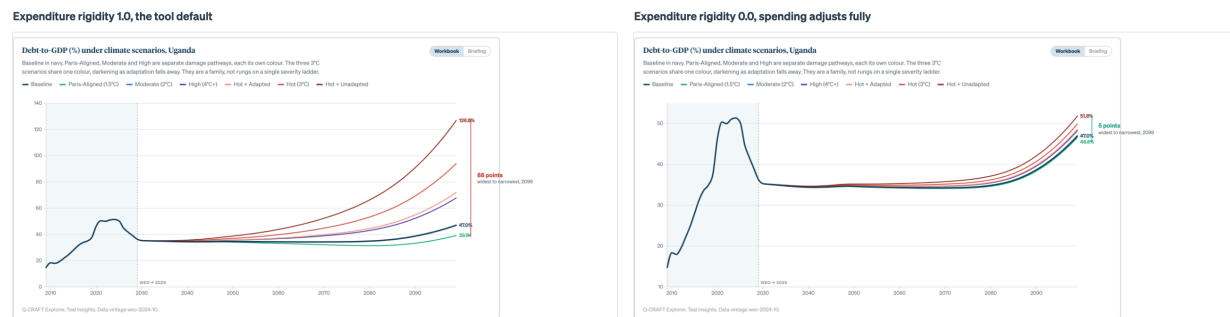
Embed the expenditure rigidity intuition widget here (lane 2, run 3: `apps/qcraft-web/widgets/*`). Iframe under Quarto for now, native React once the site moves to Astro. The integration pass happens after this restructure and run 3 both land, and this callout is the anchor point. The prose above must carry the full idea on its own, because the default state of any interactive has to work for readers who never touch it.

💡 Predict, observe, explain

Predict. On the Analysis tab you are about to compare the climate fan at rigidity 1.0 against 0.0. Write down which produces the wider fan, and by roughly what factor.

Observe. Set rigidity to 1.0 and read the 2099 gap between the baseline and Hot + Unadapted. Set it to 0.0 and read the same gap.

Explain. At 1.0 the fan spreads wide, because the whole climate revenue loss accumulates as debt, so the Paris-Aligned to Hot + Unadapted distance is large. At 0.0 the scenarios compress toward the baseline, because the government absorbs the shock through spending. The distance between the two fans is your country's fiscal exposure to expenditure rigidity, usually the largest single lever in the tool. If you predicted rigidity matters less than the choice of climate scenario, check that prediction. For most countries it is wrong.



Q-CRAFT Explorer, Verified mode, Uganda, Analysis tab, fiscal rule on. Expenditure rigidity 1.0 on the left and 0.0 on the right. The Explorer scales each chart to its own data, so the two vertical axes are not the same and the eye will read the two fans as more alike than they are. The bracket on each panel is the measurement to compare: the fan spans 88 points of GDP at rigidity 1.0 and 5 at rigidity 0.0. That collapse is what the rigidity control does.

Figure 4.6: The same country, the same scenarios, and the expenditure rigidity control moved from 1.0 to 0.0. The Explorer rescales each chart to its own data, so the bracket rather than the eye is the comparison: the scenario fan spans 88 points of GDP at rigidity 1.0 and 5 at 0.0.

! Defend your choice: rigidity

A colleague says 1.0 is alarmist: “no government holds spending flat in local currency for seventy years while the economy shrinks.” Another says anything below 1.0 assumes a fiscal discipline your country has not demonstrated. What do you do?

DRAFT FOR TEAL: defending a rigidity choice

Both colleagues are making empirical claims. Settle them with evidence rather

than adjectives.

The evidence to reach for is the composition of the budget, not a view about political will. The share of primary expenditure in wages, pensions, statutory transfers and other commitments requiring legislation to change is close to a floor on rigidity. In most low-income countries it is high, and the flexible remainder is dominated by domestically financed development spending, the spending a government cuts first.

The defensible position is to run 1.0 as the headline and a lower value as the sensitivity, saying why in one line: the default is conservative by design, and a fiscal risk statement is the document where conservatism belongs. The sensitivity run gives the second colleague their number: at rigidity 0.6, the 2099 gap narrows from X to Y percentage points.

What not to do is pick a middle value because it feels balanced. An unexplained 0.5 is weaker than 1.0 with a stated rationale, because 1.0 at least has the tool's own conservative-default argument attached. A middle number with no reasoning is the assumption that gets quietly reused for five years by people who never knew it was a guess.

The sentence for the packet: "Rigidity set at 1.0, the tool default, because roughly N percent of primary expenditure is wages, pensions and statutory commitments. The sensitivity run at 0.6 changes the 2099 climate-baseline debt gap from X to Y percentage points."

i Document it

Record: **rigidity used, the budget-composition evidence behind it, and the sensitivity run and its effect on the headline gap.** This is the entry the capstone rubric weights most heavily.

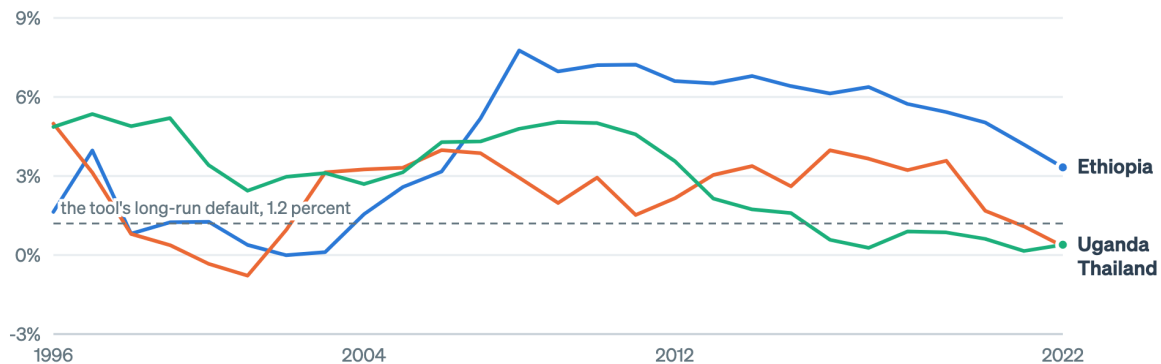
4.7 The assisted layer: growth assumptions and the interest-rate approach

The five remaining controls are assumptions about the future, and the Explorer puts the published record beside each one. Productivity growth and inflation each take a start and a long-run value. The interest-rate approach picks which of the three rate rules from Chapter 2 runs, the constant nominal rate by default. All five start at engine defaults, so an untouched run is the defaults' run, and the annex will say so.

Two of those defaults are worth seeing against the record they are meant to continue. Productivity growth slides from 5.0 percent to 1.2 percent, and inflation from 5.0 percent to 3.5 percent, for every country in the tool.

What output per worker has actually done

Growth in GDP per employed person, five-year trailing average. The engine default slides every country from 5.0 percent to 1.2 percent.

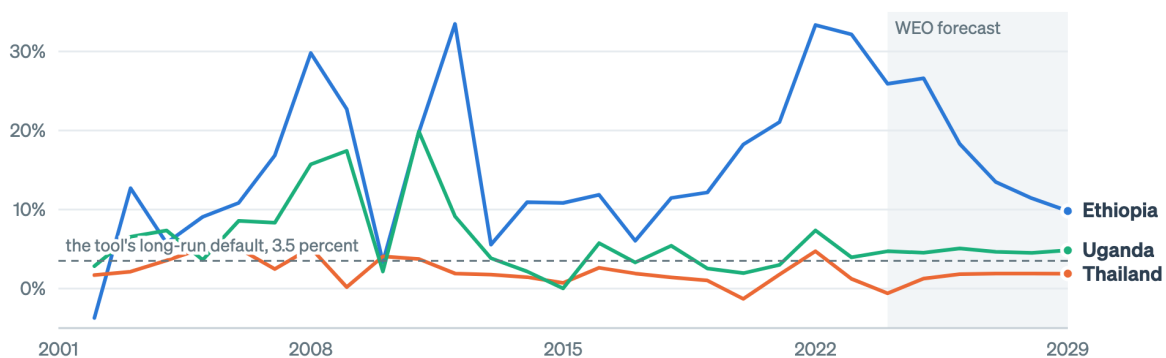


Source: World Bank World Development Indicators, as bundled with Q-CRAFT Explorer.

Figure 4.7: The engine default slides productivity growth from 5.0 percent to 1.2 percent for every country. Set that path against what output per worker has actually done before you quote a result that depends on it.

What the price path has actually done

GDP deflator growth, percent a year. The engine default slides every country from 5.0 percent to 3.5 percent.



Source: IMF World Economic Outlook, October 2024. Shaded years are the forecast.

Figure 4.8: The engine default slides inflation from 5.0 percent to 3.5 percent for every country. The three records here start in very different places, which is what the single default is averaging over.

Neither figure tells you the default is wrong. Both tell you what it is averaging over, the thing to say out loud when a result turns on either assumption. Where the gap between your country's record and the default is large, write one sentence: the run keeps the engine's global default, and your country's record sits above or below it. Name which.

In the Explorer, that comparison is one click. The CONTEXT button beside each growth assumption opens the same record for the loaded country, places your setting against it, and offers a computed sentence you can add to the rationale. The interest-rate approach gets the same treatment, and Chapter 3's interest-rules comparison is the reason to look: for Kenya, the three approaches

disagree by thirty-one points of GDP by 2099.

A changed growth assumption earns the same line in the annex as the five core calls.

An untouched default needs nothing beyond its name. A moved one is a claim about your country's future, and the record behind the CONTEXT button is where its defence starts.

4.8 What you can now do

- You can set the five core judgment calls for a country and give a reason for each that cites evidence rather than preference, and you know where the assisted layer's records live.
- You can predict each parameter's direction of effect before moving it, which tells you when the tool is behaving oddly.
- You have five **Document it** entries. Together they are the assumption-rationale section of your export packet.

Three errors recur on this material: inverting the rigidity scale (see Chapter 1, question 2), moving the debt target while the fiscal rule is off and wondering why nothing happened, and choosing a middle value to avoid an argument.

On your desk this week: find one number in a projection your department published and ask who chose it and why. If the answer takes more than a minute to find, you have made the case for the export packet.

Chapter 5

A worked example, end to end

In this module

You will run one country the whole way through, from parameter choices to publishable prose. Uganda is the worked case: the tool's golden-master verification country, every figure checkable against two published documents. The scaffolding then fades: a completion problem on Ethiopia where you finish the interpretation, then an independent problem on a third country done alone. The target is a write-up, not a chart, in the format of the IMF's C-PIMA presentation of Q-CRAFT results and Section III of Uganda's Fiscal Risk Statement.

Fast path

There is no fast path: this is the module the capstone is built from. If short of time, do Section 5.6 (the Uganda run, all seven steps) and Section 5.8 (the unscaffolded third-country problem) and skip the middle.

5.1 Warm-up

From Chapter 4 (choosing the parameters) and Chapter 2 (the warming block):

1. Which two choices change the climate result but not the baseline?
2. In which year do climate scenarios begin to diverge from the baseline?

(Answers: expenditure rigidity, and the climate scenario itself. Everything else moves both. 2030.)

5.2 Two documents already show what the output looks like

You are not inventing a format. Two published documents show how Q-CRAFT output reads when it reaches a policymaker. Both are about Uganda, where the tool has run the full cycle from workshop to published statement. The genre is not Ugandan, and your ministry's version will look much the same.

The IMF's C-PIMA write-up covers the whole assessment in a single paragraph. *Uganda:*

PFM Climate Assessment: Public Investment and Fiscal Risks Management (IMF, 2024) reports the September 2023 workshop with staff of Uganda’s Ministry of Finance, Planning and Economic Development (MoFPED, the finance ministry). Climate change looks milder in Uganda than in other sub-Saharan African countries, it says, yet still matters for growth and fiscal sustainability: by end of century GDP loss could pass 4 percentage points, with debt at 66 percent of GDP against 47.5 percent in the baseline. That one paragraph carries three numbers and a comparative judgment.

Uganda’s Fiscal Risk Statement FY 2024/25, published by MoFPED, is the fuller treatment. It carries scenario definitions, a baseline fiscal path table, GDP deviation charts, fiscal balance and debt charts, and a closing policy sentence. Its climate section runs pages 13 to 17 of a 31-page document, and three source lines under its exhibits read “QCRAFT (2023).”

Your capstone is the shorter of the two. This module builds it.

i The published section, page by page

Every line below is from the document itself: open it and check.

Page 4, the introduction. The statement groups fiscal risks into three categories and says the climate analysis “expanded its scope on fiscal risks associated with climate change by using the Quantitative Climate Risk Assessment Fiscal Tool (Q-CRAFT) for long term fiscal sustainability analysis.” That is the tool named in the ministry’s own words, in the framing section.

Page 13, the section opens. The body heading reads “III. CLIMATE CHANGE AND NATURAL DISASTER FISCAL RISKS”. The table of contents lists the same section as “CLIMATE CHANGE FISCAL RISKS”. The page is national hazard context rather than Q-CRAFT output: floods and epidemics at about 75 percent of recorded events between 1985 and 2021, footnoted to the World Bank’s Climate Risk Country Profile 2021, and annual economic losses from natural disasters estimated at US\$87 million. Table 4 is sourced to the World Bank Group (2022).

Pages 14 to 15, the set-up. “Using the Q-craft tool, an assessment of the impact of climate change on Uganda’s economic growth and fiscal sustainability was undertaken.” The tool quantifies effects “under five different climate scenarios, against a baseline”, the five named and defined on the page: Paris, Moderate, High, Hot, Vulnerable. A footnote cites [Kahn et al. \(2021\)](#) for the temperature-to-growth empirics. Page 15 carries Figure 8, the baseline decomposition of nominal GDP growth, and Table 5, the baseline fiscal path reproduced below, under the line “Source (Fig 8 & Tab 5: QCRAFT (2023))”.

Page 16, the macroeconomic effects. Figure 9, deviations from the baseline in level GDP and in labour productivity growth, “Source: QCRAFT (2023)”. The text reports “a 4 percent reduction in nominal GDP by the end of the century under the Hot scenario”, with Paris and Moderate described as benign for Uganda.

Page 17, the fiscal effects and the finding. Figures 10 and 11, balances and the debt ratio by scenario, “Source: QCRAFT (2023)”. Then the two sentences the whole section exists for: the primary deficit under Hot ends “0.7 percent worse than the baseline, thereby raising public debt by over 18 percent of GDP”, and “In the High, Hot, and Vulnerable scenarios, public debt surpasses the 50 percent of GDP fiscal rule ceiling, taking on an unsustainable upward trajectory.” The section closes on a policy sentence about meeting Paris commitments. The section runs five pages and carries one table, four figures, and one threshold finding. That is the size of the thing you are learning to write.

5.3 By the end of this module you can

- **Execute** a complete Q-CRAFT run for a country, from parameter selection to CSV export
- **Apply** the Baseline Sanity Check before interpreting any climate result
- **Interpret** the three chart families, saying what each one licenses you to claim
- **Draft** a two-paragraph fiscal risk write-up in the register of a published Fiscal Risk Statement
- **Assemble** the export packet: inputs, rationale, output

5.4 Where you are in the course

The map below lights the far end of the chain: the debt paths, and the write-up you build from them.

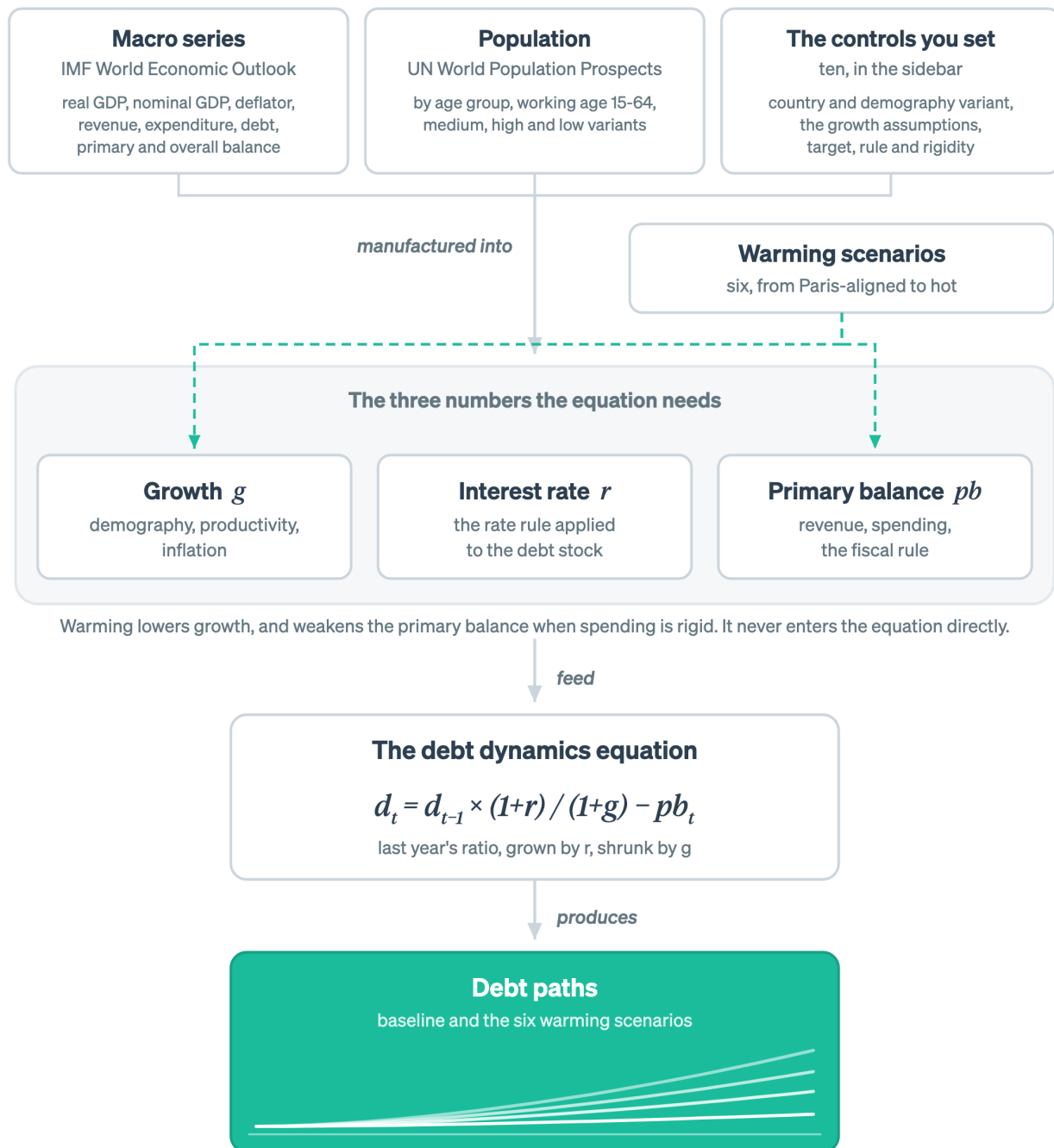


Figure 5.1: This module is the output end: the debt paths, and the write-up that comes out of them.

5.5 The target format

The Q-CRAFT part of the Statement's climate section is organised in three moves, numbered in the document itself. Copy that structure: it survived contact with its readers.

Move	What it contains	Where your material comes from
1. Set up the analysis	What the tool does, the scenarios used, and the baseline fiscal path	Chapter 4 parameters, Baseline tab
2. Macroeconomic effects	GDP deviation from baseline by scenario, and the productivity channel behind it	Climate tab for the deviation chart; Step 3 of Chapter 3 for the productivity mechanism
3. Fiscal effects	Primary and overall balance by scenario, debt-to-GDP by scenario, read against the fiscal anchor	Analysis tab for the debt chart; the exported results (the packet's CSV or its Excel file, Step 6) for the balance series

The published baseline table is worth copying too. Uganda's Table 5 reports four years, not seventy:

Summary fiscal path	2023	2050	2075	2099
Primary expenditure (% of GDP)	19.9	19.6	18.8	19.4
Interest expenditure (% of GDP)	3.3	1.3	1.8	2.4
Interest rate (%)	7.7	3.9	5.5	5.5
Primary balance (% of GDP)	-6.3	-1.7	-0.8	-1.4
Overall balance (% of GDP)	-9.6	-3.0	-2.7	-3.8
Debt-to-GDP ratio (%)	47.1	36.2	35.8	47.5

Source: Uganda Fiscal Risk Statement FY 2024/25, Table 5, sourced to QCRAFT (2023).

Four columns is a deliberate editorial choice, because a seventy-row table is not a policy document. Before any climate damage, the baseline says this: debt falls to the mid-30s by mid-century and comes back to 47.5 by 2099. The interest rate row, read against the growth path, shows why.

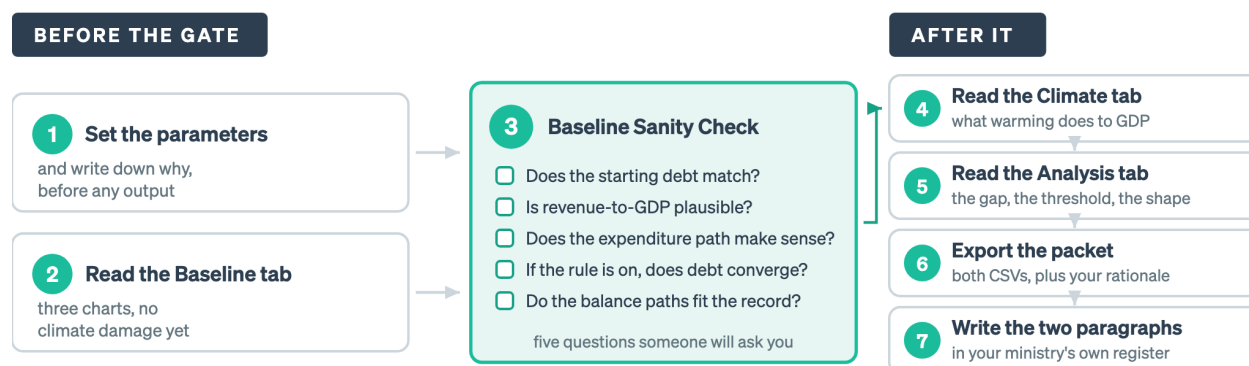
5.6 The worked case: Uganda, start to finish

Uganda is the worked case because its numbers are published and checkable, not because it is the country you work on. What transfers is the method: the seven steps below. Work through them in the app. Every step names what you are doing and what it licenses you to say.

The figure below lays the seven out, with step 3 across the middle rather than beside the others.

Step 3 gates every step after it

A climate result computed on a baseline nobody checked is a number with no owner.



The four steps on the right are only worth doing once the five boxes are answered. The gate applies to your own country, and to every rerun after a parameter changes. Note any box you skipped, and why.

Figure 5.2: Steps 1 and 2 produce a baseline. Steps 4 to 7 spend it. The five boxes in the middle are the only thing standing between the two, and each one is a question a reviewer will ask you anyway.

Everything right of that gate is only as good as the baseline beneath it, so the sanity check comes before any climate reading.

5.6.1 Step 1: set the parameters, and write down why

Open [Q-CRAFT Explorer](#) and select **Uganda**. Use these settings for the worked case:

Parameter	Setting	Rationale for the packet
Country	Uganda	Analysis subject
Demography variant	Medium	Reference variant, with Low tested as a sensitivity
Debt target	50% of GDP	The ceiling Uganda's own Fiscal Risk Statement reads results against
Fiscal rule	On for the headline run	The setting the published workbook ships, so the headline run is the one a reader can reproduce in Excel, with the rule-off run reported alongside as the sensitivity
Expenditure rigidity	1.0	Tool default, conservative, with budget composition evidence per Chapter 4 and 0.6 tested as a sensitivity

That table is the first page of your export packet. Write it before looking at any output, so the settings are choices rather than post-hoc justifications. The five controls under Growth assumptions stay at engine defaults for the worked case, and the packet's annex records they were left there.

i Going deeper: why the rule is on, and what happens when it is off

Turning the fiscal rule off looks like the more neutral choice. It is not.

With the rule off, Uganda's baseline debt ratio falls to zero by 2050 and stays there. The climate scenarios keep going, landing between -474 and -523 percent of GDP by 2099. Their order inverts on the way, Paris-Aligned finishing lowest and Hot + Unadapted highest, so the chart appears to say the hottest future is the best fiscal outcome.

Nothing has gone wrong. That is the workbook's own arithmetic. On the **Baseline** sheet the debt recursion is wrapped in a floor, `=IF((...)<0,0,...)`, so a negative baseline is written as zero. The six scenario sheets carry no floor. Section 6.7 sets out the asymmetry and the reading rule for charts that touch zero. This is what it looks like when it bites.

A rule-off baseline is not assumption-free. It is a baseline pinned to a floor, compared against scenarios that have none, and the comparison stops meaning anything. The rule-on run is also the setting the published workbook ships, at cell C33 of its Dashboard, so a reader can reproduce it in Excel step for step.

Report the rule-off run as a sensitivity if you want it, and say in the packet which years the floor is active. Do not put it in the headline.

5.6.2 Step 2: read the Baseline tab

The Baseline tab shows the no-climate-change scenario: the country's fiscal trajectory under current trends and the assumptions you set. Three charts share the tab.

Debt-to-GDP trajectory (first) is the headline chart. Follow it out of the shaded region (the WEO record and forecast) into the model's own projection. Is debt rising, stable, or falling? With the fiscal rule on, debt should converge toward your target. With it off, the trajectory reveals the underlying fiscal dynamics: favourable when growth exceeds interest rates, unfavourable when interest rates exceed growth.

Revenue and Expenditure (% of GDP) (second) shows two ratios that can drift apart. Revenue-to-GDP stays constant by assumption: revenue grows with nominal GDP. Primary expenditure-to-GDP may diverge: spending follows total population, revenue the working-age population, with productivity and inflation common to both sides. Where working-age population outgrows total (the demographic dividend), expenditure-to-GDP falls and fiscal space opens. In ageing countries it rises, a structural fiscal pressure independent of any policy choice.

Fiscal Balances (third) carries two lines: the primary balance, revenue minus non-interest expenditure, and the overall balance, which includes interest payments. The zero line is the reference: a primary surplus means non-interest revenue covers non-interest spending. The gap between the lines is the interest burden on existing debt.



Q-CRAFT Explorer, Verified mode (WEO October 2024), Uganda, demography Medium, debt target 50, expenditure rigidity 1.0, fiscal rule on. Captured from the frozen build at tag freeze-2026-08-29.

Figure 5.3: The Baseline tab for Uganda, the image the rest of this module refers back to. Three things are marked: where the shaded WEO period ends, the 2099 debt ratio of 47.0%, and the distance between the primary and overall balance, which is the interest burden on the existing stock.

5.6.3 Step 3: run the sanity check before you interpret anything

💡 Baseline Sanity Check

Before interpreting climate results, verify the baseline makes sense:

- ☐ Does the initial debt level match your country's actual current debt-to-GDP ratio?
- ☐ Is the revenue-to-GDP ratio plausible given WEO forecasts?
- ☐ Does the expenditure path reflect realistic spending trends?
- ☐ If the fiscal rule is on, does debt converge toward your target?
- ☐ Do the primary and overall balance paths make economic sense given the country's fiscal

history?

! DRAFT FOR TEAL: the sanity check applied to Uganda

Initial debt level: the published 47.1 percent of GDP for 2023 agrees with the ministry's own 46.9 at June 2023. If your Explorer run differs materially, the likely cause is the WEO vintage: Current mode runs WEO April 2026, Verified mode the workbook's own October 2024, and the published table came from the 2023 run. Note the difference rather than fixing it by overriding data.

Revenue-to-GDP: the check is whether the level is right, because the share is held constant by construction. For a country midway through a revenue mobilisation strategy, a flat revenue ratio is a conservative assumption worth one sentence in the write-up.

Expenditure path: the published table has primary expenditure drifting from 19.9 to 19.4 percent of GDP over seventy-six years, dipping to 18.8 around 2075. That is the demographic dividend: total population growth drives spending, working-age growth drives GDP, and for a while the second outruns the first. If your path rises instead of drifting down, check the demography variant.

Fiscal rule convergence: Uganda's debt converges toward the target from below. Note in your packet that convergence from below means the rule is not binding on this path, so the target is not doing the work the trajectory suggests.

Balance paths: the published primary deficit narrows from 6.3 percent of GDP in 2023 to 1.4 by 2099, with the overall deficit at 3.8. That is a large near-term consolidation built into the WEO forecast period, not something the model invented. Whether it is credible is a question about the WEO, and belongs in the write-up as a caveat rather than a correction.

The rule this step encodes: a climate result computed on a baseline you have not checked is a number with no owner. Every one of the five boxes is a question someone will ask you.

5.6.4 Step 4: read the Climate tab

The Climate tab shows how climate change affects real GDP under six warming scenarios, using empirical estimates from the FADCP Climate Dataset ([Centorrino, Massetti, and Tagklis, 2024](#)), building on [Kahn et al. \(2021\)](#).

The Briefing view, the default, shows one chart: real GDP under each scenario as a percentage deviation from the baseline path, so you read the damage rather than the growth. The Workbook view shows the same content as the workbook draws it: absolute GDP levels and a GDP index set to 100 in 2029. The six scenarios run from Paris-Aligned (below 2°C warming by 2100) to Hot + Unadapted (high warming with slow adaptation). Impacts begin in 2030, and before that every scenario sits on the baseline.

The size of the modelled loss is country-specific even though the response rate is pooled. The estimates apply historical relationships between temperature deviations and output to each country's own projected warming path and historical norm (Step 3b of Chapter 3), so the same scenario hits two economies differently. The C-PIMA summary describes Uganda's impact as milder than other sub-Saharan African countries ([IMF, 2024](#)).

A country showing a large gap between Paris and Hot + Unadapted faces high climate-fiscal vulnerability. A small gap suggests lower direct climate exposure in the model, though Q-CRAFT's conservative methodology omits indirect effects: natural disasters, sea-level rise, and spillovers such as trade disruptions.

! DRAFT FOR TEAL: what the Uganda GDP chart licenses you to say

The published result is a level GDP loss of around 4 percent by end of century under the Hot scenario.

Four percent needs a perspective sentence: on its own it reads as small. Two are available, pointing in opposite directions, so pick deliberately and say which you picked.

The deflating one: 4 percent of GDP spread over seventy-five years is a few hundredths of a percentage point off annual growth, well inside the noise of any single year's outturn.

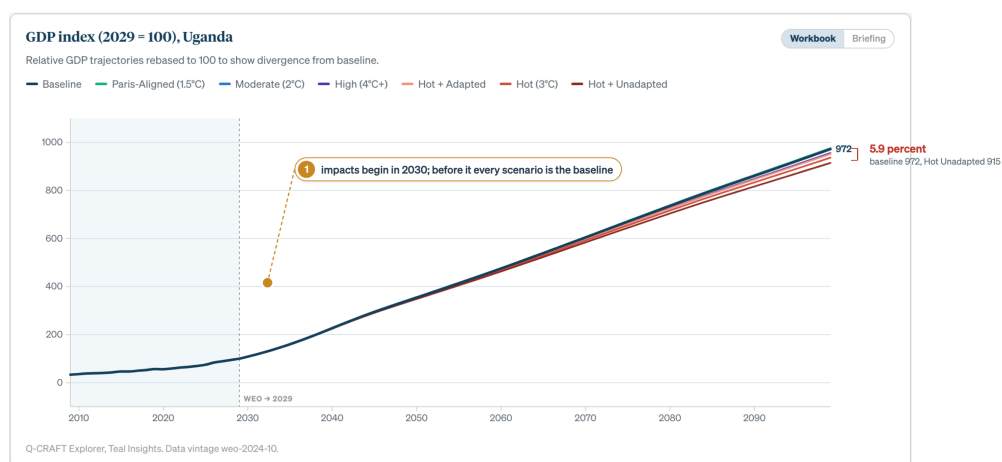
The alarming one: it is a permanent level loss that compounds into the debt ratio every year thereafter, and 4 percent of GDP is a recurring annual amount comparable to a large sector budget.

TODO: verify the perspective figure

The second framing needs a checked comparator before it ships: 4 percent of Uganda's GDP expressed against a named line of the approved budget (health, or the road sector), sourced to the Approved Estimates for the relevant year. Do not ship an approximate comparison in a document aimed at the ministry that publishes the actual number.

Both framings are true. The second is what a fiscal risk statement is for. The first is what a sceptical reader will send back. Write the paragraph so it survives both.

What the chart does not license: any statement about what Uganda's GDP will be. This is a deviation between two runs of the same model, not a growth forecast.



Q-CRAFT Explorer, Verified mode, Uganda, Climate tab. The index is set to 100 at 2029, the last year Uganda's WEO series reports. The Explorer draws the six scenarios against a legend and labels only the baseline endpoint, so the spread at 2099 is named here: the scenarios run from Paris-Aligned 976 down to Hot Unadapted 915, against a baseline of 972.

Figure 5.4: The Climate tab for Uganda. Every scenario tracks the baseline exactly until 2030 and only then separates. By 2099 the index reaches 972 on the baseline and 915 under Hot Unadapted, a level loss of 5.9 percent.

5.6.5 Step 5: read the Analysis tab

The Analysis tab is the comparison view, overlaying baseline and all climate scenario debt trajectories on a single chart. The spread between them is the additional debt burden attributable to climate change. Three factors set that spread.

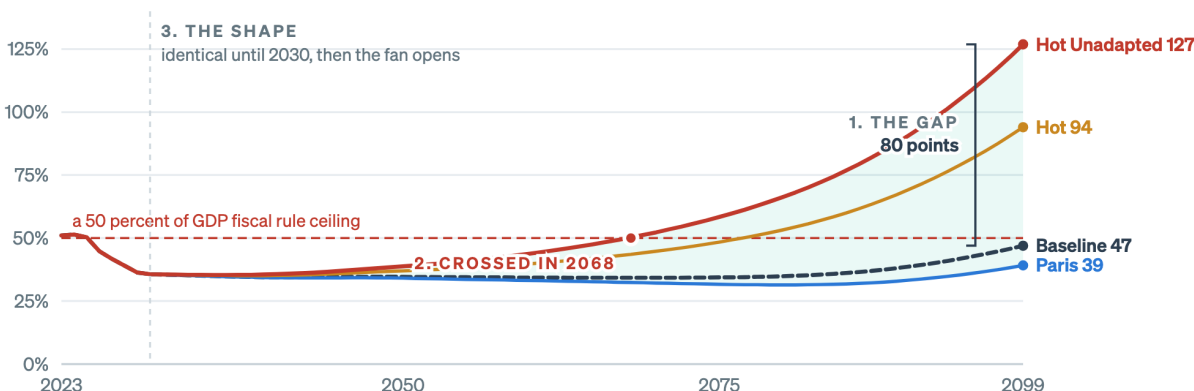
1. **Climate exposure:** how much GDP the country loses under each scenario, from the Climate tab.
2. **Expenditure rigidity:** how much of the GDP loss passes through to the primary balance and debt. Higher rigidity means a wider spread.
3. **Starting fiscal position:** countries with unfavourable baseline debt dynamics, where interest exceeds growth, see climate shocks amplified through the debt dynamics equation.

If baseline debt holds stable or falls while the Hot + Unadapted path rises rapidly, climate change could destabilize an otherwise sustainable fiscal position. That is the core finding Q-CRAFT is designed to surface: even if our fiscal policy is sound today, could climate change make it unsustainable?

The figure below marks the three readings on a real fan chart.

The threshold crossing is the reading that moves a conversation

The gap is the headline. The threshold crossing is what converts a projection into a breach of a commitment that already exists.



Source: the golden-master Uganda run in this repository, on WEO October 2024 at the tool's defaults. The shaded band is the spread across all seven runs, not a probability range.

Figure 5.5: These are the Explorer's own numbers, on the WEO October 2024 vintage at its default settings. They run well above the published 2023 workshop figures quoted in this module, and the whole of that difference is vintage and parameter choice.

The gap is the number people quote, the crossing year is the number that changes a meeting, and the flat stretch to 2030 is the reminder that none of this is a forecast of the next decade.

! TODO: two sets of Uganda numbers now coexist in this module

The figure above is the Explorer's own run on WEO October 2024 at the tool's defaults, ending around 127 percent of GDP under Hot + Unadapted. The prose in this module quotes the published 2023 workshop figures, an earlier vintage with its own parameter choices, which end

at 66 percent under Hot. Both are correct for what they are, and the difference is exactly the vintage-and-parameters point Chapter 2 makes. Whether to keep both in one module, and which leads, is an editorial call for review.

! DRAFT FOR TEAL: reading the Uganda fan chart

The gap: baseline at 47.5 percent of GDP in 2009, Hot at 66. Eighteen and a half percentage points is the number for the first sentence. Uganda’s Statement says the same from the other end: the primary deficit under Hot ends 0.7 percentage points worse than baseline, “thereby raising public debt by over 18 percent of GDP.” A 0.7 point annual deficit gap producing an 18 point debt gap is the compounding traced in Step 3 of Chapter 3, worth saying out loud because non-specialists miss it.

The threshold: the gap becomes a policy fact only when it crosses something. Uganda’s Statement reports that in the High, Hot and Vulnerable scenarios debt passes the 50 percent of GDP fiscal rule ceiling. That sentence does more work than the 18 points: it converts a projection into a breach of an existing commitment.

The shape: look at when the lines separate, as well as where they end. Scenarios are identical until 2030, then fan slowly. A path crossing the ceiling in 2085 and one crossing in 2060 imply very different urgency, a difference the fan chart shows at a glance and a 2099 table does not.

If your run has debt rising in the baseline, the climate gap is no longer the interesting number, because both paths are unsustainable. Say so, and report the year each path crosses the anchor instead.

5.6.6 Step 6: export the packet

The Data tab holds an interactive data grid, a scenario selector, and two CSV downloads.

- **Download this scenario (CSV):** the scenario selected above the grid, Baseline as the tab opens. Useful for folding the baseline projection into your own fiscal framework or DSA.
- **Download all scenarios (CSV):** baseline plus all six climate scenarios, stacked with a scenario column. For custom analysis, cross-country comparison, or integration with other tools.

Every column is a share of GDP or a rate, in percent, and both files carry the run manifest below the data: the same settings record that the packet’s run file restores. The grid runs from 2009 through 2099: the WEO record and forecast, then the projection.

The Export tab assembles the packet proper. One zip holds a read-me, the scenario report, an Excel file, the chart pack, the results CSV, the standalone charts, and a run file that restores the whole configuration on import. The assumptions annex prints all ten parameters with the rationale lines you wrote in the sidebar, and where you wrote none it says so. Add a run label and an analyst’s note before downloading: both travel into every file.

Download it. Together with the parameter table from Step 1 and the Document it entries from Chapter 4, that is the export packet: inputs, rationale, output, in one place, reproducible by someone who has never met you.

The packet does the document-assembly work for you, the fourth reason in Chapter 2.

Assumptions, your written remarks and presentation-ready output come out of the same run, so the half hour that would have gone on copying numbers goes on the Step 7 paragraph instead, the part anyone will read.

5.6.7 Step 7: write the two paragraphs

! DRAFT FOR TEAL: model two-paragraph write-up

Built from the published Uganda figures so every number in it is checkable. Learners replace them with their own run.

Climate change presents a measurable long-term risk to Uganda's fiscal position. Using the IMF's Quantitative Climate Risk Assessment Fiscal Tool, the baseline projection, which assumes no climate damage, has public debt at 47.5 percent of GDP in 2099 after falling to around 36 percent by mid-century as the demographic dividend widens the gap between growth and the effective interest rate. Under the Hot scenario, in which emissions follow SSP3-7.0 and temperatures track the 90th percentile of model projections, output is around 4 percent below baseline by the end of the century and public debt reaches 66 percent of GDP. The mechanism is indirect: higher temperatures slow labour productivity growth, slower growth reduces revenue, and with primary expenditure held at its baseline level in shilling terms the shortfall accumulates as debt. An annual primary deficit only 0.7 percentage points wider than baseline compounds into a debt ratio more than 18 percentage points higher over seventy years.

The policy significance lies in the threshold rather than the magnitude. In the High, Hot and Vulnerable scenarios, public debt passes the 50 percent of GDP fiscal rule ceiling and takes on a rising trajectory, which means climate damage alone is sufficient to move Uganda from compliance to breach without any other adverse shock. These estimates are conservative. They capture the slow productivity effects of temperature change and exclude natural disasters, sea-level rise, non-market damages and the public expenditure required to adapt, so the true fiscal exposure is larger than reported here. The results are also sensitive to the expenditure rigidity assumption: at full rigidity, the tool's default, the entire revenue loss becomes borrowing, while at 0.6 the end-century gap narrows materially. Meeting Uganda's Paris commitments and building expenditure flexibility both reduce this exposure, and the two are complements rather than alternatives.

Six things make this a Fiscal Risk Statement paragraph rather than a report of a model run:

1. It names the tool and the scenario in one clause each, then moves on.
2. Every number is a comparison between two runs, never a forecast.
3. It gives the mechanism in one sentence, so a reader who does not know the debt dynamics equation can still follow the causal chain.
4. It states the threshold, because a breach of an existing commitment is what makes a projection actionable.
5. It states the conservatism caveat before anyone else can, which makes the rest credible.
6. It ends on something the ministry can do.

One editorial query for Teal: the closing sentence of the second paragraph does policy advocacy inside a risk document. Whether that register is right is a call I should not make on

your behalf.

! Now do your own country

The seven steps are the part that transfers. Uganda supplied numbers to check each step against. Nothing in the sequence depended on the country: set the parameters and write down why, check the baseline before you interpret anything, read the three tabs in order, export the packet, write two paragraphs.

The rest of this module runs that method twice more, on countries that behave differently from Uganda and from each other, with less help each time. The capstone in Chapter 7 runs it on yours.

5.7 Completion problem: Ethiopia under the High scenario

You run a new country with the same method, and most of the setup is fixed for you. Ethiopia earns its place because its demography runs the other way from most countries in a fiscal risk conversation: the working-age population is still climbing, pushing the baseline one way while climate damage pushes the other.

Done for you:

- The method: the same seven steps, in the same order. You are not inventing a procedure.
- The scenario: **High** is SSP3-7.0 at the average of model temperature projections, rather than the 90th percentile.
- The parameter template from Step 1, with one open cell. The debt target is Ethiopia's own fiscal anchor if you can establish one, and 50 percent of GDP if you cannot. Record which you used and why.

Your turn, six tasks:

1. Run the Baseline Sanity Check on Ethiopia. A different baseline, so the box is now yours, and it comes before any climate reading.
2. Read the 2099 debt ratio under High from the Analysis tab. Record it. _____
3. Compute the gap to baseline. _____ percentage points.
4. Read the year the High path crosses your target, if it does. _____
5. Write one sentence explaining why High sits below Hot when both use the same emissions pathway.
6. Rewrite the first sentence of the model paragraph above using your Ethiopia numbers.

i DRAFT FOR TEAL: checking yourself on task 5

The distinction is the temperature distribution, not the emissions. High takes the average temperature outcome across climate models running SSP3-7.0, Hot the 90th percentile of the same set. Same policy world, different draw from the uncertainty. Which belongs in a fiscal risk statement is a real question: risk documents conventionally report adverse tail outcomes, arguing for Hot, while a central-case fiscal framework argues for High. Report both and say why you led with the one you led with.

5.8 Independent problem: adaptation speed, on a third country

This time you get a checklist only, and a country whose fiscal position looks like neither of the first two. Compare Hot + Adapted against Hot + Unadapted for one in the dropdown whose interest-growth differential runs against it. Temperatures are the same in both. The difference is how quickly the economy adjusts.

Deliver five things:

- ☐ The 2099 debt ratio under each, and the gap between them
- ☐ One sentence on what that gap represents in policy terms
- ☐ One sentence on what it does **not** represent, which is the trap in this comparison
- ☐ The same comparison at expenditure rigidity 0.6, with a note on whether the adaptation gap or the rigidity gap is larger for your country
- ☐ A two-paragraph write-up in the format of Step 7

i DRAFT FOR TEAL: the trap, if you want to check before you write

Faster adaptation in Q-CRAFT is a faster economic adjustment to a given temperature path. It is not adaptation spending. The model does not cost the investment that would produce that adjustment, and the User Guide says so directly (p. 6). So the Adapted-to-Unadapted gap is the benefit side of an adaptation programme with the cost side missing. Reporting it as the value of adaptation without that sentence attached is the likeliest way to misuse this tool in a budget discussion.

5.9 What you can now do

- You have run three countries, one with full support and one alone, and produced an export packet with rationale attached.
- You can apply the Baseline Sanity Check and say what each box protects you from.
- You can write two paragraphs that a Fiscal Risk Statement editor would recognise.

Common errors on this material: interpreting climate results before checking the baseline, quoting a 2099 level as a forecast, and reporting the adaptation gap as the value of adaptation.

On your desk this week: take the two paragraphs you drafted to whoever owns your ministry's fiscal risk chapter and ask what they would cut. The answer tells you more about the format than this module can.

Chapter 6

What the tool tells you and where it stops

In this module

What the tool does well and why the IMF built it that way. What Q-CRAFT leaves out and which direction the omissions run, so you can state the limit before a reviewer does. How to read a fan chart whose baseline is clipped at zero when the climate scenarios are not. And how to tell a Q-CRAFT question from a debt sustainability question, in both directions.

Fast path

No fast path. This is the shortest module and the one that keeps you out of trouble. If you only read one section, read Section 6.6 (the exclusions, and why the estimate is a floor).

6.1 Warm-up

From Chapter 5 (the worked example):

1. What does the gap between Hot + Adapted and Hot + Unadapted represent, and what does it leave out?
2. Which check comes before any climate interpretation?

(Answers: the benefit of faster adjustment, with the cost of achieving it not modelled. The Baseline Sanity Check.)

6.2 A senior official asks whether the fan chart shows the cost of climate change

You have presented the fan chart. The senior official who will have to defend it looks at the Hot + Unadapted line and asks: “So this is what climate change will cost us?”

Your whole credibility rests on the next thirty seconds. This module is that answer.

6.3 By the end of this module you can

- **Name** four things Q-CRAFT does well, and **say** which design choice buys each one
- **List** the specific damage channels Q-CRAFT excludes, and **state** the direction of the resulting bias
- **Explain** why baseline and climate scenarios treat negative debt differently, and **adjust** your reading of a chart accordingly
- **Compare** two countries with similar debt paths and **distinguish** the different drivers behind them
- **Decide** whether a given question calls for Q-CRAFT, the LIC-DSF, both, or neither, and **justify** the choice

6.4 Where you are in the course

The map below lights this module's two boxes.

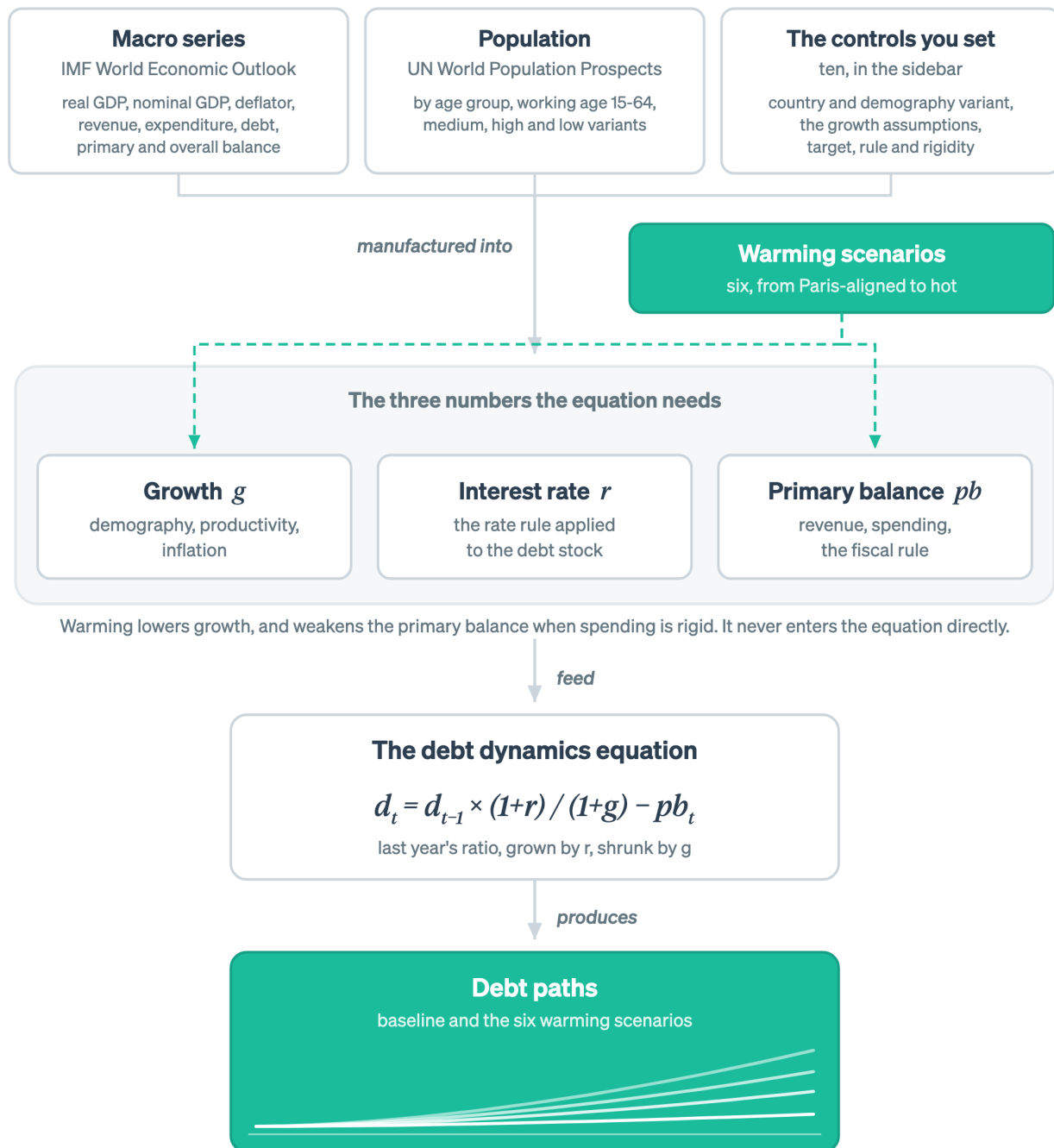


Figure 6.1: This module is about what the warming scenarios contain, and what the debt paths therefore license you to write.

The two lit nodes are the two ends of the climate channel: what the warming scenarios contain, and what the debt paths let you write.

6.5 What the tool is good at, and why it was built that way

The whole module in one breath: Q-CRAFT runs nearly everywhere on published data, and everything it leaves out makes the estimate a floor. Nearly everywhere: 171 economies, no bespoke country dataset needed, its arithmetic verified against the IMF’s original workbook. A floor: the scenarios capture the slow productivity channel and exclude the six channels the User Guide names (natural disasters, sea-level rise, tipping points, non-market damages, spillovers, adaptation spending) (User Guide, pp. 5-6), so the results read conservative, a lower bound on fiscal impact rather than a central estimate of the total.

Q-CRAFT does four things well, and each carries a cost. Strengths first: a caveat is only useful to someone who knows what it qualifies.

It isolates one channel and models it from data. Temperature affects growth, growth affects revenue, revenue affects debt. The temperature-to-growth step comes from cross-country empirical estimates using the Kahn et al. (2021) methodology, updated in the FADCP Climate Dataset (Centorrino, Massetti, and Tagklis, 2024), then runs through a standard debt dynamics equation a reviewer can check line by line. A narrower claim than an integrated assessment model makes, and one you can defend in a room.

It runs on data every country already has. The User Guide is explicit: the tool can be deployed “without any additional data, even in low-capacity settings”, building scenarios from IMF, UN and World Bank sources (p. 6). No new collection, no survey, no waiting for a mission. A team can have a first pass by week’s end. The same page is equally explicit that country expertise is still needed to make the baseline match the country’s own fiscal framework, the work of Chapter 4.

It is comparable across countries. The same growth identity, the same debt dynamics equation and the same six scenarios run for every economy in the tool. That puts two countries side by side, the basis of this module’s diagnosis exercise.

It produces the shape a fiscal risk chapter needs. The output is a comparison between scenarios rather than a forecast, and a risk chapter is a comparison between scenarios. The User Guide positions Q-CRAFT as “a starting point for climate change fiscal risk analysis” and as a foundation for deeper work with general equilibrium models where capacity allows (pp. 5-6).

All four strengths come from one brief: the Fiscal Affairs Department needed something a country team could run in a week, on existing data, producing a defensible comparison rather than a false forecast. Every limitation in this module follows from the same brief, so each reads as the cost of a choice rather than an oversight.

Use Q-CRAFT to identify which climate-fiscal channels matter most for your country, then investigate those channels with more detailed models and country-specific data. The value sits in the comparison across scenarios and the sensitivity to assumptions, never in any single projected number.

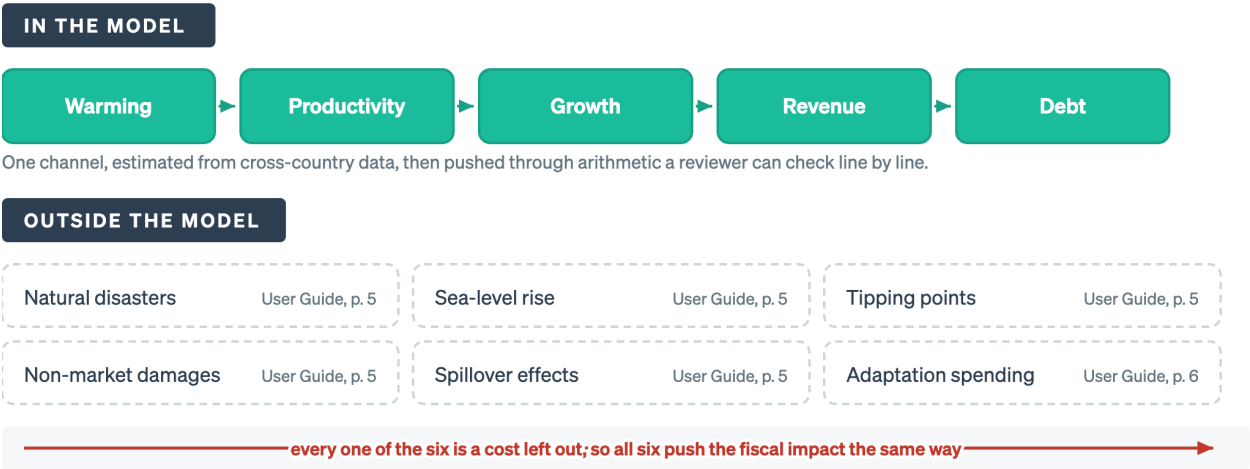
6.6 Every exclusion is a cost left out, so the estimate is a floor

The scenarios capture one channel: the slow effect of temperature change on productivity, and through it on growth. The User Guide names six exclusions, and every one of them runs in the same direction.

The figure below puts the modelled channel and the six exclusions on one page, with the direction they share drawn along the bottom.

One channel is modelled, and the six left out all point the same way

The direction is what makes the list usable: the modelled impact is a lower bound rather than a central estimate of the total.



Source: Q-CRAFT User Guide, pp. 5-6, which names all six. Say it yourself in the write-up, in the same paragraph as the headline number, before a reviewer says it for you.

Figure 6.2: Each exclusion carries the User Guide page that documents it, so the list is quotable in a footnote without further research. What runs across the top is everything that is left.

Modelling any of the six would raise the estimate, which is why it is a floor rather than a central case.

Excluded	Documented at	Why it matters for a fiscal risk statement
Climate-induced natural disasters	User Guide, p. 5	Often the channel a national risk statement leads with. Uganda’s puts floods and epidemics at about 75 percent of recorded events, 1985 to 2021
Sea-level rise	User Guide, p. 5	Absent for a landlocked economy, decisive for Bangladesh or a small island state
Tipping points	User Guide, p. 5	The estimates are fitted to historical temperature variation and cannot extrapolate to discontinuities
Non-market damages	User Guide, p. 5	Morbidity, mortality, conflict, food insecurity: real costs, absent from GDP

Excluded	Documented at	Why it matters for a fiscal risk statement
Spillover effects	User Guide, p. 5	Trade partners, remittance sources and commodity markets are hit too
Adaptation spending	User Guide, p. 6	Even the Hot + Adapted scenario models faster adjustment, not the public expenditure that buys it

The modelled impact is a lower bound you can state in one sentence. The User Guide says it in its own voice: the results “do not account for the potential impacts of climate change induced natural disasters, sea-level rise risks and other environmental risks, rendering the outcomes conservative” (p. 5). The sentence: *actual fiscal impacts are likely larger than these results suggest*.

Say it yourself, in the write-up, before a reviewer says it for you. A conservative number with the conservatism disclosed is stronger than a larger number without it.

One more limitation sits inside the estimates: fitted to historical variation in temperature, they say nothing about warming outside the lived range. The six above sit outside the model. This one sits inside it. Nonlinearities beyond that range are outside the estimate rather than judged to be small. Whether that understates damage from warming well outside the historical record is an open empirical question. Expect the question, and answer it the module’s standard way: report the number, name the channel it covers, and name the channels it does not.

Partial equilibrium is the one limit with no sign: Q-CRAFT cannot cost a policy. The User Guide describes the set-up as “essentially a partial-equilibrium” one (p. 5) and says plainly Q-CRAFT “is not a forecasting model nor a general equilibrium model of the economy” (p. 6). Fiscal policy does not feed back into growth or interest rates, so cutting spending inside the model never slows the economy down. Unlike the six above, this limit flatters a consolidation scenario and leaves out whatever growth an adaptation programme would buy. Q-CRAFT can only show the debt path under policy assumptions you supplied.

! DRAFT FOR TEAL: what conservatism does not buy you

Conservatism protects a number’s credibility. It does not make the number safe to quote out of context, and it creates one hazard worth naming.

A conservative estimate picked up as *the* estimate becomes a ceiling in someone else’s argument. “The analysis shows climate costs us 18 points of debt” starts life as your careful lower bound and ends life as a budget office’s reason to fund adaptation at only 18 points’ worth. The defence is structural rather than rhetorical: put the exclusion list in the same paragraph as the headline number, not in a footnote and not in an annex.

6.7 Only the baseline floors debt at zero

i The rule

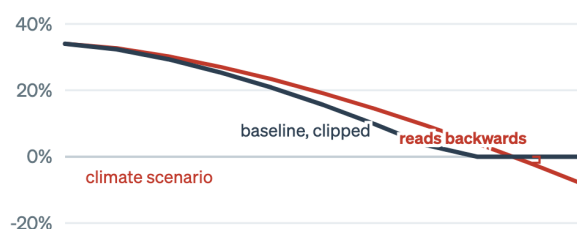
The baseline scenario applies a floor of zero to the debt-to-GDP ratio: if the equation produces a negative value, debt is set to zero. Climate scenarios do not apply this floor, and debt is allowed to go negative, representing net asset positions. The asymmetry is the workbook's own construction: on its **Baseline** sheet the debt recursion is wrapped in a floor, `=IF((...)<0,0,(...))`, while the six scenario sheets carry the bare recursion, and the workbook records no reason for the difference.

The two panels below are the same projection, drawn as the chart shows it and then as it actually is.

A floored baseline makes the vertical gap read backwards

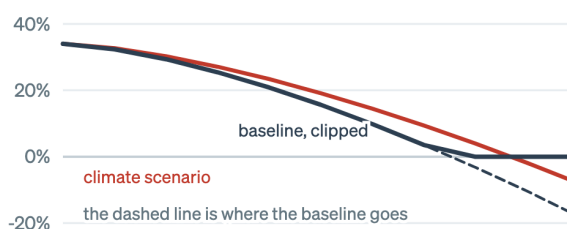
Baseline debt is floored at zero. The climate scenarios are not, and may run negative into net asset positions.

1. WHAT THE CHART SHOWS



the climate line sits below the baseline, which cannot be right

2. WHAT IS HAPPENING



one line is clipped at zero and the other is not

The reading rule: if any line in your fan touches zero, stop using the vertical gap as your headline number.

Schematic, drawn to show the rule rather than a country. It bites only where a projection drives debt to zero, which takes a strongly favourable interest-growth differential held for seventy years.

Figure 6.3: The floor is a deliberate choice rather than a bug. Flooring the climate scenarios too would compress exactly the range the tool exists to show, so the asymmetry stays and the interpretive burden lands on you.

In the left panel the climate line sits below the baseline, which cannot be true, and the right panel shows the reason: the baseline stops at zero and the climate scenario does not.

! DRAFT FOR TEAL: how the asymmetry changes what you are looking at

For a narrow set of countries the asymmetry matters: the floor bites only when a projection drives debt to zero. That happens for countries with strongly favourable interest-growth differentials and sustained primary surpluses over a seventy-year horizon. The worked case is safe: its published baseline lands at 47.5 percent of GDP.

In such a country the baseline flattens along zero while the climate scenarios continue downward into negative territory. Read naively, the chart shows climate scenarios with *lower* debt than baseline, which is nonsense: one line is clipped and the others are not.

If any line in your fan chart touches zero, stop using the vertical gap as your headline number. Check the underlying series in the Data tab, then either report the comparison over a window before the floor binds or report the primary balance gap instead, which is not clipped.

Flooring the climate scenarios too would compress the spread between them. Unfloored, they keep their full range, and the interpretive burden lands on you.

6.8 Diagnosing two countries with similar debt paths

This course is training you to recognise which situation you are in when the labels come off. You practise by comparing pairs of countries.

 Comparison exercise: same path, different drivers

Find the pair. In the Explorer dropdown, find two countries whose baseline 2099 debt ratios are within about 5 percentage points of each other but whose gaps between baseline and Hot + Unadapted differ by 10 percentage points or more. Skip any country whose six scenarios land exactly on the baseline. Eleven selectable countries carry an all-zero climate slice: the gap is zero because the climate dataset has no coverage for that economy, not because the economy faces no climate risk. The Explorer flags these on screen when you select one. The User Guide names 25 economies with no climate estimates ([p. 19, footnote 12](#)). The other 14 never reach the dropdown because they have no productivity data either.

Then diagnose. For each country, decide which of the three drivers from Chapter 5 explains the difference:

1. Climate exposure, meaning how much GDP is lost, from the Climate tab
2. Expenditure rigidity, meaning how much of that loss becomes debt
3. Starting fiscal position, meaning whether the interest-growth differential amplifies or damps it

Write one sentence per country saying which driver dominates, and one sentence saying what policy conclusion follows. The conclusions should differ. If they do not, you have picked a pair that is too similar.

Then reverse it. Find two countries with similar climate exposure on the Climate tab whose debt outcomes diverge. The driver is now somewhere in the fiscal position, and finding it is the exercise.

 **TODO:** seed the comparison with verified pairs

This exercise currently sends learners hunting. Two or three verified country pairs, run and checked against the current data vintage, would make it usable in a 20-minute workshop breakout. Needs a real run rather than an assumption about which pairs work.

 **DRAFT FOR TEAL:** why the search is the exercise

The instinct is to hand over the pair and the answer, but the search is the exercise: the skill is diagnosis from surface features that mislead, and a pair you were handed arrives with the

diagnosis already done.

Carry one principle out of this module: a debt path is an outcome, not a diagnosis. Two countries arriving at 60 percent of GDP in 2099, one because it is highly exposed to warming and one because it borrows at rates above its growth, need different policies and different chapters in different documents. Q-CRAFT will show you the outcome. Deciding which country you are looking at is the part of your job that cannot be automated.

6.9 Q-CRAFT and the LIC-DSF answer different questions

Q-CRAFT and the IMF-World Bank Low-Income Country Debt Sustainability Framework answer different questions, even though both produce debt projections for low-income countries. They are not substitutes, and using one where the other belongs is a common and expensive error.

	Q-CRAFT	LIC-DSF
Question it answers	How much does climate damage change the long-run fiscal path?	Is this country's debt sustainable, and at what risk rating?
Horizon	To 2099	Medium term, with longer-run tails
Output	Scenario comparison, no rating	A risk rating that carries operational consequences
Climate	Central: six scenarios	Not the organising principle
Debt detail	One aggregate stock	External and public splits, currency, creditor composition
Used for	Fiscal risk disclosure, C-PIMA annexes, long-term planning	Lending decisions, concessionality terms, programme design

💡 Self-check: five questions, which tool?

For each, name Q-CRAFT, LIC-DSF, both, or neither, and say why in one line.

1. "The Fiscal Risk Statement needs a section on climate. What do we run?"
2. "Are we still at moderate risk of debt distress?"
3. "If we take this Eurobond instead of an IDA credit, what happens to our risk rating?"
4. "What is the fiscal cost of the National Adaptation Plan?"
5. "Does climate risk change our debt sustainability assessment?"

DRAFT FOR TEAL: self-check answers

The test in every case is which question the tool was built to answer, never which tool you happen to have open.

1. **Q-CRAFT.** This is its designed use. Uganda used it exactly this way in the FY2024/25 Fiscal Risk Statement.
2. **LIC-DSF.** Q-CRAFT produces no risk rating and has no thresholds to breach. Answering this question from a Q-CRAFT chart is the error this section exists to prevent.

3. **LIC-DSF.** The question is about creditor composition and terms, which Q-CRAFT does not represent, because it carries one aggregate debt stock and one effective interest rate.
4. **Neither.** Q-CRAFT does not cost adaptation, by design ([User Guide, p. 6](#)), and the LIC-DSF does not either. This question needs a costed adaptation plan, and the honest answer is to say so rather than to produce a number from a tool that cannot see adaptation spending. The economics of folding adaptation into fiscal analysis is set out in [Bellon and Massetti \(2022\)](#).
5. **Both, in sequence.** Q-CRAFT establishes whether the climate channel is large enough to matter for the fiscal path. If it is, the finding belongs alongside the DSF assessment as a qualitative risk consideration rather than as an input to it, because the two run on different horizons and different definitions of debt. Anyone claiming to have mechanically fed Q-CRAFT output into a DSF has done something that needs explaining.

Analysts usually skip the reverse check: a question that arrives framed for the DSF may be a Q-CRAFT question. “Our risk rating is fine, so climate is not a debt issue” is a DSF answer to a long-horizon question the DSF does not ask. Uganda’s own result makes the point: a baseline that is comfortable on any medium-term view still crosses the fiscal rule ceiling by end of century once climate damage is applied.

6.10 What you can now do

- Say what the tool does well and which design choice buys each strength.
- List what the scenarios exclude and state the direction of the bias without looking it up.
- Spot a clipped baseline in a fan chart and say why the vertical gap no longer means what it usually does.
- Take a question and say which tool it belongs to, in both directions.

Common errors on this material: quoting a conservative estimate as a total, reading a floored baseline as a favourable climate result, and answering a sustainability-rating question from a Q-CRAFT chart.

The thirty-second answer, for reference: “No. It is what climate change does to the debt ratio through one channel, growth, under stated assumptions about how spending responds. It excludes disasters, sea-level rise and the cost of adapting, so it is a floor rather than a total. What it tells us is that the channel is large enough to breach the ceiling on its own.”

On your desk this week: find a projection in circulation in your ministry and write down one question it cannot answer. Circulate that sentence with it.

Chapter 7

The capstone

In this module

The capstone: an export packet plus a two-paragraph fiscal risk draft for a country you choose. Marked on whether you can defend your assumptions, never on whether your numbers match anyone else's. The module closes the loop on Chapter 1 to show what changed.

Fast path

Short on time? Read the brief and the rubric, then do the work. The retake at the end takes ten minutes and is the cheapest way to find out whether the economics landed or only the clicking did.

7.1 Warm-up

From across the course, without looking:

1. Which two of the three numbers does climate damage move?
2. What does expenditure rigidity 1.0 mean?
3. Name one thing the climate scenarios exclude.

(Answers: *g* and *pb*. Spending is sticky, so the revenue loss becomes borrowing. Natural disasters, sea-level rise, tipping points, non-market damages, spillovers, or adaptation spending.)

7.2 By the end of this module you can

- **Produce** a complete export packet for a country of your choosing
- **Defend** each parameter choice against a specific challenge
- **Assess** your own draft against the rubric before anyone else sees it
- **Compare** your current capability against your Chapter 1 self-assessment

7.3 Where you are in the course

The map below is lit end to end for the first and last time: the capstone runs the whole chain.

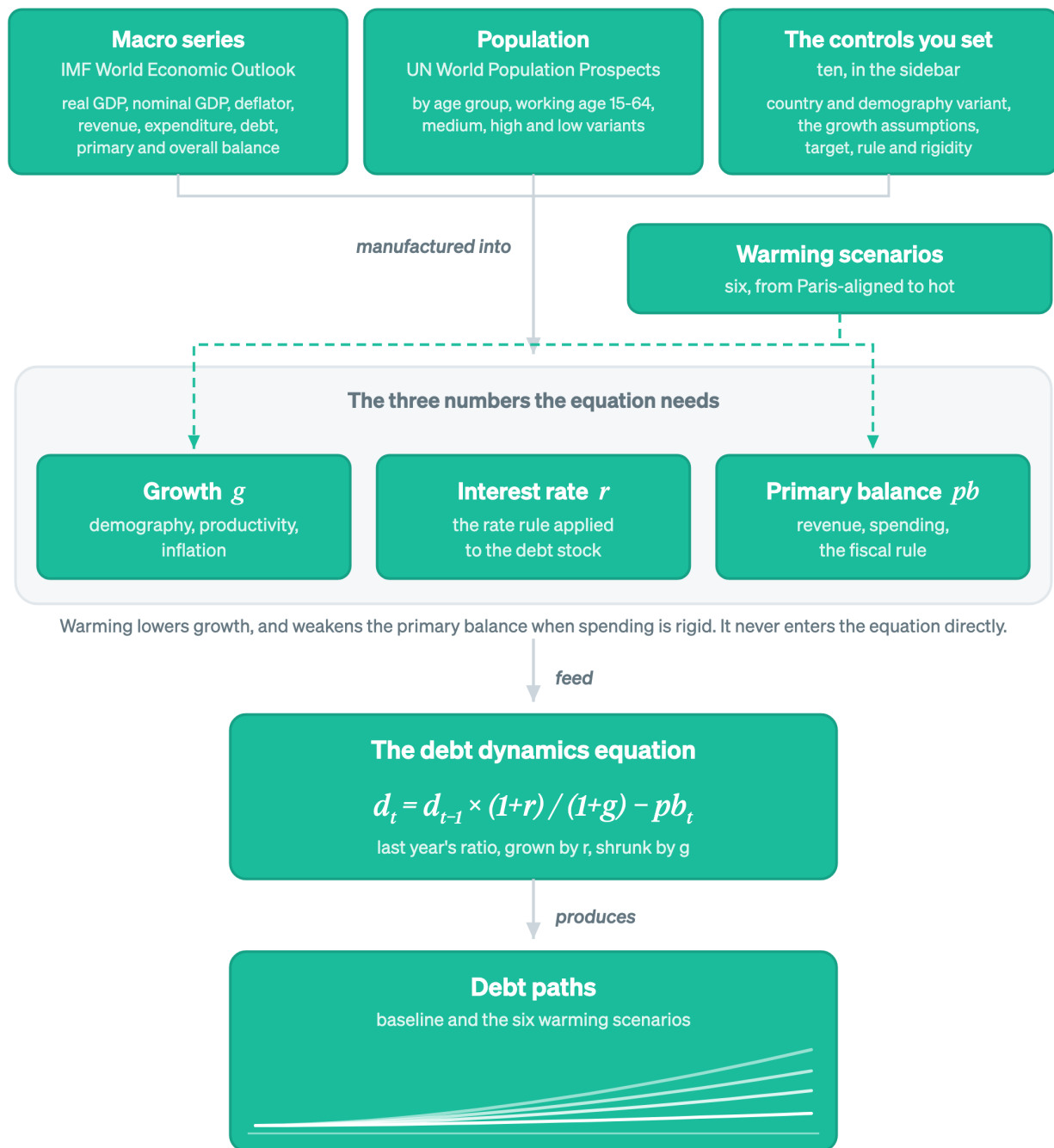


Figure 7.1: The capstone runs the whole chain, from the sources the tool loads to the paths you hand over.

7.4 The capstone brief

You have been asked for a climate-fiscal risk section for your ministry’s next fiscal-risk document, and you will defend it to the senior officials who sign it off. You have the Explorer, the data mode you choose to run, and whatever you know about the country. You have one week and half a page.

You hand in three parts.

1. **The export packet.**
 - The parameter table: every setting, with a one-line rationale each (the five **Document it** entries from Chapter 4)
 - The data vintage, stated
 - Both CSV exports, baseline and all scenarios
 - At least two sensitivity runs, with what changed and by how much
2. **The two-paragraph draft**, in the format of Chapter 5, Step 7.
3. **Three challenge answers.** Written responses of two to three sentences each to:
 - “Is this a forecast?”
 - “Why did you choose that rigidity value?”
 - “So this is what climate change will cost us?”

The packet puts assumptions, rationales, and presentation-ready output into one handover: the fourth reason from Chapter 2 in finished form. That frees the week for the policy question instead of assembling the evidence by hand.

Pick the country you work on: the capstone rehearses the document you must write. Pick Uganda instead to compare your run against the published section Chapter 5 works through. Avoid Ethiopia and your country from the Chapter 5 independent problem, both already worked. A handful of countries do not project: the published record lacks the projection’s starting figure. The Explorer refuses those with a notice rather than inventing the number. If yours is refused, record the notice and take a second country for the packet. A few others project from an earlier anchor year than the data release runs to. Where the Explorer names that year on screen, carry it into your write-up. Neither case is a fault in the run. Both are facts about what the source publishes for your country.

Matching anyone else’s published figures is not the skill assessed. You are on a later WEO vintage, possibly with different parameters.

7.5 The rubric

! DRAFT FOR TEAL: capstone rubric

The rubric scores four criteria. The weights are a proposal, not a decision.

Criterion	Weight	Meets the standard	Falls short
Assumptions defended	40%	Every parameter has a rationale citing country evidence. Rigidity in particular has budget-composition reasoning behind it, and a sensitivity run.	Parameters listed without reasons, or reasons that restate the setting (“rigidity 1.0 because spending is rigid”).
Baseline checked	20%	The five sanity-check boxes are answered against the actual run, including any that were skipped and why. Differences from the ministry’s own figures are explained rather than reconciled away.	Climate results interpreted with no evidence the baseline was examined.
Interpretation	25%	Numbers presented as comparisons between runs. The mechanism stated in one sentence. A threshold identified. The conservatism caveat in the same paragraph as the headline number.	A 2099 level quoted as a prediction, a debt gap described as a cost, or the caveat relegated to a footnote.
Written for the reader	15%	Reads like your ministry’s own risk chapter. A non-specialist can follow the causal chain. Two paragraphs, not five.	Reads like a description of a model run. Jargon undefined. Over length.

One error disqualifies on its own, and it recurs often enough to name separately: reporting the Adapted-to-Unadapted gap as the value of adaptation. It is the benefit side with the cost side missing (Chapter 5, independent problem).

In the live workshop, the rubric doubles as the peer-review instrument, run in three passes. First pass, assumptions only. Second, interpretation only. Third, the writing. Split passes stop reviewers spending all their attention on prose, as they do when reviewing everything at once.

The figure below pairs the three hand-in parts with their marking weights.

Sixty percent of the mark is the assumptions and the baseline

The capstone is marked on whether you can defend the analysis, never on whether your numbers match anyone else's.

WHAT YOU HAND OVER

1 The export packet

- the parameter table, one rationale a line
- the data vintage, stated
- both CSVs, baseline and all scenarios
- at least two sensitivity runs

2 The two-paragraph draft

- in your ministry's own register
- every number a comparison, never a forecast
- the caveat beside the headline number

3 Three challenge answers

- "Is this a forecast?"
- "Why that rigidity value?"
- "So this is what climate will cost us?"

HOW IT IS MARKED

Assumptions defended

40%

Baseline checked

20%

Interpretation

25%

Written for the reader

15%

sixty percent is marked before anyone reads a number you produced

Weights as proposed in the rubric above, which is a draft rather than a decision. The disqualifying error sits outside the weights: reporting the adaptation gap as the value of adaptation.

Figure 7.2: Nobody is marking whether your 2099 number matches a published one. The one thing that fails a capstone outright sits outside the weights entirely: reporting the adaptation gap as the value of adaptation.

Under the proposed weights, assumptions and the baseline check settle sixty percent of the mark before anyone reads a number you produced.

7.6 Common errors to watch for

⚠️ TODO: replace with pilot-observed errors

This list comes from the misconceptions the course was designed against, not from watching people make them. The redesign plan makes a pilot with two or three target-profile testers non-negotiable. Until that pilot's error log replaces this list, treat it as a hypothesis.

- Reading rigidity backwards, so the headline risk number comes out inverted
- Interpreting climate results before checking the baseline
- Quoting a 2099 level as a forecast
- Adjusting the debt target while the fiscal rule is off, then concluding the target does nothing
- Describing a debt-ratio gap as a cost in local currency
- Reporting the adaptation gap as the value of adaptation
- Answering a debt-sustainability-rating question from a Q-CRAFT chart

7.7 Where you started, where you are

Go back to your three answers from Chapter 1. Answer them again now, before reading on.

On long-term fiscal projection: A1 (not worked with a DSA) / A2 (can follow one) / A3 (could build one this week)

On climate-fiscal analysis: B1 (climate side only) / B2 (can describe the use) / B3 (have run or interpreted one)

On the tool itself: C1 (never opened it) / C2 (clicked around) / C3 (ran it and explained the output)

If your A or B answer moved up, the course worked. Capstone done, C3 is true by construction: you ran the tool and wrote the explanation. C3 with A and B unmoved means you can drive the tool while the economics underneath never landed. That deserves a conversation, not a shrug.

 Retake the three questions from Module 0

Same three items, unchanged. Answer from memory before opening anything.

1. Zero primary balance, $r = 9$ percent, $g = 6$ percent. Debt ratio next year: falls, flat, or rises?
2. Expenditure rigidity 1.0 means what about spending under a climate scenario?
3. Debt at 66 percent under Hot against 47.5 percent baseline. What is the defensible statement of what that means?

DRAFT FOR TEAL: the answers, and what a wrong one means now

A wrong answer here costs more than in Chapter 1: it is attached to a document you are about to circulate.

1. **Rises**, to about 51.4 from 50. Still wrong? Step 1 of Chapter 3 failed, and nothing in your capstone's interpretation section can be trusted: every climate result in the tool is this arithmetic with a lower g .
2. **Spending stays at its baseline level in local currency terms**, so the whole revenue loss becomes borrowing. Get this wrong now and your headline number may have the wrong sign on its sensitivity, the error most likely to reach print.
3. **Under stated assumptions, climate damage of this magnitude adds about 18.5 percentage points of GDP to the debt ratio by 2099 relative to the same projection without it.** A comparison between two runs, with the assumptions attached. If you reached for the forecast framing or the cost framing, reread Chapter 6 before you circulate anything.

7.8 What to use on your desk this week

- The debt-stabilizing primary balance for your country: three numbers, one division, and it opens most credible fiscal risk write-ups.
- The **Document it** habit: every assumption gets a line saying why. A minute per parameter separates an analysis from numbers nobody can defend a year later.
- The which-tool question. When a request arrives, name the question it is asking before opening anything.

7.9 Spaced follow-up

Two follow-ups are designed into the course, to ship with the workshop materials.

- **Day 3:** one retrieval question by email. No preparation, no marking.
- **Week 2:** an application check. What did you run, what did you find, what got used.

Spacing beats massing, and the IMF’s independent evaluators found that training with hands-on follow-up sticks while the standalone workshop decays ([IEO, 2022](#)).

 **TODO:** workshop materials

The live sessions need artefacts this guide does not yet contain: breakout task cards, the anonymous poll items, the peer-review sheet derived from the rubric above, and the two follow-up emails (Day 3 retrieval question, Week 2 application check). Tracked separately from the book.

7.10 The whole course in six lines

- One equation decides the debt path, and the rest of the tool builds its three inputs. (Chapter [2](#))
- Two things move the ratio: the interest-growth gap, and the primary balance. (Step 1, Chapter [3](#))
- Every parameter is a judgment call, and the record of the call is the deliverable. (Chapter [4](#))
- The output is a comparison between two runs, written up in your ministry’s own format. (Chapter [5](#))
- The estimate is a floor, the baseline is floored at zero and the scenarios are not, and some questions belong to a different tool. (Chapter [6](#))
- What you hand over is an analysis you can defend. (this module)

Appendix A

Running the official workbook

i The User Guide is the official manual

The IMF's User Guide ([Tim and Rahman, 2024](#)) is the authoritative manual for operating the workbook. Where this appendix differs, the Guide is right. This appendix is the quick orientation: download the workbook, run a country, find the outputs, and see each control mapped to its course concept.

Everything the course taught transfers unchanged: concepts, parameter judgments and the reading of the output. The workbook and the Explorer run the same model (Section 2.7 is the check). What differs is operational: where controls live and results come out.

A.1 Download and open

The Fund posts the workbook as one file that opens without installation. Download the [Excel workbook](#) from the IMF's [Q-CRAFT page](#), with the [User Guide \(PDF\)](#) beside it. A plain .xlsx with no macros, the file needs nothing enabled and opens under standard ministry IT policies. This appendix describes version 1.0, dated 15 November 2024 on its own Dashboard.

Read the *Read Me* sheet first, and expect #VALUE! in many output cells on first open. It carries the colour conventions (green Dashboard, blue input sheets, red scenario calculations, yellow outputs; green cells are inputs, black are calculations) and the ten steps of a full run. The errors trace to the shipped selection, Afghanistan, which has gaps in its WEO series. Nothing is broken: select a country with a complete series and every cell fills in.

A.2 The Dashboard runs the whole tool



Q-CRAFT

Quantifying the macro economic fiscal risks of climate change

Reuse of this Q-CRAFT Tool, its and IMF data does not imply any endorsement of any resulting research and/or product. Any research presented should not be reported as rep
Version 1.0 11-15-2024

Select Country
Uganda

BASELINE SCENARIO		
Demography	Medium	Page 10 - 11
Productivity		Page 11 - 13
Start	5.0	
End	1.2	
Inflation		Page 13
Start	3.5	
End	3.5	
Interest rate		Page 14 - 15
Constant	Nominal interest rate	
(a)real interest rate, if chosen	1.0	
FISCAL RULE		
Rule	Yes	Page 16
Debt Target	60.0	
CLIMATE SCENARIOS		
Expenditure Rigidity	1.0	Page 18

For more information User Guide

Figure A.1: The Dashboard with Uganda selected and every other control at its shipped value. Column D points into the User Guide, page by page, beside each control.

Select a country in cell C12 and the workbook does the rest. Macro series, population projections and climate estimates follow the selection, and every sheet recalculates. C12 is the workbook's equivalent of the Explorer's country dropdown. The *Macrofiscal* sheet then shows that country's diagnostic charts, the same look-before-you-project habit Chapter 2 teaches with the data banner and the Baseline tab.

The controls beneath C12 are the course's controls under their workbook names. Chapter 3 built the concepts, and Chapter 4 taught you to choose and defend each value:

On the Dashboard	Ships as	What it sets
Demography (C17)	Medium	the population variant behind employment growth
Productivity, start and end (C20, C21)	5.0 and 1.2	the productivity convergence path inside g

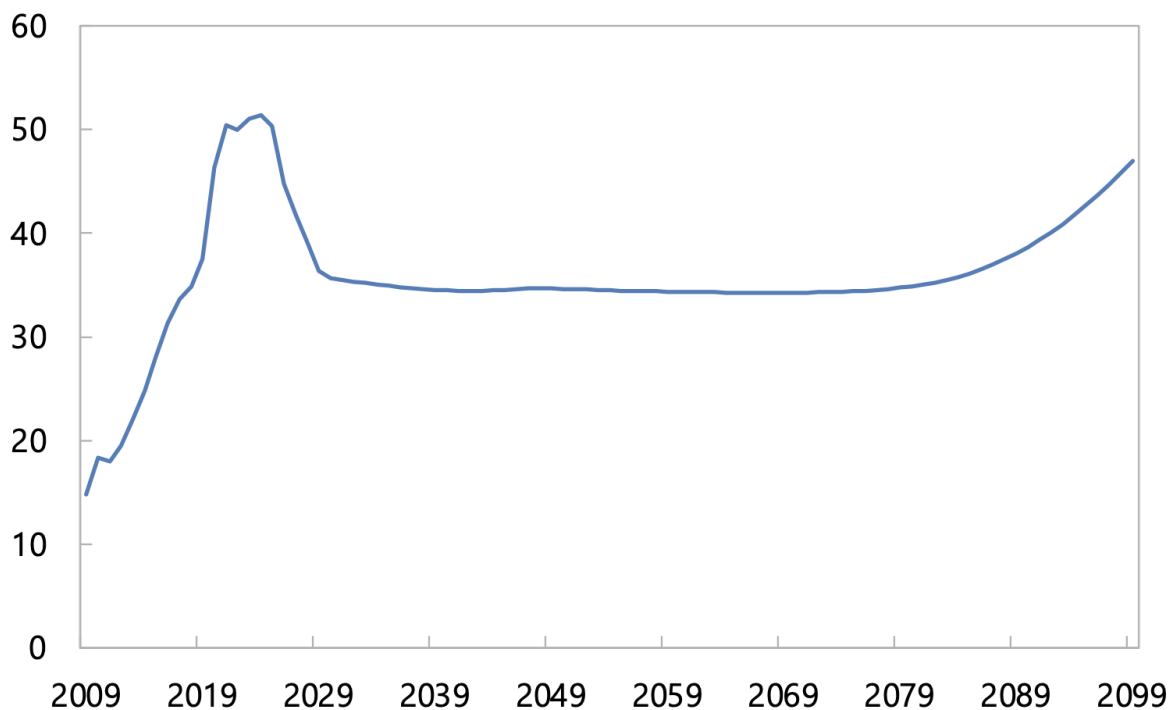
On the Dashboard	Ships as	What it sets
Inflation, start and end (C24, C25)	3.5 and 3.5	the deflator path that turns real growth nominal
Interest rate (C28, C29)	Nominal interest rate	the r applied to the debt stock
Rule and Debt Target (C33, C34)	Yes and 60	the fiscal rule acting on the primary balance
Expenditure Rigidity (C38)	1.0	how fully spending holds on when growth disappoints

The shipped settings are a real run: fiscal rule on, debt target 60 percent of GDP. The User Guide’s worked example instead starts with the rule off (p. 18). The one in Chapter 5 keeps the shipped setting and documents both readings.

A.3 Where the outputs live

Two yellow sheets carry the results: *Output Baseline* for the no-climate run, *Output Scenarios* for the comparison. The Read Me has you assess the baseline before any scenario, the same order the course works in.

Debt-to-GDP (%)



Source: QCRAFT (2024)

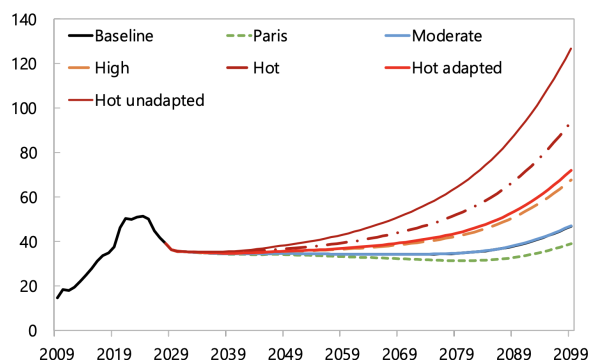
Figure A.2: Output Baseline, Uganda: the no-climate debt path every scenario is compared against.

Output Scenarios opens with summary tables, then charts. The debt table reads the ratio at 2023, 2050, 2075 and 2099 for the baseline and all six scenarios, and the debt chart matches the Explorer's Analysis tab. With Uganda selected and all controls at shipped values, the baseline reads 47.0 percent of GDP at 2099, the figure the worked example in Chapter 5 builds on.

Uganda					
Summary tables					
Primary balance (% of GDP)	2023	2050	2075	2099	
Baseline	-1.6	0.4	0.8	0.2	
Paris		0.4	0.8	0.3	
Moderate		0.4	0.8	0.2	
High		0.3	0.6	-0.1	
Hot		0.2	0.3	-0.5	
Hot adapted		0.3	0.5	-0.2	
Hot unadapted		0.0	0.0	-0.9	
Debt-to-GDP ratio (%)					
	2023	2050	2075	2099	
Baseline	51.0	34.6	34.4	47.0	
Paris		34.1	31.7	39.2	
Moderate		34.6	34.4	47.2	
High		35.7	40.6	67.8	
Hot		37.0	48.4	94.0	
Hot adapted		35.8	41.7	72.0	
Hot unadapted		38.8	58.4	126.9	

(a) The summary tables at the top of Output Scenarios, Uganda.

Debt-GDP ratio, %



Sources: Q-CRAFT (2024)

(a) The debt chart on the same sheet: a baseline and six warming scenarios.

A.4 The ten steps, and where the course taught each

Of the ten steps, the course covered every concept in the eight core ones.

The workbook's step	Where the concept lives
1. Choose country	the ten-minute run in Chapter 2
2. Choose demography scenario	Chapter 4
3. Choose productivity assumptions	Chapter 4
4. Choose inflation assumption	Chapter 4
5. Choose interest rate assumptions	Chapter 4
6. Assess the baseline	Chapter 5
7. Analyse the macroeconomic effects of climate change	Chapter 3, Step 3
8. Analyse the climate scenarios	Chapter 5
9. Add discrete risks (optional)	outside this course; the User Guide covers it
10. Adjust expenditure rigidity (optional)	Chapter 4

A.5 Three operational differences from the Explorer

The workbook fixes the data vintage, carries no rationale field, and shows #VALUE! where the Explorer would explain the gap in plain language. That vintage is WEO October 2024 with the 2022 UN population revision. The data ages until the Fund posts a new file. Saving your assumptions means saving the file: no export packet, so the write-it-down discipline from Chapter 4 is yours to keep by hand.

Appendix B

Co-design and the SovTech vision

This appendix is not part of the course

The modules teach running the tool and defending the analysis. This appendix covers the project behind it: why the tool was built and what it is a proof of concept for. Nothing in the capstone depends on it. Read it to influence what the next version does.

This is not an IMF product

Q-CRAFT Explorer, this course and everything proposed below are an independent project by [Teal Insights](#) and [NatureFinance](#). None of it is an official IMF product and none of it carries IMF endorsement. The work is meant to be complementary to the IMF's own training materials and to the Q-CRAFT User Guide ([Tim and Rahman, 2024](#)), which remain the authoritative references. Nothing in this appendix should be read as a criticism of a tool the IMF Fiscal Affairs Department built for its own technical assistance and made available to everyone.

What you need to know:

- Q-CRAFT is a proof of concept for a larger goal: making the LIC-DSF more usable through human-centered design
- The biggest barrier to effective fiscal projection tools is ergonomics, not economics
- We are looking for co-design partners, not endorsements

B.1 Ergonomics is the biggest barrier to using fiscal projection tools

Tool design stops people who can handle the economics. The tools do not guide them through the decisions they must make.

Years of conversations with MoF economists, central bank staff, and IFI economists all pointed the same way. So we ran structured design sprints and discussions with practitioners who use these tools daily.

B.2 We rebuilt Q-CRAFT as a web app and verified it against the Excel workbook

Q-CRAFT Explorer reimplements the IMF’s Q-CRAFT Excel workbook as an open-source web application. The [calculation engine](#) is a standalone Python package, developed and tested separately from the interface, with automated parity tests against the original Excel tool.

The testing pipeline is public and automated. Baseline parity is exact for 147 of 147 tested countries: zero percentage-point deviation across debt-to-GDP, revenue, primary balance and primary expenditure as shares of GDP, with a further 25 parameter sensitivity combinations passing. Climate-scenario parity is confirmed for ratio metrics only, meaning debt-to-GDP, revenue and expenditure as percent of GDP. Higher-stakes tools need the same transparent, automated verification. Chapter 2 has the fuller statement of what the parity claim covers.

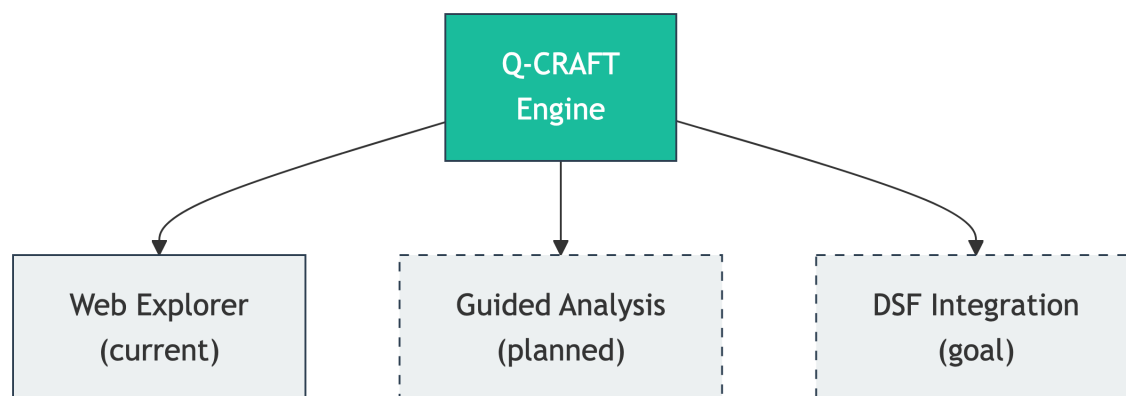
The deployed app carries that verification into the workflow and ships two data modes. Current, the default, bundles the latest WEO and UN population releases. Verified uses the workbook’s own vintages, so any number can be checked against the original. Source-record context panels sit beside six of the ten controls, showing what the underlying sources publish while you choose. Any control moved off its engine default gets a CHANGED badge and a one-line rationale field, and the export annex records all ten settings either way. Three teaching widgets are linked from the header. The export packet contains the charts, the CSVs, the assumptions annex and a run file that restores the whole configuration, twelve files in one download. Because the app builds to a static bundle, a ministry can serve it inside its own network with one command, no outside connection required.

B.3 Q-CRAFT is the proof of concept

Q-CRAFT is deliberately small: ten user-facing controls, one core equation. It demonstrates the approach in miniature.

The real goal is the LIC-DSF, the Low-Income Country Debt Sustainability Framework. It is far more complex, far more consequential, and used for every low-income country. We already work with LIC-DSF end-users and want to work with its designers too.

B.4 One engine powers three interfaces



One engine, multiple interfaces

Each front-end serves a different need:

1. **Web Explorer (current).** Select a country, choose the assumptions with the source records beside you, compare scenarios, export the packet.
2. **Guided Analysis Tool (planned).** AI-assisted help deciding what to enter and why. Always available, cheaper than a technical assistance mission.
3. **DSF Integration (the goal).** Q-CRAFT and [DIGNAD](#) feeding into LIC-DSF assumptions through the same ergonomic layer.

Today, if you do not know what to enter for a parameter, you search a user guide and hope. The context panels and rationale field are the first pass at design that makes the right thing easy and the wrong thing hard.

B.5 The co-design invitation

The current version is live, and practitioner input shapes the next one in five areas.

1. **Parameter guidance.** What would make the six context panels more useful, and what heuristics do experienced practitioners use that the panels do not carry?
2. **Default assumptions.** What are sensible defaults by country type? Should defaults vary by region?
3. **Capacity development integration.** How would this tool fit into an actual CD workshop or IMF mission?
4. **Validation priorities.** Which countries and scenarios should we prioritize for verification?
5. **Missing features.** What is the first thing you would want that is not here?

We are asking for a few hours from a few people: design feedback, a continuing conversation as we build, and word on whether we are making something useful.

We are not asking for official endorsement, confidential information, or money: we understand institutional constraints and work with publicly available materials. Everything is permissively licensed open source, free to use, copy and improve.

B.6 Q-CRAFT delivers value now and proves the approach

We start with Q-CRAFT for two reasons:

1. **Immediate value.** A co-designed Q-CRAFT tool could be in use within weeks, costs nothing, and requires minimal commitment to evaluate. It complements, not replaces, the Excel workbook.
2. **Pioneer the approach.** We prove the design process works on a small tool before applying it to the one that matters most.

B.7 The SovTech vision

Central banks have been applying technology to supervisory functions for years. The BIS documented “SupTech generations”, a framework for technology in supervision and regulation ([di Castri et al., 2019](#)). The open question is when sovereign debt management will catch up.

We call this SovTech: SupTech principles applied to sovereign debt analysis. Three principles guide this work:

Modularity respects how people work. Components fit together like building blocks, so a team can swap one module without rebuilding everything.

Open source makes the tool institutional infrastructure. Anyone can examine how the analysis is done, and the permissive MIT license puts no barrier to adoption.

AI has made analytical software dramatically cheaper to build, and a trust layer makes it safe to rely on. Sovereign debt analysis serves policy decisions worth billions of dollars. At those stakes, verification pipelines, automated testing, and human-in-the-loop governance are requirements. Hence the parity verification pipeline described in Chapter 2.

Q-CRAFT Explorer is our attempt to show what this looks like. The source code is on [GitHub](#). We welcome your input.

Glossary

- Climate scenarios** Six warming pathways used in Q-CRAFT, based on IPCC Shared Socioeconomic Pathways (SSPs). The coolest is Paris-aligned (below 2°C warming by 2100, SSP1-2.6). The hottest is Hot + Unadapted (high warming with slow adaptation, SSP3-7.0 at the 90th percentile). Climate impacts on GDP are derived from the FADCP Climate Dataset (Centorino, Massetti, and Tagklis, 2024), building on the empirical estimates of Kahn et al. (2021). See the IMF User Guide, Section IV.B for detailed methodology.
- C-PIMA (Climate Public Investment Management Assessment)** An IMF framework for evaluating how well a country’s public investment management institutions account for climate change. C-PIMA engagements can include Q-CRAFT modelling of the fiscal impacts of climate scenarios for the assessed country. Uganda’s 2024 assessment reports one such analysis. See the C-PIMA Handbook (IMF, 2025) for the full framework.
- DSA (Debt Sustainability Analysis)** A framework used by the IMF and World Bank to assess whether a country’s debt is sustainable (whether the government can meet its current and future debt obligations without debt relief or arrears). Q-CRAFT can complement a DSA with long-term climate-adjusted fiscal trajectories.
- Debt dynamics equation** The core equation in Q-CRAFT: $d_t = d_{t-1} \times \frac{1+r_t}{1+g_t} - pb_t$. Next year’s debt-to-GDP ratio equals the current ratio, adjusted for the interest-growth differential, minus the primary balance. When interest rates exceed growth ($r > g$), debt rises automatically: the government must run a primary surplus even to hold debt still. See the IMF User Guide, Section IV.A on debt dynamics for detailed derivation and corollaries.
- Debt-to-GDP ratio** Total government debt expressed as a percentage of nominal GDP. The standard measure of a country’s debt burden, since it scales debt relative to the economy’s ability to service it. Q-CRAFT projects this ratio through 2099 under baseline and climate scenarios.
- Expenditure rigidity** The degree to which government spending resists downward adjustment when climate shocks reduce GDP. Measured on a 0-1 scale. A value of 1.0 means spending is fully sticky: it stays at the baseline level in local currency terms even as GDP falls, the worst case for debt accumulation. A value of 0.0 means spending adjusts fully with GDP, maintaining the baseline expenditure-to-GDP ratio. This parameter only affects climate scenarios, not the baseline. See the IMF User Guide, pp. 20 and 35-36.
- Fiscal rule** A mechanism in Q-CRAFT that adjusts primary expenditure to steer debt toward a target ratio. When enabled, the model computes the debt-stabilizing primary balance and adjusts spending by the fiscal gap (the difference between the actual and debt-stabilizing primary balance). The adjustment is additive in levels (not a rate) and depends on the prior year’s state, so the model must compute it year by year. See the IMF User Guide, pp. 15-18.
- DIGNAD (Debt, Investment, Growth, and Natural Disasters)** An IMF dynamic general equilibrium model of natural disasters and public investment in developing countries. As a

general equilibrium model it carries far more structure than Q-CRAFT's single recursion.

Discussed in the co-design appendix as a tool that could benefit from the SovTech approach.

Golden master In software verification, a reference output used to detect unintended changes. Q-CRAFT's golden master tests compare the Python engine's outputs against values extracted from the IMF's Excel workbook for the same country and parameter inputs. Parity with the Excel tool (not any independent calculation) is the acceptance criterion.

LIC-DSF (Low-Income Country Debt Sustainability Framework) The IMF-World Bank framework for assessing debt sustainability in low-income countries, built around country-specific debt burden thresholds. Compared against Q-CRAFT in Module 5, and discussed in the co-design appendix as a more complex tool that could benefit from the SovTech approach.

Interest-growth differential The difference between the nominal interest rate on government debt (r) and nominal GDP growth (g), expressed as $(r - g)/(1 + g)$. When positive ($r > g$), debt dynamics are unfavorable: the debt ratio rises automatically. When negative ($g > r$), debt dynamics are favorable. This single variable determines whether debt is self-stabilizing or self-reinforcing.

Primary balance Government revenue minus non-interest expenditure (primary expenditure), expressed as a share of GDP. The primary balance is the fiscal variable a government can control in the near to medium term. A primary surplus means the government generates enough revenue to cover its non-interest spending. A primary deficit means it does not.

WEO (World Economic Outlook) The IMF's flagship publication providing analysis and projections of the global economy. Q-CRAFT uses the WEO database as the source for macroeconomic and fiscal data through the medium-term forecast horizon: the Explorer's Verified mode runs the October 2024 release the workbook ships, and Current mode, the default, runs April 2026, with the horizon marked on every chart. A country's projection starts from the last year that carries the data the model needs, which for a few countries is earlier than the release boundary. The Explorer names that anchor year on screen where the two differ. Beyond the anchor year, Q-CRAFT's own projection methodology takes over.

SovTech The application of SupTech (supervisory technology) principles (modularity, open source, human-centered design) to sovereign debt analysis and public financial management tools. Q-CRAFT Explorer is a proof of concept for this approach.

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